



Splashery

The catalog: 319 toys made of 3D Gaussian
splats

Every toy runs in your browser, free, with nothing to install. Each has a page here with what it is, how to play with it, what you can change, and where it comes from. Scan a toy's code, or follow its link, to play with it. The contents on the next page link to every toy, and your PDF reader's bookmarks list them by shelf.



ryanjosephkamp.github.io/splashery/site/

October 8, 2026. Splashery's code is under the MIT license; each toy's assets keep their own licenses, credited on its page.

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Cactus

Tap it to open a ring of flowers around its top.



Scan to play with it.

Learn more: [Cactus on Wikipedia](#)

A cactus is a plant built to live where rain is rare. Its thick green stem stores water, and its spines are leaves that have changed shape: they shade the stem and keep hungry animals away. The stem, not leaves, does the work of turning sunlight into food.

Tap it and nine magenta flowers open one after another in a ring around the top, then close again. Many cacti bloom for only a few days a year, and some flowers open for just one night.

HOW TO PLAY

- **Tap:** Tiny flowers bloom on the spines, then close.
- **Sound:** Soft pluck with a dry rattle.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/cactus/

CREDITS

- **Scan:** [Cactus \(3DGS sample data\)](#) by steam studio / 3D SCAN STUDIO iris, CC0 1.0.

Strawberry

Tap it to pop out its seeds.



Scan to play with it.

Learn more:

[Strawberry on Wikipedia](#)

A strawberry is not quite a berry. The red part is the swollen tip of the flower's stem, and the tiny yellow specks on the outside are its real fruits, each holding one seed. A single strawberry carries about 200 of them.

Tap it and the seeds pop out in a wave and glow, the red flesh blushes deeper, and then everything settles back. Strawberry plants spread by sending out runners, long thin stems that take root and grow new plants.

HOW TO PLAY

- **Tap:** Seeds pop out and sparkle, then settle back; the berry blushes redder.
- **Sound:** Juicy squish.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/strawberry/

CREDITS

- **Scan:** [Strawberry](#) by Dany Bittel, CC BY 4.0.



Heart cookie

Tap it to make the jam heart beat.



Scan to play with it.

A jam cookie like this one is two thin, crumbly cookies with a layer of fruit jam between them. The top cookie has a heart cut out of it, so the red jam shows through like a window.

Tap it and the jam heart glows and swells twice, like a heartbeat, in time with two soft thumps. Cookies with a window of jam are baked in many countries, often for holidays, and the top is often dusted with powdered sugar.

HOW TO PLAY

- **Tap:** The jam heart glows and throbs twice like a heartbeat.
- **Sound:** Short crunch and a soft double thump.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/cookie/

CREDITS

- **Scan:** [Heart Cookie](#) by Dany Bittel, CC BY 4.0.

Honeybee

Tap it to make it buzz its wings and lift off.



Scan to play with it.

Learn more:

[Western honey bee](#)
[on Wikipedia](#)

Honeybees live together in large colonies of tens of thousands, with one queen who lays the eggs. Worker bees fly from flower to flower drinking nectar and gathering pollen, and back at the hive they turn the nectar into honey. As they go, they carry pollen between flowers, which helps plants make fruit and seeds.

Tap it and its wings buzz in a blur as it lifts off, tilts and lands again. A honeybee's wings beat about 200 times a second: that is what makes the buzz.

HOW TO PLAY

- **Tap:** Wings buzz in a blur and the bee lifts off a little, then lands.
- **Sound:** Honeybee buzz (higher than the bumblebee).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/bee/

CREDITS

- **Scan:** [Japanese Bee](#) by YUMA Co., Ltd., CC BY 4.0.



Cluster fly

Tap it to make it groom its legs and face.



Scan to play with it.

Learn more: [Pollenia rudis on Wikipedia](#)

Cluster flies are a little larger than house flies, with golden hairs on their backs. They get their name from the way hundreds of them gather in attics and window frames in autumn to wait out the winter. Their young live in the soil, feeding on earthworms.

Tap it and it grooms like a real fly: the front legs rub together, reach up and wipe its face while its head dips, and then it gives a little shake. Flies clean themselves often because their legs and eyes need to stay free of dust to taste and see.

HOW TO PLAY

- **Tap:** Front legs rub together, then the wings flick.
- **Sound:** Short thin fly buzz.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/cluster-fly/

CREDITS

- **Scan:** [Cluster Fly](#) by Dany Bittel, CC BY 4.0.



May beetle

Tap it to lift its wing cases and fan its antennae.



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Learn more:
[Cockchafer on Wikipedia](#)

The May beetle, or cockchafer, is a big brown beetle that flies on warm evenings in late spring. Its hard wing cases cover a pair of thin flying wings folded underneath, and its antennae end in little fans that it spreads to smell the air.

Tap it and the wing cases lift and part with a shiver, and the fans spread, then everything folds away again. A cockchafer lives most of its life underground as a white grub, eating roots for three or four years before it digs out as a beetle.

HOW TO PLAY

- **Tap:** Wing cases lift and the fanned antennae spread, then fold.
- **Sound:** Low clicking whirr.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/may-beetle/

CREDITS

- **Scan:** [MAY-BEETLE-HIRES](#) by trilithon, CC BY 4.0.



Millipede

Tap it to send a wave down its legs.



Scan to play with it.

Learn more:
[Millipede on Wikipedia](#)

A millipede is a long, slow animal with a body made of many rings, and most of the rings carry two pairs of legs. The name means "thousand feet," though most kinds have far fewer. Millipedes munch on dead leaves and rotting wood, which helps turn them back into soil.

Tap it and a wave ripples down its legs as it stretches out straight, then it curls up tighter and relaxes. The leggiest millipede known, found in Australia in 2021, has more than 1,300 legs.

HOW TO PLAY

- **Tap:** A leg wave ripples down the body while it stretches and curls.
- **Sound:** Soft many-footed patter.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/millipede/

CREDITS

- **Scan:** [millipede spec](#). by scant3d, CC BY 4.0.



Carder bumblebee

Tap it to make it hover.



Scan to play with it.

Learn more: [Brown-banded carder bee](#)
on [Wikipedia](#)

This is a brown-banded carder bumblebee, a small, fuzzy bumblebee of Europe's grassy meadows. Carder bees make their nests on the ground in thick grass and cover them with bits of moss and grass that they comb into place.

Tap it and the fuzzy body shivers, the wings hum, and it hovers up with a slow, clumsy wobble before settling down. Bumblebees can shiver their flight muscles to warm up, so they can fly on cool mornings when many other bees stay home.

HOW TO PLAY

- **Tap:** Fuzzy body shivers and the wings hum; a slow, clumsy hover.
- **Sound:** Deep, lazy bumblebee drone.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/bumblebee/

CREDITS

- **Scan:** [Macroscan - Brown-banded carder bumblebee \(Bombus humilis\)](#) by macroscans, CC BY 4.0.



Raspberry

Tap it to drop its drupelets and put them back.



Scan to play with it.

Learn more: [Raspberry on Wikipedia](#)

A raspberry is made of many tiny round fruits called drupelets, each with its own juicy skin and one small seed inside. They grow stuck together around a soft core. When you pick a raspberry the core stays on the plant, which is why the berry is hollow.

Tap it and the drupelets break off one after another, tumble to the table and hop back into place. A drupelet only grows if its own part of the flower got pollen, so bees help make a full, round berry.

HOW TO PLAY

- **Tap:** The drupelets bounce one by one in a ripple.
- **Sound:** A real juicy squish as the drupelets break off, then a soft tap for each one landing on the table, and a small squish as they come back (no bubbles).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/raspberry/

CREDITS

- **Scan:** [Raspberry](#) by Dany Bittel, CC BY 4.0.
- **Sound:** [Impact Sounds \(impactSoft_medium_000.ogg\)](#) by Kenney, CC0 1.0.
- **Sound:** [Squish impact](#) by Bertsz, CC0 1.0.



Blackberry

Tap it to burst off its drupelets.



Scan to play with it.

Learn more: [Blackberry on Wikipedia](#)

A blackberry, like a raspberry, is a cluster of many tiny, shiny fruits called drupelets, each with a seed inside. Blackberries grow wild on thorny, tangled bushes called brambles, and ripen from green to red to deep black.

Tap it and half the glossy drupelets burst off, bounce on the table and glint, then spring back. Unlike a raspberry, a blackberry keeps its soft white core when you pick it, so it is solid all the way through instead of hollow.

HOW TO PLAY

- **Tap:** Drupelets gleam in a wave, like light running over it.
- **Sound:** A darker, deeper real squish, the drupelets tapping down on the table as they bounce, and a soft squish as they spring back (no bubbles).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/blackberry/

CREDITS

- **Scan:** [Blackberry](#) by Dany Bittel, CC BY 4.0.
- **Sound:** [Impact Sounds \(impactSoft_medium_000.ogg\)](#) by Kenney, CC0 1.0.
- **Sound:** [Squish impact](#) by Bertsz, CC0 1.0.



Blueberry

Tap it to peel back its skin.



Scan to play with it.

Learn more:
[Blueberry on Wikipedia](#)

Blueberries are small round berries that grow on bushes; the main cultivated kinds come from North America. A fresh one wears a dusty, pale coat called bloom: a thin layer of wax the berry makes to protect itself. At the bottom sits a little five-pointed crown, the leftover end of the flower.

Tap it and the dark skin peels back toward you in five strips to show the pale green flesh, then closes again. Almost all the blue is in the skin; the inside of most blueberries is pale.

HOW TO PLAY

- **Tap:** The bloom (dusty coat) wipes off in a swirl, showing deep blue, then returns.
- **Sound:** Small round pop.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/blueberry/

CREDITS

- **Scan:** [Blueberry](#) by Dany Bittel, CC BY 4.0.



Grape

Tap it to peel back the skin.



Scan to play with it.

Learn more: [Grape on Wikipedia](#)

A real grape, captured from many photos and drawn with many thousands of tiny, soft splats. Grapes grow in bunches on woody vines; people eat them fresh, dry them into raisins and press them for juice.

Tap it and four strips of the dark skin peel back from the stem end to show the pale flesh underneath, then the skin closes again. The color of a grape is almost all in its skin: in most table grapes, the flesh beneath is pale and see-through, though some varieties have colored flesh.

HOW TO PLAY

- **Tap:** The skin peels back in strips to show the pale flesh, then closes.
- **Sound:** Slippery peel.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/grape/

CREDITS

- **Scan:** [Grape](#) by Dany Bittel, CC BY 4.0.



Cinnamon star cookie

Tap it to crumble it and put it back together.



Scan to play with it.

Learn more:
[Zimtstern on Wikipedia](#)

Cinnamon stars are small star-shaped cookies baked at Christmas, above all in Germany, Switzerland and Austria. They are made with ground almonds or hazelnuts, sugar, egg whites and cinnamon, often with little or no flour, and are topped with a crisp white glaze of beaten egg white and sugar.

Tap it and it crumbles into pieces and crumbs that fall away, then puts itself back together. The glaze goes on before baking, and a low oven keeps it snowy white.

HOW TO PLAY

- **Tap:** Crumbles into pieces and crumbs, then reassembles.
- **Sound:** A real crisp snap and crumble as it breaks, and a small soft crumble as it puts itself back together.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/star-cookie/

CREDITS

- **Scan:** [Cinnamon Star Cookie](#) by Dany Bittel, CC BY 4.0.
- **Sound:** [Crisp Crunch.m4a](#) by Yin_Yang_Jake007, CC0 1.0.



Tomatoes

Tap it to make the tomatoes hop and rock. With 🖐️ on, drag a tomato to lift it and set it down.



Scan to play with it.

Learn more: [Tomato on Wikipedia](#)

Tomatoes are the fruit of a plant that first grew wild in western South America. They are cooked and eaten like vegetables, but to a botanist a tomato is a berry: a juicy fruit that grows from a flower and holds many seeds.

Tap it and the plate gives a little shake: a wave of small hops runs around it, and the tomatoes land in four groups, rock outward and settle with a wobble. There are thousands of kinds of tomato, from tiny cherry tomatoes to ones bigger than a fist, in red, yellow, orange, green and purple. With 🖐️ Hands-on on, drag any tomato to lift it on its own and set it down on the plate or on another; reset puts them all back.

HOW TO PLAY

- **Tap:** The tomatoes roll and jostle each other, then settle.
- **Sound:** Soft thuds rolling on a table.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

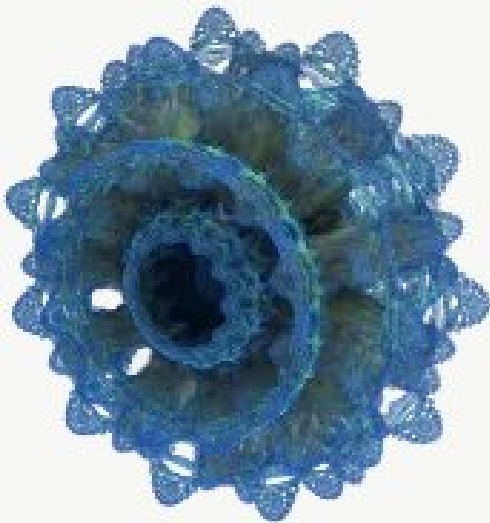
SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/tomatoes/

CREDITS

- **Scan:** [Tomatoes - No Postshot!](#) by simonbethke, CC BY 4.0.



Mandeltorus

Tap it to spin its rings opposite ways.



Scan to play with it.

The mandeltorus is a fractal: a shape whose small parts repeat the pattern of the whole, over and over, smaller and smaller. It is a piece of math art, made by a computer from a formula related to the Mandelbrot set and bent into a ring, or torus.

Tap it and its rings turn opposite ways in bands with a cyan glow, then lock back into place. The mathematician Benoit Mandelbrot made up the word "fractal" in 1975; fractal patterns turn up in nature too, in ferns, coastlines and snowflakes.

HOW TO PLAY

- **Tap:** Rings of the fractal counter-rotate in bands, then lock back.
- **Sound:** Like the Mandelbulb: a low drone with a ratchet as the rings start to turn, and again as they lock back.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/mandeltorus/

CREDITS

- **Scan:** [MandelTorus 4,12,8,1.5 \(Spirula Studio\)](#) by [harry7557558](#), CC BY 4.0.

Basket

Tap it to shake the shells.



Scan to play with it.

Learn more: [Basket](#)
on [Wikipedia](#)

A woven basket full of seashells from the beach: sea urchins, sand dollars, starfish and a scallop. Baskets are made by weaving strips of plants, like willow, reed or grass, over and under each other, and people have woven them for thousands of years.

Tap it and the basket shakes: ten of the biggest shells bounce up one after another, spinning, and drop back. A sand dollar is a kind of flat sea urchin; alive, it is covered in short, soft spines that it uses to move through the sand.

HOW TO PLAY

- **Tap:** Shells hop inside the basket one after another.
- **Sound:** Clattering shells.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

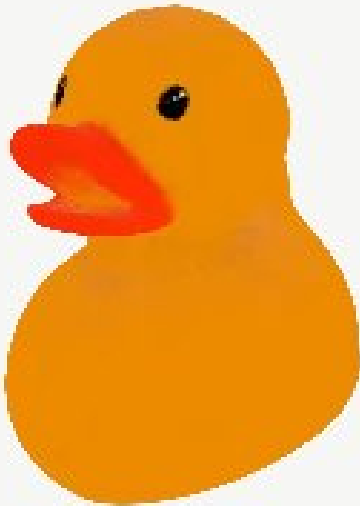
Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/basket/

CREDITS

- **Scan:** [Basket](#) by hollmar, CC BY 4.0.

Real rubber duck

Tap it to squeeze it.



Scan to play with it.

Learn more: [Rubber duck on Wikipedia](#)

The rubber duck is a bath toy: a hollow yellow duck that floats. Squeeze it and air rushes out through a tiny hole in its base, which often holds a small whistle that squeaks.

Tap it and it squeezes flat with a high squeak, then springs back up taller, wobbles and settles. Most bath ducks today are made of soft plastic rather than real rubber, but the name has stuck.

HOW TO PLAY

- **Tap:** Squeezes flat and springs back.
- **Sound:** Squeaky squeeze (high whistle) — the other duck quacks.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/rubber-duck-real/

CREDITS

- **Scan:** [Rubber Duck Toy](#) by Plat251, CC0 1.0.



Garden gnome

Tap it to make it hop and light its lantern.



Scan to play with it.

Learn more: [Garden gnome on Wikipedia](#)

Garden gnomes are little statues of bearded men in pointed hats, set among the flowers to look after the garden. In old European tales, gnomes were small, shy folk who lived underground and guarded treasure. The first garden gnomes were made of clay in Germany in the 1800s.

Tap it and the gnome gives a little hop, and the lantern in its hand lights up with a warm glow, then goes out again. This one wears the classic outfit: a red cap, a white beard and a belt around a green shirt.

HOW TO PLAY

- **Tap:** The lantern lights with a warm glow and the gnome gives a little hop.
- **Sound:** A soft stone thud as the gnome lands from its hop, and a match-like flare and soft flame as its lantern lights (no chirp, no click).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/garden-gnome/

CREDITS

- **Scan:** [Garden Gnome](#) by Bhargav Kubal, CC0 1.0.
- **Sound:** [Match Lighting Candle](#) by devilqube, CC0 1.0.



Wooden elephant

Tap it to raise its trunk and trumpet.



Scan to play with it.

A carved wooden elephant, the kind of figure woodcarvers have made for a very long time, shaped by hand with knives and chisels. Elephants are the largest animals on land. Their trunk is a nose and an upper lip joined together, strong enough to lift a log and gentle enough to pick up a single peanut.

Tap it and the trunk uncurls straight up to trumpet with a few toots, then curls back down while the elephant rocks on its feet. An elephant's trunk has no bones at all, but tens of thousands of muscle bundles.

HOW TO PLAY

- **Tap:** Trunk lifts and it rocks on its wooden feet.
- **Sound:** A light wooden knock, then a real elephant trumpeting as the trunk lifts.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/wooden-elephant/

CREDITS

- **Scan:** [Carved Wooden Elephant](#) by Greg Zaal, CC0 1.0.
- **Sound:** [Elephant Trumpets Growls.flac](#) by D.jones, CC0 1.0.



Marble bust

Tap it to make it turn and say hello in Latin.



Scan to play with it.

Learn more: [Bust \(sculpture\) on Wikipedia](#)

A bust is a statue of a person's head, neck and shoulders. The ancient Romans carved many busts of emperors, leaders and families in marble, a stone that is soft enough to carve but polishes to a smooth, pale shine.

Tap it and the head turns on its neck to look at you, the jaw moves with each syllable, and a speech bubble says "SALVE, AMICE!": Latin for "Hello, friend!" Latin was the language of ancient Rome, and many English words grew from it.

HOW TO PLAY

- **Tap:** The head turns to look at you; a speech bubble says a short Latin line.
- **Sound:** Stone grind and a low murmur.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/marble-bust/

CREDITS

- **Scan:** [Marble Bust 01](#) by Rico Cilliers, CC0 1.0.



Ukulele

Tap it to strum a chord.



Scan to play with it.

Learn more: [Ukulele on Wikipedia](#)

The ukulele is a small, four-stringed guitar from Hawaii. It grew from little guitars that Portuguese settlers brought to the islands in the 1870s, and its name is often said to mean "jumping flea," for the way a player's fingers hop over the strings.

Tap it and a pick sweeps across the strings four times, and each string shakes back and forth between the two ends where it is held. A string's note depends on how long, how tight and how heavy it is: a shorter, tighter string plays higher.

HOW TO PLAY

- **Tap:** Strings vibrate and it plays a short strum pattern.
- **Sound:** Plucked ukulele chord (synth, bright, nylon).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/ukulele/

CREDITS

- **Scan:** [Ukulele 01](#) by Joseph Burgan, CC0 1.0.



Real alarm clock

Tap it to ring the bells.



Scan to play with it.

Learn more: [Alarm clock on Wikipedia](#)

An old wind-up alarm clock runs on a spring: winding the key tightens the spring, and as it slowly unwinds it turns the gears and the hands. Two metal bells sit on top, and when the alarm goes off a little hammer rattles back and forth between them.

Its hands show the real time on this device, in its local time zone, and the red second hand ticks once a second. Tap it and the twin bells ring while the clock rattles across the table. Before alarm clocks were common, some people paid a "knocker-up" to tap on windows with a long pole and wake people for work.

HOW TO PLAY

- **Tap:** The hands show the real time and the second hand ticks; tap rattles it off the table while it rings.
- **Sound:** A real mechanical twin-bell alarm ringing while the clock rattles (2.3 s).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/alarm-clock/

CREDITS

- **Scan:** [Alarm Clock 01](#) by Yann Kervran, James Ray Cock, CC0 1.0.
- **Sound:** [old alarm clock ringing](#) by tuberatanka, CC0 1.0.



Vintage camera

Tap it to take a photo with a flash.



Scan to play with it.

Learn more: [Camera on Wikipedia](#)

An old film camera. Inside, a roll of film coated with chemicals that change when light hits them sits behind the lens. Pressing the button opens a shutter for a split second, the lens lets in a picture of the world, and the film keeps it. Then a lever winds the film on to a fresh frame.

Tap it and the flash bursts, the whole scene whites out for a moment, and you hear the shutter and a stylized whoomph from the flash. Early cameras needed people to sit still for minutes; a modern shutter can open for less than a thousandth of a second.

HOW TO PLAY

- **Tap:** Flash burst from the camera and the view briefly whites out.
- **Sound:** A real vintage press camera: a soft shutter and the whoomph of the flash powder.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/vintage-camera/

CREDITS

- **Scan:** [Camera 01](#) by Rajil Jose Macatangay, CC0 1.0.
- **Sound:** [Vintage Camera Flash Powder and Shutter](#) by Werra, CC0 1.0.



Boombox

Tap it to play a beat.



Scan to play with it.

Learn more:
[Boombox on Wikipedia](#)

A boombox is a portable stereo: a radio, a cassette tape player and two big speakers in one box with a handle, run on batteries so music could go anywhere. They were hugely popular in the 1970s and 1980s, carried to parks, streets and beaches.

Tap it and it plays two bars of a beat: the speakers pump and glow blue on each kick drum, and the tape reels turn. A cassette holds a thin plastic tape covered in tiny magnetic grains, and the music is stored in the pattern of their magnetism.

HOW TO PLAY

- **Tap:** Speakers pulse to a beat and the tape reels turn.
- **Sound:** A short synth beat loop.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/boombox/

CREDITS

- **Scan:** [Boombox](#) by Thomas Paul Mouilleron, CC0 1.0.



Real croissant

Tap it to tear it open.



Scan to play with it.

Learn more:
[Croissant on Wikipedia](#)

A croissant is a flaky, buttery pastry shaped like a crescent moon, and "croissant" is the French word for crescent. Its dough is folded around a slab of butter again and again, which makes dozens of thin layers. In the hot oven, water in the butter turns to steam and puffs the layers apart.

Tap it and it tears in half to show the soft, layered inside, a puff of steam rises, and then it closes up again. Croissants are a favorite French breakfast, but the idea came to France from Vienna, in Austria.

HOW TO PLAY

- **Tap:** Tears in half and steam puffs out.
- **Sound:** Flaky tear.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/croissant-real/

CREDITS

- **Scan:** [Croissant](#) by Greg Zaal, Dario Barresi, CC0 1.0.



Carrot cake

Tap it to lift out a slice.



Scan to play with it.

Learn more: [Carrot cake on Wikipedia](#)

Carrot cake is a spiced cake made with grated carrots, which bake into the crumb and keep it moist. It is often baked with cinnamon and nuts and topped with a thick, tangy cream cheese frosting.

Tap it and a slice facing you lifts out of the cake, showing the layers of crumb and frosting on its cut sides and in the gap. Carrots are sweet enough to bake with because they are among the sweeter vegetables.

HOW TO PLAY

- **Tap:** A slice cuts out and lifts away, showing layers.
- **Sound:** A softer knife slide, a moist crumb as the slice lifts out, and the same soft plate thud.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/carrot-cake/

CREDITS

- **Scan:** [Carrot Cake](#) by Greg Zaal, James Ray Cock, CC0 1.0.

Pomegranate

Tap it to split it open.



Scan to play with it.

Learn more:

[Pomegranate on Wikipedia](#)

A pomegranate is a round fruit with a tough, leathery skin and a little crown on top. Inside, it is packed with hundreds of seeds, each wrapped in a juicy, jewel-red coat, in pockets separated by a thin, bitter, pale skin. It first grew in the lands around Iran and has been eaten for thousands of years.

Tap it and it splits open like a book on a hinge at the bottom, showing faces packed with glowing red seeds, and a few tumble out. People eat the juicy seed coats and often leave out the white skin.

HOW TO PLAY

- **Tap:** Splits open to show glowing red seeds that spill a little.
- **Sound:** Crack and a scatter of beads.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/pomegranate/

CREDITS

- **Scan:** [Food Pomegranate 01](#) by Oliver Harries, CC0 1.0.



Lantern

Tap to light the lantern; tap again to put it out.



Scan to play with it.

Learn more: [Lantern on Wikipedia](#)

An old oil lantern, the kind people carried on farms, on boats and on camping trips before flashlights were common. A cloth wick soaks up oil from the tank at the bottom and burns at its tip, and the glass globe keeps the flame safe from the wind while letting the light out.

Tap it to light a flickering flame inside that warms the glass with an orange glow; tap again to put it out. A metal frame and a wire handle let it hang from a hook or swing from a hand.

HOW TO PLAY

- **Tap:** A flame lights inside with a warm glow and flicker.
- **Sound:** A real match strike and a soft flame catching when it lights; a real puff blowing it out.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/lantern/

CREDITS

- **Scan:** [Lantern 01](#) by Rajil Jose Macatangay, CC0 1.0.
- **Sound:** [blowing out candle.wav](#) by Reitanna, CC0 1.0.
- **Sound:** [Match Lighting Candle](#) by devilqube, CC0 1.0.



Cat statue

Tap it to make it look around and mew.



Scan to play with it.

A concrete statue of a sitting cat, wearing a red collar with a little bell. Cats have lived alongside people for about 10,000 years, since farmers first stored grain and cats came to hunt the mice it drew. People have made statues of cats for thousands of years; in ancient Egypt, cats were loved and honored.

Tap it and the cat comes to life: its head turns right around to look at you, tilts one way as it meows and then the other way, and turns back. Its tail stays still.

HOW TO PLAY

- **Tap:** The head turns and the tail swishes once.
- **Sound:** A soft stone scrape as the head turns, then a real cat's meow as it tilts.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/cat-statue/

CREDITS

- **Scan:** [Concrete Cat Statue](#) by Rico Cilliers, Riley Queen, CC0 1.0.
- **Sound:** [cat meow short](#) by skymary, CC0 1.0.

Horse statue

Tap it to make it rear up.



Scan to play with it.

A white marble statue of a horse. Horses have been carved in stone for thousands of years, from ancient Greek temples to city squares. A real horse is strong and fast, and can sleep standing up: special locks in its legs hold it steady while it dozes.

Tap it and the whole horse above the base rears up onto its hind hooves, prances at the top and lands again with a small bounce. Horses rear up when they are startled or excited, or to show off to other horses.

HOW TO PLAY

- **Tap:** Rears up on its hind legs, then settles.
- **Sound:** A soft stone scrape, a real horse's whinny as it rears, and a stone thud as it lands.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/horse-statue/

CREDITS

- **Scan:** [Horse Statue 01](#) by Rico Cilliers, CC0 1.0.
- **Sound:** [Renill de cavall / Horse Neigh](#) by Salsero_classic, CC0 1.0.

Real pencil

Tap it to spin it on the desk.
Pick its color under Look in
the Toy tab.



Scan to play with it.

Learn more: [Pencil on Wikipedia](#)

A wooden pencil. The "lead" inside is not lead at all but graphite, a soft gray form of carbon, mixed with clay and baked into a thin stick. More clay makes a harder, lighter line and less clay a softer, darker one; the letters on a pencil tell which: H for hard and B for black, with HB in between.

Tap it to flick it: it spins flat on the desk about its middle for two whole turns and slows to a stop where it began. Under Look in the Toy tab, pick yellow, red, blue, green, black, plain wood or the original scan.

HOW TO PLAY

- **Tap:** Flicked, it spins flat on the desk and slows to a stop.
- **Sound:** A wooden flick, then a soft scrape that slows.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/pencil-real/

CREDITS

- **Scan:** [Kleiner Bleistift](#) by Tim Reckmann, CC BY 2.0.



Real tin can

Tap it to knock it into a spin.
Pick its label under Look in the
Toy tab.



Scan to play with it.

Learn more: [Steel and tin cans on Wikipedia](#)

A tin can is a steel can with a thin coat of tin that keeps it from rusting. Food is sealed inside and heated to kill germs, so it stays good for years. Cans were invented in the early 1800s, but a handy can opener came only about 50 years later; before that, people used a hammer and chisel.

Tap it to knock it onto its bottom rim: it spins around on the rim faster and faster, like a coin settling, and drops flat with a clank. Under Look in the Toy tab, pick a peaches label, a tomatoes label, plain metal or the original scan.

HOW TO PLAY

- **Tap:** Knocked onto its rim, spins round on it faster and faster, and drops flat.
- **Sound:** One real recording of an empty can rolling on a hard floor, slowing and settling; no drum hit, no coin ring.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/tin-can-real/

CREDITS

- **Scan:** [Can opened with side opener](#) by LI1324, CC0 1.0.
- **Sound:** [Rolling Can Sound 2](#) by cower, CC0 1.0.



Torus

Tap it to spin it on its edge.
Dress it as a donut, bagel or swim ring in the Toy tab.



Scan to play with it.

Learn more: [Torus on Wikipedia](#)

A torus is the shape of a ring: a circle swept around a line beside it, like a hoop made of tube. Donuts, bagels, swim rings and bicycle inner tubes are all tori. In topology, the math of shapes that can stretch but not tear, a torus has exactly one hole, so a coffee mug with its handle counts as one too.

Tap it and it pops up onto its edge and wobbles around like a coin spun on a table, leaning lower and circling faster until it drops flat. Real rings and coins do this: as one leans lower, its low point runs around faster, so a settling coin whirs quicker just before it stops. Under Dress it as in the Toy tab, pick a plain torus, a donut, a bagel or a swim ring.

HOW TO PLAY

- **Tap:** Pops onto its edge, wobbles around like a coin spun on a table, and settles flat.
- **Sound:** A soft pop, a roll that circles faster, a settle.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Dress it as:** Plain torus, Donut, Bagel, Swim ring

In the Toy tab of the app.

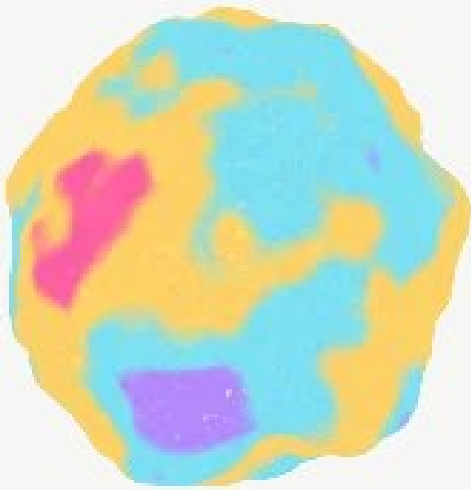
Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/torus/

CREDITS

- Built from a recipe in the browser; no outside files.

Jelly blob

Tap it to split it into three.
Make a shape of your own
in the Make tab.



Scan to play with it.

A wobbly jelly blob, built by the computer from a few rules rather than from photos. Thousands of splats, tiny soft blobs of color, are scattered over a lumpy round shape, and a pattern called noise gives it its bumps and swirls of candy color.

Tap it and it splits into three smaller blobs that wobble apart, then merge back with a jelly bounce. Real jelly wobbles because it is mostly water, held in a loose net of gelatin strands. In the Make tab you can pick a shape, a palette and a seed to build a blob of your own. With 🖐️ Hands-on on, drag it to stretch it; let go and it wobbles back.

HOW TO PLAY

- **Tap:** Splits into three small blobs that merge back.
- **Sound:** Gloopy wobble.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/blob/

CREDITS

- Built from a recipe in the browser; no outside files.

Neon knot

Tap it to make it twist and writhe.



Scan to play with it.

A glowing tube tied in a trefoil knot, the simplest true knot. Tie a knot in a string and join the two ends, and you get a loop that can never be untangled without cutting it; the trefoil crosses itself three times. Mathematicians study knots like this in a field called knot theory.

Tap it and waves of swelling loops run around the tube, lifting and twisting it as it glows pink, then it settles back into its trefoil. Real neon signs glow because electricity makes the gas inside the glass tube shine; pure neon glows red-orange.

HOW TO PLAY

- **Tap:** The neon flows along the knot and it ties tighter, then relaxes.
- **Sound:** The torus knot's sound, timed to this knot: it strains as the loops tie tighter, then lets go with a soft thump and swish as it relaxes; no jelly.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/knot/

CREDITS

- **Sound:** [highvoltagearc.wav](#) by Sciolex, CC0 1.0.



Basketball

Tap it to dribble it. Hands-on: hold it and flick up to spin it on your fingertip.



Scan to play with it.

Learn more:
[Basketball on Wikipedia](#)

A basketball is a bouncy ball with a pebbled skin for grip and dark grooves, called channels, running around it. The game was invented in 1891 by James Naismith, a teacher in Springfield, Massachusetts, who nailed up two peach baskets as the first hoops.

Tap it for three fast, low dribbles, then a toss onto the fingertip of a robot hand, where it spins until the hand drops away and it bounces down. A full-size ball is about 75 centimeters (29.5 inches) around. Pick its color in the Toy tab.

HOW TO PLAY

- **Tap:** Dribble: fast low bounces, then a spin on a fingertip.
- **Sound:** Hollow basketball boing.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colour:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/basketball/

CREDITS

- Built from a recipe in the browser; no outside files.



Soccer ball

Tap it for keepy-uppy: three kicks in the air, then it bounces to a stop.



Scan to play with it.

Learn more: [Ball \(association football\) on Wikipedia](#)

The classic soccer ball is stitched from 32 panels: 12 black pentagons and 20 white hexagons, a shape mathematicians call a truncated icosahedron. In most of the world the game is called football, and it is the most popular sport on Earth.

Tap it for keepy-uppy, the trick of keeping the ball in the air with your feet: three small kicks, each with its own spin, the last one higher, then it drops and bounces lower each time until it settles. Pick the colors of the panels and the base in the Toy tab.

HOW TO PLAY

- **Tap:** Keepy-uppy: three small kicks with spin.
- **Sound:** Firm thump.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Panels:** pick a color
- **Base:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/soccer-ball/

CREDITS

- Built from a recipe in the browser; no outside files.



American football

Tap it to throw a spiral pass.



Scan to play with it.

Learn more:

[American football](#)
[on Wikipedia](#)

An American football is a pointed oval ball of leather with a row of white laces along one side for the fingers to grip. It is often called a pigskin, though today it is made of cowhide or rubber.

Tap it to throw a spiral: it flies up nose first, spinning fast around its long middle line, tips over at the top and lands with a wobble. The spin is what makes a good pass: like a spinning top, a spinning ball holds its direction and cuts cleanly through the air. Pick the color of the leather in the Toy tab.

HOW TO PLAY

- **Tap:** A tight spiral spin in place.
- **Sound:** Leathery whoosh.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Leather:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/american-football/

CREDITS

- Built from a recipe in the browser; no outside files.



Tennis ball

Tap it to slam it down and watch it bounce high.



Scan to play with it.

Learn more: [Tennis ball on Wikipedia](#)

A tennis ball is a hollow rubber ball filled with air under pressure and covered in fuzzy felt. The felt slows the ball in the air and helps the racket's strings grip it for spin. Bright yellow balls came in during the 1970s, because they are easier to see on television.

Tap it and it is slammed onto the floor: it squashes hard, shoots up high with topspin, the fuzz fluffing out at each hit, and bounces a little lower each time. Pick the color of the felt in the Toy tab.

HOW TO PLAY

- **Tap:** Fast high bounce with the fuzz fluffing out.
- **Sound:** A real tennis ball slammed down, then a real tennis-ball bounce on each bounce, softer each time.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Felt:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/tennis-ball/

CREDITS

- **Sound:** [bola de tenis caindo.wav](#) by Negraovictor, CC0 1.0.
- **Sound:** [ball_hit_ground.wav](#) by Kyanite_, CC0 1.0.

Baseball

Tap it to pitch a curveball and hit it back.



Scan to play with it.

Learn more:

[Baseball \(ball\) on Wikipedia](#)

A baseball has a small cork and rubber center, wound with long strands of yarn and covered with two figure-eight pieces of white leather. They are sewn together with 108 double stitches of red thread.

Tap it and it is pitched as a curveball: it spins hard, and the spin makes the air push it down and to the side late in its flight, which is why curveballs are so hard to hit. Then the crack of a bat sends it looping back to its spot. A fast pitch reaches the batter in less than half a second.

HOW TO PLAY

- **Tap:** Curveball: it spins hard and swerves.
- **Sound:** The same pitch whoosh, then the crack of a real wooden bat.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/baseball/

CREDITS

- **Sound:** [Bat Hit 9 FF095.aif](#) by martinimeniscus, CC0 1.0.



Softball

Tap it to pitch it underhand in a slow, high arc.



Scan to play with it.

Learn more: [Softball on Wikipedia](#)

A softball is bigger than a baseball, about 30 centimeters (12 inches) around, and in spite of its name it is not soft. The game was first played indoors in Chicago in 1887, and today it is played all over the world.

In softball the pitcher throws underhand, swinging the arm down and forward past the hip. Tap it and it is pitched just like that: a swing back and through, then a high, slow arc with a little backspin. It lands with a soft thud, hardly bounces, and a gentler toss brings it back.

HOW TO PLAY

- **Tap:** Underhand arc and a softer thud.
- **Sound:** The same whoosh, then the real dull thump of a heavy ball landing (away and back home).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/softball/

CREDITS

- **Sound:** [ball_hit_ground.wav](#) by Kyanite_, CC0 1.0.



Beach ball

Tap it to toss it up. Hands-on: toss it and it floats and drifts down.



Scan to play with it.

Learn more: [Beach ball on Wikipedia](#)

A beach ball is a thin plastic ball blown up with air, usually with bright stripes of color. It weighs so little for its size that the air slows it down almost at once, which makes it easy and safe to play with.

Tap it and it is punched up: it rises, slows, then floats down slowly, drifting and turning lazily, lands softly with a wobble and bobs to a stop. Pick its three colors in the Toy tab. Air pushes on every ball as it moves, but a light, big ball like this one feels it the most.

HOW TO PLAY

- **Tap:** Floats up slowly, drifts and bobs down.
- **Sound:** A real plastic beach-ball bounce each time it lands (no boing), and a light slap as it's punched up.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colour 1:** pick a color
- **Colour 2:** pick a color
- **Colour 3:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/beach-ball/

CREDITS

- **Sound:** [footballPlast.wav](#) by blindmanonacid, CC0 1.0.

Golf ball

Tap it to chip it; the backspin pulls it back.



Scan to play with it.

Learn more: [Golf ball on Wikipedia](#)

A golf ball is small and hard, covered in hundreds of little dents called dimples. The dimples help it fly: they stir up a thin layer of air around the ball, which lets it slip through the air more easily, so it flies about twice as far as a smooth ball would.

Tap it for a chip, a short, high shot played close to the hole. It pops up with heavy backspin, lands, checks with a tiny hop, and then the spin grips the ground and pulls it back to where it started.

HOW TO PLAY

- **Tap:** Tiny chip hop and a spin-back.
- **Sound:** Click of a club.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/golf-ball/

CREDITS

- Built from a recipe in the browser; no outside files.



Rugby ball

Tap it to punt it end over end.



Scan to play with it.

Learn more: [Rugby ball on Wikipedia](#)

A rugby ball is a big oval ball, rounder and larger than an American football. Rugby is named after Rugby School in England, where the game grew up in the 1800s. Players may only pass the ball backward or sideways with their hands, but they can kick it forward.

Tap it for a punt, a kick of the ball dropped from the hands: it tumbles end over end, up and down, lands on a point and takes an odd, awkward bounce before settling, as oval balls do. Pick the color of its bands in the Toy tab.

HOW TO PLAY

- **Tap:** Tumbles end over end with an odd bounce.
- **Sound:** Dull thud.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Bands:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/rugby-ball/

CREDITS

- Built from a recipe in the browser; no outside files.



Volleyball

Tap it to set it up and spike it down.



Scan to play with it.

Learn more:
[Volleyball on Wikipedia](#)

A volleyball is a light ball of smooth leather panels. Volleyball was invented in 1895 by William G. Morgan, a teacher in Holyoke, Massachusetts. In ordinary indoor play, a team has three hits to return the ball; a block does not count as one of them.

Tap it for a set and a spike, the classic attack: a soft touch sends it straight up without spin, then a spike drives it down hard with topspin, so it slams into the floor, kicks up high and bounces out. Pick its two colors in the Toy tab.

HOW TO PLAY

- **Tap:** A set then a spike straight down.
- **Sound:** Slap.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colour 1:** pick a color
- **Colour 2:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/volleyball/

CREDITS

- Built from a recipe in the browser; no outside files.



Water polo ball

Tap it to toss it in. Hands-on: drop it on the water and it bobs and floats.



Scan to play with it.

Learn more: [Water polo on Wikipedia](#)

Water polo is a team game played in a swimming pool, often in water too deep to stand in, so the players tread water the whole time. Its ball is about the size of a soccer ball, with a grippy, bumpy rubber skin so it can be held in one wet hand.

Tap it and it is tossed up and plunges into the water with a splash. Because it is full of air it pops straight back up, then bobs on the surface, sending out rings of ripples that spread and fade.

HOW TO PLAY

- **Tap:** Bobs on an invisible water line with ripples.
- **Sound:** Splash.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/water-polo-ball/

CREDITS

- Built from a recipe in the browser; no outside files.

Ping-pong ball

Tap it to flick it up and let it bounce to a buzz.



Scan to play with it.

Learn more: [Table tennis on Wikipedia](#)

A ping-pong ball, used in table tennis, is a hollow plastic ball 40 millimeters across that weighs less than 3 grams, about as much as a small coin. The name ping-pong comes from the sound it makes on the bat and the table.

Tap it and it is flicked up, then bounces on and on, each bounce a little lower and quicker, with a tik for every one, until it buzzes to a stop. A light, springy ball keeps most of its speed at each bounce. Pick a white or an orange ball in the Toy tab.

HOW TO PLAY

- **Tap:** Rapid tiny bounces getting faster.
- **Sound:** Tik-tik-tik.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colour:** White, Orange

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/ping-pong-ball/

CREDITS

- Built from a recipe in the browser; no outside files.



Cricket ball

Tap it to flick it seam up; it skids with backspin, then rolls home.



Scan to play with it.

Learn more: [Cricket ball on Wikipedia](#)

A cricket ball is a hard ball with a cork center, wound tight with string and covered in leather. A raised seam of stitching runs around its middle. Red balls are used in long matches that can last days, and white ones in shorter games.

Bowlers hold the ball with the seam upright, because a seam that lands upright can jump sideways off the ground. Tap it for a seam-up flick: the seam stands tall while the ball spins backward, then it comes down, skids on with its backspin until the spin grips, and rolls home.

HOW TO PLAY

- **Tap:** Seam-up spin and a skid.
- **Sound:** Crisp leather knock.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/cricket-ball/

CREDITS

- Built from a recipe in the browser; no outside files.



Bowling ball

Tap it to bowl it down the lane with a hook.



Scan to play with it.

Learn more:
[Bowling ball on Wikipedia](#)

A bowling ball is a heavy ball with three finger holes, rolled down a long wooden or plastic lane at ten pins. The heaviest balls allowed weigh 16 pounds (about 7 kilograms).

Tap it to bowl it: it drops onto the lane with a thud, rolls away straight and then hooks across, the finger holes turning over as it goes; the pins crash far off, and it rolls back home. Good bowlers spin the ball so it curves into the pins from the side, which knocks down more of them. Pick its two colors in the Toy tab.

HOW TO PLAY

- **Tap:** Heavy roll with a hook.
- **Sound:** A real bowling ball rolling down the lane, real pins crashing far off, and the roll back.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

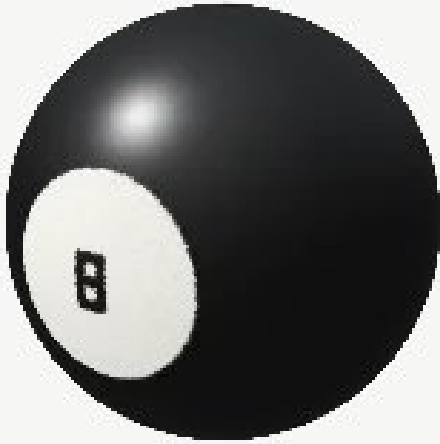
- **Colour 1:** pick a color
- **Colour 2:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/bowling-ball/

CREDITS

- **Sound:** [B_2 Bowling ball striking pins.wav](#) by Yarmonics, CC0 1.0.
- **Sound:** [Bowling Ball.wav](#) by driftworks, CC0 1.0.



Pool ball

Tap it for a draw shot: it slides out and spins back. Pick its number in the Toy tab.



Scan to play with it.

Learn more: [Billiard ball on Wikipedia](#)

Pool is played on a cloth-covered table with pockets. A set has a white cue ball and 15 numbered balls: 1 to 8 are solid colors and 9 to 15 have stripes, and the 8 ball is black. Players hit the cue ball with a long stick, the cue, to knock the others into the pockets.

Tap it for a draw shot: struck low, it slides forward while spinning backward, stops, and the backspin pulls it back until it rolls home. Pick the cue ball or any of the 15 numbers in the Toy tab.

HOW TO PLAY

- **Tap:** Clicks forward and spins with backspin.
- **Sound:** The real solid tock of a cue tip striking the ball.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Number:** Cue ball, 1, 2, 3, 4, 5, 6, 7 and 8 more

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/pool-ball/

CREDITS

- **Sound:** [pool_break.wav](#) by reg7783, CC0 1.0.



Pickleball

Tap it to pop it up off a paddle.



Scan to play with it.

Learn more:
[Pickleball on Wikipedia](#)

Pickleball is a game played with solid paddles and a light plastic ball full of holes, over a low net on a small court. It was invented in 1965 on Bainbridge Island, near Seattle in Washington, by three fathers looking for a game for their families.

Tap it and an unseen paddle pops it up twice. The holes make the ball slow down fast in the air, and it wobbles as it flies instead of spinning much. It lands with a hollow click and a small, dead bounce. Pick its color in the Toy tab.

HOW TO PLAY

- **Tap:** Light pop-up with air whistling through the holes.
- **Sound:** Its hollow plastic pock, without the whistle.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

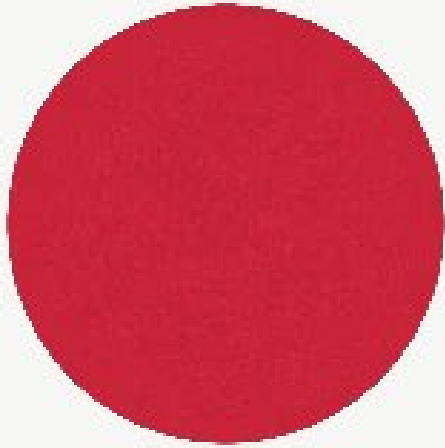
- **Colour:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/pickleball/

CREDITS

- Built from a recipe in the browser; no outside files.



Dodgeball

Tap it to slam it down and watch it squash.



Scan to play with it.

Learn more:
[Dodgeball on Wikipedia](#)

A dodgeball is a soft, light ball of rubber or foam, made to be thrown at other players in a game of dodging, catching and throwing. Its soft material cushions an impact.

Tap it and it is lifted and slammed down: the soft rubber squashes flat and wobbles, then it bounces up again and squashes at each landing. A soft ball squashes more than a hard one, and it loses more of its bounce while it is squashed. Pick its color in the Toy tab.

HOW TO PLAY

- **Tap:** Squashes hard on a bounce.
- **Sound:** Rubbery bwong.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colour:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/dodgeball/

CREDITS

- Built from a recipe in the browser; no outside files.



Medicine ball

Tap it to heave it up and drop it with a thud.



Scan to play with it.

Learn more:
[Medicine ball on Wikipedia](#)

A medicine ball is a heavy ball for exercise, used for throwing, catching and lifting to build strength. Most weigh from about 1 to 10 kilograms (2 to 20 pounds), and they are made of leather, rubber or tough cloth, often with a grippy surface.

Tap it and it is heaved up just a little, slowly, because it is heavy. It drops with a heavy thud and a big, slow squash, does not bounce at all, and a puff of dust spreads out across the floor.

HOW TO PLAY

- **Tap:** Barely lifts and lands with a heavy squash.
- **Sound:** Heavy thud.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/medicine-ball/

CREDITS

- Built from a recipe in the browser; no outside files.

Lacrosse ball

Tap it to slam it down; it bounces hard and fast.



Scan to play with it.

Learn more:
[Lacrosse on Wikipedia](#)

A lacrosse ball is a small ball of solid, hard rubber. In lacrosse, players catch, carry and throw it with a stick that has a net pocket on the end. The game began with the Indigenous peoples of North America, who played it long before Europeans arrived.

Tap it and it is slammed down: it rockets up and bounces hard and fast, keeping more of its speed at each bounce than any other ball on the shelf. Solid rubber stores the energy of a bounce and gives most of it back. Pick its color in the Toy tab.

HOW TO PLAY

- **Tap:** Very fast, hard bounces.
- **Sound:** Hard rubber knock.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colour:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/lacrosse-ball/

CREDITS

- Built from a recipe in the browser; no outside files.



Squash ball

Tap it to warm it up; the warmer it gets, the higher it bounces.



Scan to play with it.

Learn more: [Squash \(sport\) on Wikipedia](#)

A squash ball is a small, hollow rubber ball that hardly bounces when it is cold. Players hit it against the walls of the court for a few minutes to warm it up: as the rubber and the air inside warm, the ball gets bouncier. The ball for the best players has two yellow dots and is the least bouncy of all.

Tap it and see: cold, it is dropped and barely bounces. Hit over and over, it warms, glows faintly and bounces higher and faster. Left alone, it bounces out and cools again.

HOW TO PLAY

- **Tap:** Warms up: glows faintly as it bounces faster.
- **Sound:** Dead, low squash thock.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/squash-ball/

CREDITS

- Built from a recipe in the browser; no outside files.



Bouncy ball

Tap it to throw it down and watch it ricochet.



Scan to play with it.

Learn more: [Bouncy ball on Wikipedia](#)

A bouncy ball is a small ball of very springy, hard rubber. It gives back almost all the energy of each bounce, so it can bounce nearly as high as it was dropped from. The first ones of this kind were made in the 1960s by a chemist experimenting with new rubbers.

Tap it and it is thrown down hard: it ricochets all over the floor, keeping nearly all its speed, its spin flipping at every bounce, then comes home and settles. Pick its three colors in the Toy tab.

HOW TO PLAY

- **Tap:** Wild ricochet bounces all over, then home.
- **Sound:** Springy boing that rises.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colour 1:** pick a color
- **Colour 2:** pick a color
- **Colour 3:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/bouncy-ball/

CREDITS

- Built from a recipe in the browser; no outside files.



Marble

Tap it to roll it around a little circle.



Scan to play with it.

Learn more: [Marble \(toy\) on Wikipedia](#)

A marble is a small, hard ball, usually of glass, with twists of color inside. Children have played games with little balls of clay and stone for thousands of years; glass marbles became common in the 1800s. A twist of colored glass is set in the middle of the ball while the glass is hot and soft.

Tap it and it rolls around a little circle, turning the way it rolls, so the swirl inside turns too, under a polished glass shell that catches the light at its edge. After one lap it is back exactly as it was. Pick the color of the swirl in the Toy tab.

HOW TO PLAY

- **Tap:** Rolls around a little circle; the swirl inside turns.
- **Sound:** The same real marble roll, about 6 dB quieter.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Swirl:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/marble/

CREDITS

- **Sound:** [Marble \(single\) rolling on wooden floor.wav](#) by LiezelDippenaar, CC0 1.0.



Hockey puck

Tap it for a slap shot across the ice.



Scan to play with it.

Learn more: [Hockey puck on Wikipedia](#)

A hockey puck is a flat disk of hard black rubber, 3 inches (about 7.6 centimeters) across and 1 inch thick. In ice hockey, pucks are frozen before a game, so they bounce less and slide better on the ice.

Tap it for a slap shot, the hardest shot in hockey, when the player swings the stick back high and slaps the puck. Sparkling ice chips spray from where the stick hit, and the puck glides flat across the ice, spinning fast, around a wide loop and back to its spot.

HOW TO PLAY

- **Tap:** Slides and spins flat, throwing up ice sparkle.
- **Sound:** Stick slap and ice scrape.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/hockey-puck/

CREDITS

- Built from a recipe in the browser; no outside files.



Shuttlecock

Tap it to hit it up; it flips over and floats down cork first.



Scan to play with it.

Learn more:
[Shuttlecock on Wikipedia](#)

A shuttlecock, or shuttle, is used in badminton instead of a ball. It has a rounded cork base with a skirt of 16 feathers, or a plastic skirt. The skirt drags on the air, so the shuttle always turns to fly cork first and slows down quickly.

Tap it and it is hit up: it flips over cork first and flies up spinning, turns over at the top and floats back down cork first, spinning slower as it falls. Even so, a hard smash can send a shuttle off the racket faster than 400 kilometers (250 miles) an hour.

HOW TO PLAY

- **Tap:** Flips over and floats down feathers-up.
- **Sound:** Light racket tock.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/shuttlecock/

CREDITS

- Built from a recipe in the browser; no outside files.



Flying disc

Tap it to throw it around a loop and back.



Scan to play with it.

Learn more: [Flying disc on Wikipedia](#)

A flying disc is a plastic disc with a curled rim, thrown with a flick of the wrist. It flies like a spinning wing: its curved shape makes the air lift it, and its spin keeps it flat and steady, just as a spinning top stays upright.

Tap it to throw it: it spins fast and flat, tilts into a curve and glides around a loop, like a throw that comes back to you, then settles in its place. Pick its color in the Toy tab. The first plastic flying discs were sold in the late 1940s.

HOW TO PLAY

- **Tap:** Spins fast and hovers, tilting.
- **Sound:** Whirring whoosh.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colour:** pick a color

In the Toy tab of the app.

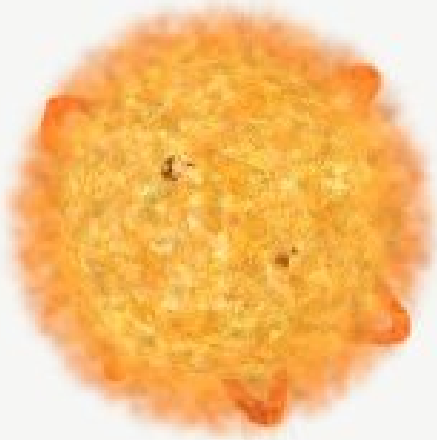
Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/flying-disc/

CREDITS

- Built from a recipe in the browser; no outside files.

Sun

Tap it to set off a solar flare.



Scan to play with it.

Learn more: [Sun on Wikipedia](#)

The Sun is the star at the center of our solar system, a huge ball of hot, glowing gas. About 1.3 million Earths could fit inside it, and its light takes about 8 minutes to reach us.

This Sun churns by itself: bright cells of hot gas swell and fade, and loops of glowing gas called prominences rise and sink at its edge. Tap it to set off a solar flare: its feet flash white, a loop of hot gas climbs off the top edge, swells, and its top breaks away into space. Real flares are sudden bursts of energy that can disturb radio signals on Earth, and the gas thrown off with them can light up auroras.

HOW TO PLAY

- **Tap:** Idle: stronger churning and pulsing.
- **Sound:** Deep roaring whoosh.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/sun/

CREDITS

- Built from a recipe in the browser; no outside files.

Solar system

Tap to line up the planets and watch Mercury cross the Sun.



Scan to play with it.

Learn more: [Solar System on Wikipedia](#)

The solar system is the Sun and everything that circles it: eight planets, their moons, and many asteroids and comets. All the planets go around the Sun the same way, in orbits that lie nearly in one flat plane. The inner planets go around fastest: Mercury takes 88 days, Neptune about 165 years.

This is a model, not to scale: sizes and distances are squeezed so no two planets touch. Tap it and every planet swings into one row. The model tips until you look along its plane, and Mercury passes in front of the Sun as a small dark dot, which is called a transit. Then the planets spread out again. Real transits of Mercury happen about 13 times a century.

HOW TO PLAY

- **Tap:** Planets swing into a line and one eclipses the Sun, then orbits resume.
- **Sound:** No synth chord: a soft whoosh as the planets swing into line, a warm flare as the corona shows, and a whoosh as they swing on.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/solar-system/

CREDITS

- Built from a recipe in the browser; no outside files.



Mercury

Tap it to spin it fast in the sunlight.



Scan to play with it.

Learn more: [Mercury \(planet\) on Wikipedia](#)

Mercury is the smallest planet and the closest to the Sun, a gray rocky world covered in craters, a little bigger than our Moon. It races around the Sun once every 88 days.

Tap it and it spins once, fast, while the side facing the Sun glows red-hot and shimmers, then cools. The real Mercury turns slowly, and it has almost no air to hold heat, so its day side gets hot enough to melt lead, about 430 degrees Celsius, while its night side drops to about minus 180 degrees.

HOW TO PLAY

- **Tap:** Quick orbit-like spin and heat shimmer on the day side.
- **Sound:** A quick spin whoosh, then a low roar of heat and a soft searing sizzle while the day side glows, fading as it cools (no tone).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/mercury/

CREDITS

- Built from a recipe in the browser; no outside files.



Venus

Tap it to whip the clouds around.



Scan to play with it.

Learn more: [Venus on Wikipedia](#)

Venus is the second planet from the Sun and nearly the same size as the Earth. Its dense atmosphere, mostly carbon dioxide, traps heat beneath thick clouds, so it is the hottest planet of all, about 465 degrees Celsius at the ground, even hotter than Mercury.

Venus spins backward compared with most planets, and very slowly, but its cloud tops race all the way around in about four Earth days. Tap it and the clouds whip around backward: the wide band at the middle goes around twice and the caps at the poles once, so the patterns stretch where they meet, then line up again.

HOW TO PLAY

- **Tap:** The thick clouds swirl fast around it.
- **Sound:** Muffled wind.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/venus/

CREDITS

- Built from a recipe in the browser; no outside files.



Earth

Tap it to turn it through one day and night.



Scan to play with it.

Learn more: [Earth on Wikipedia](#)

The Earth is the third planet from the Sun and our home, the only world known to have life. Oceans cover about 71 percent of its surface, and it turns once on its axis about every 24 hours, which gives us day and night.

Tap it to turn it through one day, with the Sun off to the left. Night falls over one half, the globe turns once, and city lights come on as the land turns into the dark, then go out again at dawn.

HOW TO PLAY

- **Tap:** Night side lights up with city lights as it turns; clouds move.
- **Sound:** The same ocean wash (a little softer) and chime, with the wind turned right down (softer, shorter and quieter).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/earth/

CREDITS

- Built from a recipe in the browser; no outside files.



Moon

Tap to land a lunar module; tap again to pack up and lift off. Pick the flag in the Toy tab.



Scan to play with it.

Learn more: [Moon on Wikipedia](#)

The Moon is the Earth's only natural satellite, a gray world of craters about a quarter as wide as the Earth. Its dark patches are wide plains of old, cooled lava. Its gravity is about one sixth of the Earth's, so things weigh much less there.

Tap it and a lunar module lands, an astronaut climbs down, bounds over in low-gravity hops and plants a flag, which you can pick in the Toy tab. Tap again and the astronaut packs up and lifts off. Between 1969 and 1972, twelve astronauts walked on the Moon.

HOW TO PLAY

- **Tap:** A lunar module lands; an astronaut plants a flag.
- **Sound:** Engine rumble, touchdown thud, lift-off roar.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Flag:** flag

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/moon/

CREDITS

- Built from a recipe in the browser; no outside files.



Mars

Tap it to raise a dust storm.



Scan to play with it.

Learn more: [Mars](#)
on [Wikipedia](#)

Mars is the fourth planet from the Sun, a cold, dry, rocky world about half as wide as the Earth. It looks red because its dust and rocks are full of iron that has rusted. It has two small moons, Phobos and Deimos, and the biggest volcano in the solar system, Olympus Mons.

Tap it to raise a dust storm: billowing orange dust sweeps across, hides the dark markings, then settles and clears. Real dust storms on Mars can grow until they cover the whole planet for weeks.

HOW TO PLAY

- **Tap:** A dust storm sweeps across it.
- **Sound:** The dust storm is quieter and darker: a softer hiss of dust and a lower wind.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/mars/

CREDITS

- Built from a recipe in the browser; no outside files.



Jupiter

Tap it to race the cloud bands and spin up the Great Red Spot.



Scan to play with it.

Learn more: [Jupiter on Wikipedia](#)

Jupiter is the biggest planet, a giant ball made mostly of hydrogen and helium gas, with no solid surface to stand on. More than 1,300 Earths could fit inside it. Its stripes are bands of clouds blown by strong winds, and the Great Red Spot is a storm wider than the Earth that has raged for at least 150 years.

Tap it and the cloud bands race, each one the opposite way to its neighbors, while the Red Spot's swirl spins up; then it all settles. Jupiter spins faster than any other planet: a day there lasts about 10 hours.

HOW TO PLAY

- **Tap:** Bands race and the Great Red Spot spins up.
- **Sound:** Low gas-giant drone.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/jupiter/

CREDITS

- Built from a recipe in the browser; no outside files.



Saturn

Tap it to send ripples across the rings.



Scan to play with it.

Learn more: [Saturn on Wikipedia](#)

Saturn is the sixth planet from the Sun, a giant ball of gas with the brightest rings in the solar system. The rings are made of countless pieces of ice and rock, from specks of dust to chunks as big as a house, and they are very wide but mostly very thin.

Here the rings turn all the time, the inner rings faster, as real ones do, with dark spokes and bright clumps that show the motion. Tap it to send two sparkling waves rippling out across the rings. Saturn is so light for its size that it would float in a big enough bathtub of water.

HOW TO PLAY

- **Tap:** Idle: rings turn.
- **Sound:** Two soft icy sweeps as the waves ripple out across the rings, with a faint glitter (no synth shimmer).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/saturn/

CREDITS

- Built from a recipe in the browser; no outside files.



Uranus

Tap it to roll it on its side like a wheel.



Scan to play with it.

Learn more: [Uranus on Wikipedia](#)

Uranus is the seventh planet from the Sun, an ice giant colored blue-green by methane gas in its air. It is tipped right over on its side, with its axis tilted about 98 degrees, so it rolls around the Sun like a ball. In 1781 William Herschel found it, the first planet discovered with a telescope.

Here its thin rings turn slowly all the time. Tap it and it rolls like a wheel, to the right and back again. A year on Uranus lasts about 84 Earth years, so each pole gets about 42 years of sunlight and then about 42 years of dark.

HOW TO PLAY

- **Tap:** Idle: slow ring motion.
- **Sound:** A soft low rumble as it rolls to the right, and again as it rolls back (no bell).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/uranus/

CREDITS

- Built from a recipe in the browser; no outside files.

Neptune

Tap it to race the clouds around it.



Scan to play with it.

Learn more:
[Neptune on Wikipedia](#)

Neptune is the eighth and farthest planet from the Sun, an ice giant of pale greenish blue, a little bluer than Uranus; the deep blue of older spacecraft pictures came from processing. It has the strongest winds in the solar system, faster than 2,000 kilometers an hour. It was the first planet found by math: astronomers worked out where it must be before they saw it in 1846.

Tap it and belts of white cloud race around it twice while a band with a dark storm drifts the other way. The spacecraft Voyager 2 saw a great dark storm like this when it flew past in 1989. Neptune takes about 165 years to go around the Sun once.

HOW TO PLAY

- **Tap:** A dark storm and fast white clouds race by.
- **Sound:** Deep icy wind.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

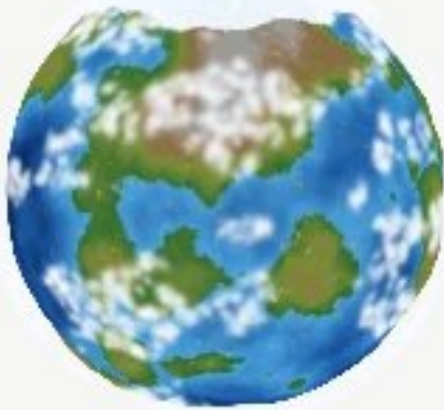
SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/neptune/

CREDITS

- Built from a recipe in the browser; no outside files.



Tiny planet

Tap it to sweep night across it. Pick a planet under Planet in the Toy tab.



Scan to play with it.

A tiny, made-up planet built by the computer from a sphere and a palette of planet colors: blue seas, green land and white clouds for Earth, rusty deserts and white polar caps for Mars, gray craters for the Moon, and banded storms for Jupiter and Neptune. It is a tribute to the very first version of Splashery.

Tap it and a band of night sweeps across it while any clouds race once around it. Day and night happen because a planet spins: at any moment, the half facing its star has day and the other half has night. Earth turns once about every 24 hours, Jupiter in under 10. Pick Earth, Mars, the Moon, Jupiter or Neptune under Planet in the Toy tab.

HOW TO PLAY

- **Tap:** Clouds race around and a day-night band sweeps over it.
- **Sound:** A quiet, darker breeze and a soft low whoosh as the clouds race round.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/planet/

CREDITS

- Built from a recipe in the browser; no outside files.



Aurora world

Tap it for a burst of bright auroras.



Scan to play with it.

An aurora is the glowing light that ripples across the sky near the poles, called the northern and southern lights on Earth. It happens when particles from the Sun are steered by the planet's magnetic field toward the poles and hit the gases high in the air, making them glow, mostly green, with pink and violet at the top.

This is the night side of an Earth-like world, its pole tipped toward you, with town lights below. Its curtains of light move all the time. Tap it to set off a storm of light: the ring flares, and a wider curtain bursts out toward the equator, then calms.

HOW TO PLAY

- **Tap:** Idle: auroras ripple as curtains.
- **Sound:** The same aurora air, much lower and gentler.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/aurora-planet/

CREDITS

- Built from a recipe in the browser; no outside files.



Asteroid

Tap it to break it apart; its pieces pull back together.



Scan to play with it.

Learn more: [Asteroid on Wikipedia](#)

Asteroids are rocky leftovers from when the planets formed, about 4.6 billion years ago. Most circle the Sun in the main asteroid belt, between Mars and Jupiter. They come in all sizes, from small boulders to worlds hundreds of kilometers wide, and most are lumpy, not round.

Tap it and glowing cracks flash, then it falls apart into 18 pieces that drift off, tumbling, with a puff of dust. Then gravity pulls them back together. Some real asteroids are rubble piles like this: loose heaps of rock held together only by their own weak gravity.

HOW TO PLAY

- **Tap:** Cracks and breaks into pieces that drift apart, then pull back together.
- **Sound:** A rock crack and a muffled boom, real rocks crumbling apart, a soft grinding as the rubble comes back, and the stone settling.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/asteroid/

CREDITS

- **Sound:** [Rocks.wav](#) by adamgryu, CC0 1.0.
- **Sound:** [Muffled Distant Explosion](#) by NenadSimic, CC0 1.0.

Comet

Tap it to swing it past the Sun and flare its tails.



Scan to play with it.

Learn more: [Comet on Wikipedia](#)

A comet is a lump of ice, dust and rock, often only a few kilometers wide, that travels around the Sun on a long, stretched orbit. As it comes near the Sun, its ice turns to gas, making a glowing cloud around it called a coma, and two tails: a straight blue tail of gas and a curved, pale tail of dust. The tails point away from the Sun.

Tap it to swing it past the Sun: jets of gas burst from the sunny side, the coma swells, and both tails flare longer and brighter, then fade back. Halley's Comet comes back about every 76 years.

HOW TO PLAY

- **Tap:** Tail flares longer and brighter as it swings by.
- **Sound:** A real burning torch roaring as the jets and tails flare, fading as they die back.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/comet/

CREDITS

- **Sound:** [Torch.wav](#) by DanielVega, CC0 1.0.



Meteor

Tap it to send it streaking in to burst in a fireball.



Scan to play with it.

Learn more:
[Meteoroid on Wikipedia](#)

A meteor, or shooting star, is the streak of light made when a bit of space rock rushes into the air at tens of kilometers a second and burns up. Most are no bigger than a grain of sand. A very bright one is called a fireball, and a piece that survives to reach the ground is a meteorite.

Tap it and it streaks in and bursts in a fireball, its rock flying apart in glowing pieces that burn out. Then the next one streaks in to take its place. Many tons of space dust fall on the Earth every day.

HOW TO PLAY

- **Tap:** Streaks in and bursts in a fireball.
- **Sound:** The streak's fire, then the star's own burst (over a short crack) as it explodes at 0.37 s, in time with the fireball.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/meteor/

CREDITS

- **Sound:** [Torch.wav](#) by DanielVega, CC0 1.0.
- **Sound:** [Muffled Distant Explosion](#) by NenadSimic, CC0 1.0.
- **Sound:** [Waving Torch.wav](#) by spookymodem, CC0 1.0.



Star

Tap to run the star's life, sped up. Pick the kind of star in the Toy tab.



Scan to play with it.

Learn more: [Star on Wikipedia](#)

A star is a huge ball of hot gas that shines because, deep in its core, it squeezes hydrogen into helium and gives off energy. Its color shows how hot it is: red stars are the coolest and blue stars the hottest. The Sun is a yellow star, about halfway through its life.

Tap it to run the life of a star like the Sun, sped up. It swells into a red giant, puffs off its outer layers as a glowing shell and shrinks to a tiny white dwarf; then a new star lights up. A real star like the Sun takes about 10 billion years to do this. Pick the kind of star in the Toy tab; a tap always plays the Sun's story.

HOW TO PLAY

- **Tap:** Tap runs a life cycle: swells into a red giant, puffs off its shell and leaves a white dwarf, then reforms.
- **Sound:** A deep stellar roar as it swells, low throbs as the red giant pulses, a real explosion as it puffs off its shell, and a soft roar as a new star lights.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Type:** Red dwarf, Yellow, like the Sun, Blue giant, White dwarf

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/star/

CREDITS

- **Sound:** [Muffled Distant Explosion](#) by NenadSimic, CC0 1.0.
- **Sound:** [Deep Boom](#) by Za-Games, CC0 1.0.

Pulsar

Tap it to spin it up. Set its spin speed in the Toy tab.



Scan to play with it.

Learn more: [Pulsar on Wikipedia](#)

A pulsar is a neutron star, the crushed core left after a giant star explodes. It is only about 20 kilometers across, the size of a city, yet it holds more matter than the Sun. It spins fast and sends out two beams of light from its magnetic poles.

Like a lighthouse, it seems to flash each time a beam sweeps past. Tap it to spin it up until the flashes blur into a strobe, each with a deep pulse, then it winds down. Set its spin speed in the Toy tab. The first pulsar was found by Jocelyn Bell Burnell in 1967, and the fastest spin hundreds of times a second.

HOW TO PLAY

- **Tap:** Beams sweep faster and faster with strobing flashes.
- **Sound:** A deep boom and rumble as it spins up, and a deep pulse with each flash instead of a tick.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Spin speed:** slider

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/pulsar/

CREDITS

- **Sound:** [Deep Boom](#) by Za-Games, CC0 1.0.



Black hole

Tap it to feed it a star.



Scan to play with it.

Learn more: [Black hole on Wikipedia](#)

A black hole is a place where gravity is so strong that nothing that crosses its edge, the event horizon, can get out again, not even light. Gas that falls toward it swirls into a flat disk and gets hot enough to glow.

Tap it to feed it a star: the star spirals in faster and faster, is stretched into a streak and plunges in, and the disk flares. In 2019 astronomers showed the first picture of a black hole's shadow, at the heart of the galaxy M87, made with radio telescopes spread across the Earth.

HOW TO PLAY

- **Tap:** A small star falls in, gets stretched into a spiral and swallowed; the disk flares.
- **Sound:** A deep growing roar as the star spirals in, a rushing whoosh, a huge low boom as it plunges, and the gas swirling down after it.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

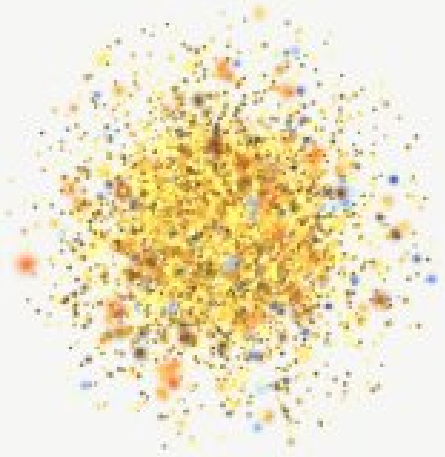
Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/black-hole/

CREDITS

- **Sound:** [Deep Boom](#) by Za-Games, CC0 1.0.

Star cluster

Tap it to make the cluster breathe in and out.



Scan to play with it.

Learn more: [Star cluster on Wikipedia](#)

A globular cluster is a ball of hundreds of thousands of stars held together by their own gravity, packed tightest at the middle. Its stars are very old, among the oldest in our galaxy, and the Milky Way has about 150 of these clusters circling it.

Tap it and the cluster breathes: it draws in, swells out and settles, the core first and the outer stars a moment later, while a wave of sparkle runs out from the middle. Near the middle of a real cluster, the stars are packed far more tightly than the stars around the Sun.

HOW TO PLAY

- **Tap:** Stars twinkle in a wave and the cluster breathes in and out.
- **Sound:** Two faint, soft swells of air; much quieter, no hiss.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

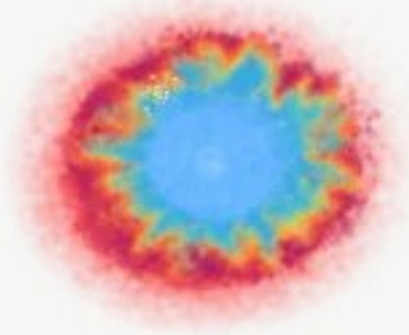
Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/star-cluster/

CREDITS

- Built from a recipe in the browser; no outside files.

Ring nebula

Tap it to blow out a new shell of gas.



Scan to play with it.

Learn more: [Ring Nebula on Wikipedia](#)

A planetary nebula is a glowing shell of gas puffed off by a dying star like the Sun. The hot little star left at the middle, a white dwarf, lights up the gas so it glows. The name is an old mistake: through early telescopes these round clouds looked a bit like planets, but they have nothing to do with planets.

This one is shaped like the Ring Nebula, about 2,500 light-years away. Tap it and the star at the middle flares and blows out a thin, bright ring of gas; when it reaches the main ring, the ring glows brighter and is pushed outward, then settles.

HOW TO PLAY

- **Tap:** The ring expands and glows brighter, then settles.
- **Sound:** A small burst as the star flares, a whoosh as the shock ring expands, and a deep soft boom as it pushes the main ring out.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/planetary-nebula/

CREDITS

- **Sound:** [Muffled Distant Explosion](#) by NenadSimic, CC0 1.0.
- **Sound:** [Deep Boom](#) by Za-Games, CC0 1.0.



Nebula

Tap it to light new stars in the clouds. Pick its colors in the Toy tab.



Scan to play with it.

Learn more: [Nebula on Wikipedia](#)

A nebula is a huge cloud of gas and dust in space. In some, gravity pulls the gas into clumps that heat up and become new stars, so they are called star nurseries. The young, hot stars light up the gas around them so it glows. Tall pillars of dust, like the famous Pillars of Creation in the Eagle Nebula, are places where stars are forming.

Tap it to light seven new stars, one after another, three at the tips of the pillars: each flares up with spikes and settles to a bright point in a glow of lit gas, then later fades. Pick its colors in the Toy tab.

HOW TO PLAY

- **Tap:** New stars ignite inside the clouds.
- **Sound:** A soft fiery flare for each of the seven new stars as it lights (no chimes).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

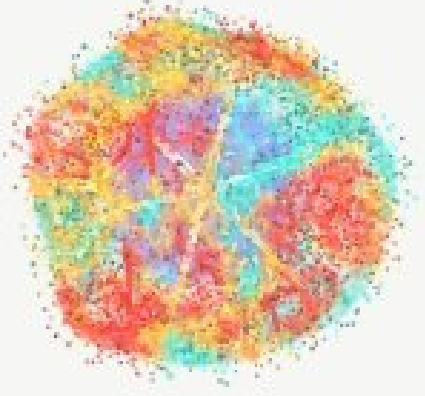
- **Colours:** Pink and teal, Gold and teal, Crimson, Blue ice

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/nebula/

CREDITS

- **Sound:** [Waving Torch.wav](#) by spookymodem, CC0 1.0.



Supernova

Tap it for a flash and a blast that blows the glowing shell apart.



Scan to play with it.

Learn more:
[Supernova on Wikipedia](#)

A supernova is the explosion of a star, one of the biggest blasts in the universe. For a few weeks one exploding star can shine brighter than a whole galaxy of billions of stars. It throws the star's gas out into space as a glowing, spreading shell.

Tap it and the core flashes and the shell flies apart in chunks that fade, then it glows back in place. Supernovas spread elements such as iron into space, where they end up in new stars and planets, and even in the blood in your body. People saw one in the year 1054; its remains are the Crab Nebula.

HOW TO PLAY

- **Tap:** Explode
- **Sound:** A real deep explosion with a long rumble, over the fiery roar.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

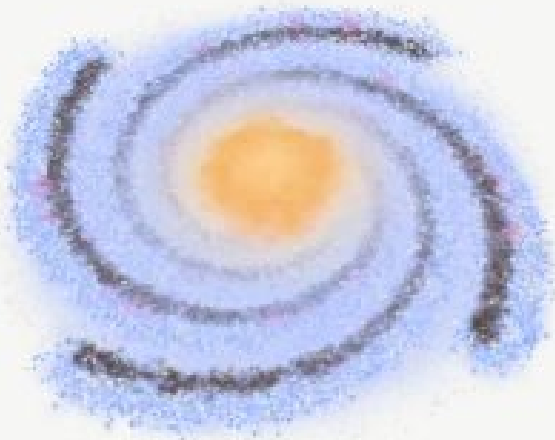
Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/supernova/

CREDITS

- **Sound:** [Deep Boom](#) by Za-Games, CC0 1.0.

Spiral galaxy

Tap to swirl the arms. Pick a galaxy style in the Toy tab.



Scan to play with it.

Learn more: [Spiral galaxy on Wikipedia](#)

A galaxy is a huge family of stars, gas and dust held together by gravity. In a spiral galaxy, bright arms curl out from a glowing center. Our own galaxy, the Milky Way, is a spiral about 100,000 light-years across with hundreds of billions of stars, and the Sun takes about 230 million years to go around it once.

Here the spiral turns slowly as one piece while its stars orbit, the inner ones faster. The toy illustrates one idea for spiral arms: a moving density pattern that stars pass through, like traffic through a jam. Real galaxies can also have changing arms. Tap it to swirl the arms once around, fast, while the core flares. Pick the Milky Way or Andromeda style in the Toy tab.

HOW TO PLAY

- **Tap:** Idle: slow rotation.
- **Sound:** Deep swirling hum.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

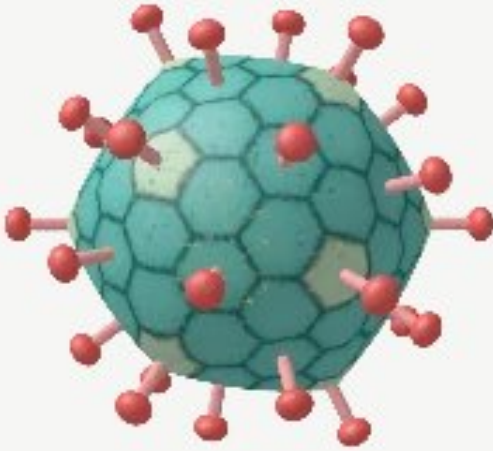
VIEWS

- **Style:** Milky Way-like, Andromeda-like

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/spiral-galaxy/

CREDITS

- Built from a recipe in the browser; no outside files.



Virus

Tap it to make copies that bud off and drift away.



Scan to play with it.

Learn more: [Virus on Wikipedia](#)

A virus is a tiny bundle of genes inside a shell of protein, usually smaller than a bacterium. It cannot grow or copy itself on its own: it has to get inside a living cell, which then makes new copies of it. The spikes on its shell help it latch on to the right kind of cell.

This one's spikes sway gently. Tap it and a wave ripples through them, and two smaller copies bud out from behind it, drift apart and fade away. Most viruses are so small that only an electron microscope can show them.

HOW TO PLAY

- **Tap:** Spikes wave and it splits into copies that fade away.
- **Sound:** The same ripple, squish and pops as the copies bud, without the shimmer at the end.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Capsid:** pick a color
- **Spikes:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/virus/

CREDITS

- Built from a recipe in the browser; no outside files.



Bacteriophage

Tap it to land and inject its DNA.



Scan to play with it.

Learn more:
[Bacteriophage on Wikipedia](#)

A bacteriophage, or phage, is a virus that infects only bacteria, not people. This tailed phage has a head full of DNA, a hollow tail and thin tail fibers, like legs, and it lands on a bacterium a bit like a spacecraft landing on the Moon.

Tap it to play the injection: the legs swing out, the sheath around the tail snaps to about half its length, the tail pokes through the base plate and a glowing strand of DNA coils out. Phages are thought to be the most numerous biological things on Earth, and doctors are studying them as a way to treat infections.

HOW TO PLAY

- **Tap:** Make the injection obvious: the legs splay, the sheath contracts a lot further and a glowing DNA strand shoots out of the tail.
- **Sound:** Spring-loaded thunk and a squirt.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

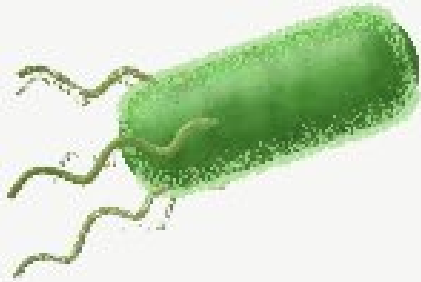
- **Head:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/bacteriophage/

CREDITS

- Built from a recipe in the browser; no outside files.



Bacterium

Tap it to make it divide in two. Try the Swim slider in the Toy tab.



Scan to play with it.

Learn more:
[Bacteria on Wikipedia](#)

A bacterium is a living thing made of just one small cell, with no nucleus: its DNA floats inside it. This one is rod-shaped, with long whips called flagella that spin like propellers to push it along. Most bacteria are harmless, and many help us, like the ones in our gut that help digest food.

Tap it and it divides in two, as bacteria do: it pinches in at the middle as a new wall closes across it, its DNA is shared between the halves, and the two cells come apart. Then the toy slides them back together. Some bacteria can divide every 20 minutes. Swim controls its swimming motion and Color sets its tint.

HOW TO PLAY

- **Tap:** Divides in two (binary fission), then merges back.
- **Sound:** Wet pop.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colour:** pick a color
- **Swim:** slider

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/bacterium/

CREDITS

- Built from a recipe in the browser; no outside files.



Red blood cell

Tap it to curl it into a sickle shape and back.



Scan to play with it.

Learn more: [Red blood cell on Wikipedia](#)

Red blood cells carry oxygen from the lungs to every part of the body. Each is a soft disc, dented in on both sides, that bends to squeeze through the tiniest blood vessels. A mature human red blood cell has no nucleus and is packed with hemoglobin, the red protein that holds the oxygen.

Tap it and it sickles: it stretches and curls into a stiff crescent, then relaxes back into a disc. This happens in sickle cell disease, where a change in the hemoglobin makes cells stiffen when oxygen is low. The body makes millions of new red blood cells every second.

HOW TO PLAY

- **Tap:** Deforms into a sickle shape (as in sickle-cell disease at low oxygen), then relaxes back to a disc.
- **Sound:** A soft, gel-like squish as it stretches into a crescent, and a soft wobble as it relaxes (no squeaks).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colour:** pick a color

In the Toy tab of the app.

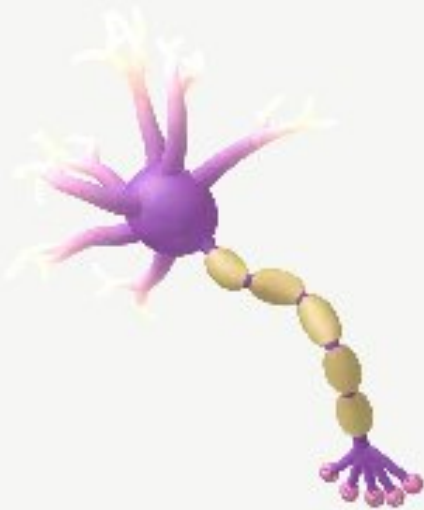
Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/red-blood-cell/

CREDITS

- Built from a recipe in the browser; no outside files.

Neuron

Tap it to fire a signal. Try the Signal slider in the Toy tab.



Scan to play with it.

Learn more: [Neuron on Wikipedia](#)

A neuron is a nerve cell, the kind of cell that carries messages around the brain and body. Its branching dendrites pick up signals, and a long wire called the axon carries them on to the tips, where they pass to the next cell.

Tap it to fire a signal: a bright spark runs from the cell body down the whole axon and bursts into light at the tips. Real nerve signals are tiny electrical pulses, and some travel at over 100 meters a second. The human brain has about 86 billion neurons.

HOW TO PLAY

- **Tap:** A bright pulse runs from the dendrites down the whole axon and bursts at the terminals.
- **Sound:** A real crackling electric arc running down the axon, and a brighter burst at the terminals.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Cell:** pick a color
- **Signal:** slider

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/neuron/

CREDITS

- **Sound:** [highvoltagearc.wav](#) by Sciolex, CC0 1.0.



Astrocyte

Tap it to send a calcium wave out along its arms.



Scan to play with it.

Learn more: [Astrocyte on Wikipedia](#)

An astrocyte is a star-shaped cell in the brain and spinal cord; its name means "star cell." Astrocytes look after the neurons around them, and the ends of their arms, called end-feet, wrap around tiny blood vessels and help control how much blood flows by.

Astrocytes pass messages to each other with waves of calcium, which scientists can watch with dyes that glow. Tap it to send one: a front of green light spreads from the middle out along every arm to the end-feet, and the blood vessel they hold widens for a moment. Faint waves pass by themselves now and then.

HOW TO PLAY

- **Tap:** Calcium wave: a glow ripples out along its arms.
- **Sound:** A soft airy swell as the calcium wave spreads (no synth shimmer), then the same soft wash as the vessel widens.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Cell:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/astrocyte/

CREDITS

- Built from a recipe in the browser; no outside files.



Animal cell

Tap it to divide it in two.
Switch off the Cutaway in
the Toy tab to see it whole.



Scan to play with it.

Learn more: [Cell \(biology\) on Wikipedia](#)

Cells are the tiny building blocks of every living thing, and the human body is made of tens of trillions of them. Inside an animal cell's soft skin, its membrane, are many parts: the nucleus holds the DNA, mitochondria turn food into energy the cell can use, and the Golgi packs up things the cell makes.

A wedge is cut away so you can see inside. Tap it and the cell divides: the nucleus splits in two, then the cell pinches in two, sharing out its parts. This is how bodies grow and repair themselves. Cutaway opens or closes the viewing wedge.

HOW TO PLAY

- **Tap:** Cell division: the nucleus splits and the cell pinches into two, then rejoins.
- **Sound:** Gloopy stretching pop.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

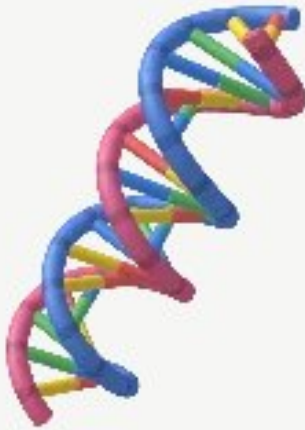
- **Membrane:** pick a color
- **Cutaway:** switch

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/animal-cell/

CREDITS

- Built from a recipe in the browser; no outside files.



DNA

Tap it to unzip it and zip it back up.



Scan to play with it.

Learn more: [DNA on Wikipedia](#)

DNA is the long molecule that carries the instructions for building and running a living thing. It is shaped like a twisted ladder, a double helix: two strands wind around each other, joined by pairs of bases, A with T and C with G, like the rungs.

Tap it and it unzips almost to the bottom, the bases lighting up in pairs as it opens, then zips back up. A cell unzips its DNA like this to copy it. In 1953 scientists worked out its shape, helped by Rosalind Franklin's X-ray pictures. The DNA in a typical human cell with a nucleus would stretch about 2 meters if pulled straight. Partner sets the matching strand's color.

HOW TO PLAY

- **Tap:** Unzips all the way, bases light up in pairs, then zips back.
- **Sound:** The real zipper timed to the strands: a quick zip as they fly open, a slower one as they close up (3.45 s).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Strand:** pick a color
- **Partner:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/dna/

CREDITS

- **Sound:** [Zipper Unzip 3 \(Slow\) .wav](#) by RutgerMuller, CC0 1.0.



White blood cell

Tap it to catch a bacterium and eat it up.



Scan to play with it.

Learn more: [White blood cell on Wikipedia](#)

White blood cells are the body's defenders: they find and clear away germs. This one is a neutrophil, the most common kind, which eats bacteria by wrapping itself around them. This is called phagocytosis, which means "cell eating" in Greek.

Tap it and a bacterium swims in from the side. The cell reaches out a cup of its skin around it, draws it inside and closes over it; then tiny grains inside the cell gather on it and break it down until it is gone. Élie Metchnikoff first saw cells eating germs like this, and won a Nobel Prize in 1908.

HOW TO PLAY

- **Tap:** Chases and engulfs a small bacterium (phagocytosis).
- **Sound:** A soft squishy membrane reaching out (no flutter clicks), the same gulp as it swallows, and a quieter fizz as it digests.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colour:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/white-blood-cell/

CREDITS

- Built from a recipe in the browser; no outside files.

Microglia

Tap it to reach out its arms and sweep them to and fro.



Scan to play with it.

Learn more:
[Microglia on Wikipedia](#)

Microglia are the brain's cleaners and guards. They are small cells with thin, branching arms, spread all through the brain, and they keep moving their arms to check the space around them, clearing away waste and worn-out parts.

This one's six arms feel about gently all the time. Tap it and every arm stretches out, sweeps to and fro twice, and draws back in. Real microglia can sweep over their whole patch of brain in a few hours, and they help keep the connections between neurons tidy.

HOW TO PLAY

- **Tap:** Arms reach out and sweep, then retract.
- **Sound:** Soft scratchy rustle.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

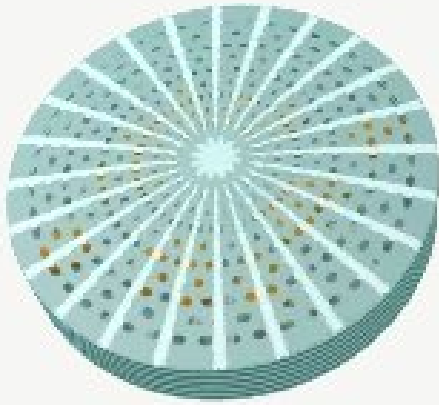
- **Cell:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/microglia/

CREDITS

- Built from a recipe in the browser; no outside files.



Diatom

Tap it to make the glass shell glint and open.



Scan to play with it.

Learn more: [Diatom on Wikipedia](#)

A diatom is a single-celled alga with a shell made of silica, the material in glass. Its patterned shell, called a frustule, has two overlapping halves, rather like a box and its lid. Diatoms live in fresh and salt water and use sunlight to make food, releasing oxygen.

Tap it and a glint of light runs across the glass, then the two halves lift apart to show the golden cell inside and close again; real diatoms don't open like this. Diatoms make about a fifth of the oxygen on Earth.

HOW TO PLAY

- **Tap:** The glass shell glints and splits into two halves.
- **Sound:** Glassy tink.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Glass:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/diatom/

CREDITS

- Built from a recipe in the browser; no outside files.



Tardigrade

Tap it to make it wiggle.



Scan to play with it.

Learn more:

[Tardigrade on Wikipedia](#)

Tardigrades, or water bears, are tiny animals, most about half a millimeter long, with a plump body and eight stubby legs that end in claws. They live all over the world, in moss, soil, ponds and the sea.

When their home dries out, they pull in their legs, curl into a dry little ball called a tun and wait, sometimes for years, until water brings them back. In 2007 some were even taken into space on the outside of a satellite, and a few came back alive. Tap it and it wiggles its body and legs.

HOW TO PLAY

- **Tap:** Wiggle
- **Sound:** Tiny squeaky step.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

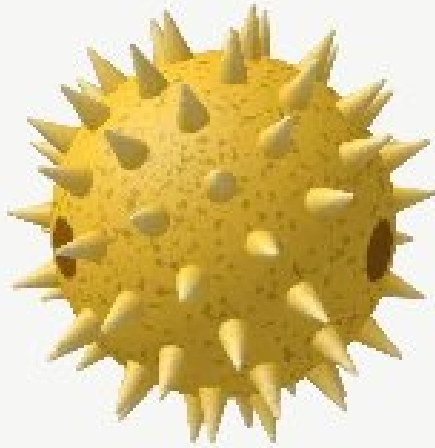
- **Colour:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/tardigrade/

CREDITS

- Built from a recipe in the browser; no outside files.



Pollen grain

Tap it to burst out a puff of tiny grains. Pick the Plant in the Toy tab.



Scan to play with it.

Learn more: [Pollen on Wikipedia](#)

Pollen is a fine powder that flowers and cones make. Each tiny grain carries what a plant needs to make seeds, and wind, bees and other animals carry it from plant to plant. Its tough outer wall has a different pattern for every kind of plant.

Tap it and it bursts, as pollen can when it soaks up rain: it swells, then a puff of tiny grains jets out and drifts down. In the Toy tab, pick the plant: a spiky sunflower grain, a pine grain with two air sacs that help it float on the wind, or an oval lily grain.

HOW TO PLAY

- **Tap:** Bursts a puff of tiny grains.
- **Sound:** The same puff and chirp, quieter.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Plant:** Sunflower, Pine, Lily
- **Colour:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/pollen/

CREDITS

- Built from a recipe in the browser; no outside files.



Snowflake

Tap to melt it and grow a new flake. Pick the kind of crystal in the Toy tab.



Scan to play with it.

Learn more:
[Snowflake on Wikipedia](#)

A snowflake starts high in a cloud, when water vapor deposits as ice onto a tiny speck of dust and grows into an ice crystal. It has six sides because of the way water molecules fit together as they freeze. How cold and damp the air is decides its shape, from simple plates to feathery stars.

Tap it and the arms melt back to the middle, then a new flake grows out, a new pattern each time. In the Toy tab, pick the kind of crystal. In 1885 Wilson Bentley became the first person to photograph a single snowflake.

HOW TO PLAY

- **Tap:** Arms regrow into a new pattern each tap.
- **Sound:** Crystalline tinkle.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Crystal:** Stellar dendrite, Fern, Sectoried plate, Simple star
- **Variant:** slider

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/snowflake/

CREDITS

- Built from a recipe in the browser; no outside files.



Chromosome

Tap it to pull its two halves apart.



Scan to play with it.

Learn more:
[Chromosome on Wikipedia](#)

A chromosome is a long thread of DNA, wound up tight and packed small. Most human cells have 46 of them, in 23 pairs. Before a cell divides, each chromosome copies itself, and the two copies, called sister chromatids, stay joined at a pinched spot, the centromere, making an X shape.

Tap it to watch them part, as they do when a cell divides: thin fibers reach in from two bright poles, hook on at the middle and pull the two halves apart to either side, arms trailing. Then the toy puts them back together. This way each new cell gets a full set.

HOW TO PLAY

- **Tap:** The sister chromatids pull apart at the centromere, then rejoin.
- **Sound:** A soft sticky tear as the chromatids pull apart, and a soft squish as they come back together (no string, no clacks).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colour:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/chromosome/

CREDITS

- **Sound:** [Velcro](#) by paulocorona, CC0 1.0.



Mitochondrion

Tap it to power it up. Switch off the Cutaway in the Toy tab to see it whole.



Scan to play with it.

Learn more: [Mitochondria on Wikipedia](#)

A mitochondrion is a tiny part inside a cell, often called the powerhouse of the cell. It uses oxygen to turn the sugar from food into ATP, the small molecule that carries energy to every part of the cell. Its inner skin is folded into ridges called cristae, which gives it lots of room to work.

The top is cut away to show the ridges. Tap it and light runs along them, it swells, and bright sparks of ATP pop out and drift off. Mitochondria even have their own DNA, which humans usually inherit from their mothers. Cutaway opens or closes the view into the cristae.

HOW TO PLAY

- **Tap:** The cristae light up and energy sparks (ATP) pop out.
- **Sound:** A real gas furnace lighting with a soft whoomp and a warm roar as it powers up, with the ATP sparks' soft pops; no motor, vroom or buzz (candidate A; B is a gas burner's steadier roar).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

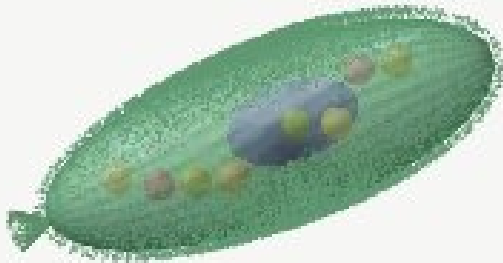
- **Colour:** pick a color
- **Cutaway:** switch

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/mitochondrion/

CREDITS

- **Sound:** [Gas furnace - Ignition happy](#) by Idezem, CC0 1.0.
- **Sound:** [Gas burner 01.wav](#) by PegasusCZ, CC0 1.0.



Paramecium

Tap it to swim a loop. Try the Swim slider in the Toy tab.



Scan to play with it.

Learn more:

[Paramecium on Wikipedia](#)

A paramecium is a living thing made of a single cell, shaped a bit like a slipper, that lives in ponds and puddles. It is covered in thousands of tiny hairs called cilia, which beat in waves like oars to row it through the water. A groove in its side sweeps in bacteria to eat.

Tap it and its cilia beat hard as it swims a full loop and comes back to where it was. A paramecium is just big enough to see with a magnifying glass, as a tiny moving speck.

HOW TO PLAY

- **Tap:** Cilia beat in waves and it swims a loop.
- **Sound:** A hand moving gently through water: soft strokes through its loop, no shuffle or wind.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colour:** pick a color
- **Swim:** slider

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/paramecium/

CREDITS

- **Sound:** [Water, Gentle Movement.wav](#) by Daen23, CC0 1.0.



Amoeba

Drag it to pull out a pseudopod; let go and it oozes back. Tap it to make it crawl.



Scan to play with it.

Learn more: [Amoeba on Wikipedia](#)

An amoeba is a living thing made of one soft cell with no fixed shape. It lives in ponds and damp soil and moves by pushing out a bulge called a pseudopod, which means "false foot," and flowing into it. It eats by wrapping its body around bits of food.

Drag any part of it and it stretches toward your finger like a new pseudopod; let go and it oozes back. Tap it and it crawls: a pseudopod pushes out to one side, the grains inside stream into it and the cell oozes over; then it pushes out the other way and oozes back. Even at rest, its pods stretch a little.

HOW TO PLAY

- **Tap:** Pushes out a pseudopod and oozes over.
- **Sound:** Louder: a soft squelch and a low rubbery wobble as each pseudopod pushes out; no whistle or ding.
- **Drag:** stretches it; let go and it springs back.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colour:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/amoeba/

CREDITS

- Built from a recipe in the browser; no outside files.



Electron orbital

Tap it to excite the electron.
Pick any orbital up to 4f, or
lobes, in the Toy tab.



Scan to play with it.

Learn more: [Atomic orbital on Wikipedia](#)

An electron does not circle the center of an atom like a planet. It spreads out as a cloud, called an orbital, that shows where it is most likely to be found. Orbitals come in set shapes: a ball, a dumbbell, a dumbbell with a ring, and more. The two colors show where the electron's wave is plus and where it is minus, not electric charge.

Pick an orbital from 1s to 4f, as a cloud or as lobes, in the Toy tab. Tap it and a tiny packet of light, a photon, comes in. The electron takes it in, jumps up to a bigger orbital and glows, then drops back with a flash and gives the light out again. This is how glowing gases give off light of their own colors.

HOW TO PLAY

- **Tap:** Jumps to a higher orbital shape with a glow, then drops back and flashes.
- **Sound:** A short electric crackle and a rising swish as the photon is absorbed, and a brighter crackle and falling swish as it is given out.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

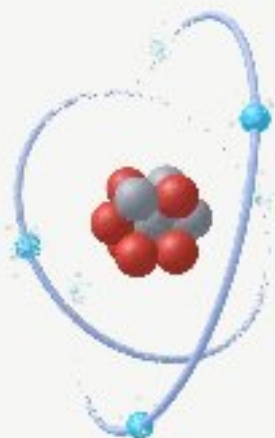
- **Orbital:** 1s, 2s, 2p (z), 2p (x), 2p (y), 3s, 3p (z), 3p (x) and 22 more
- **Look:** Cloud, Lobes
- **Plus phase:** pick a color
- **Minus phase:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/orbital/

CREDITS

- **Sound:** [highvoltagearc.wav](#) by Sclolex, CC0 1.0.



Atom

Tap it to speed up the electrons.
Pick any of the 118 elements in the Toy tab.



Scan to play with it.

Learn more: [Atom on Wikipedia](#)

Everything around you is made of atoms. Each one has a tiny heavy center, the nucleus, made of protons and (in almost every atom) neutrons, with electrons around it. The number of protons decides which element it is: carbon has 6, oxygen 8 and gold 79. Pick any of the 118 elements in the Toy tab.

This model draws the electrons in rings called shells, an idea of Niels Bohr's from 1913; pick the Cloud style for a truer picture, or Every nucleon to see all its protons and neutrons. Tap it and the electrons whirl faster until each shell blurs into a glowing ring, then slow down. The nucleus is tens of thousands of times smaller than the whole atom, which is mostly empty space.

HOW TO PLAY

- **Tap:** Electrons speed up into blurred shells.
- **Sound:** The periodic table's atom sound: a soft electric blip as the electrons speed up and a ping as they blur into rings; no motor.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

VIEWS

- **Style:** Bohr, Cloud

SETTINGS

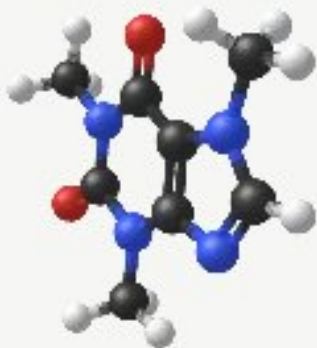
- **Element:** 1 Hydrogen, 2 Helium, 3 Lithium, 4 Beryllium, 5 Boron, 6 Carbon, 7 Nitrogen, 8 Oxygen and 110 more
- **Nucleus:** Simple, Every nucleon
- **Electrons:** slider

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/atom/

CREDITS

- **Element data:** [Atomic Weights and Isotopic Compositions](#); [Atomic Spectra Database ground levels](#) by NIST Physical Measurement Laboratory; PubChem; IUPAC (elements 95 to 118), Public domain (facts).



Molecule

Tap it to heat it up. Pick a molecule, or type your own, in the Toy tab.



Scan to play with it.

Learn more: [Molecule on Wikipedia](#)

A molecule is a group of atoms held together by chemical bonds. This one is a ball-and-stick model: each ball is an atom, colored by its element (carbon dark gray, hydrogen white, oxygen red, nitrogen blue), and each stick is a bond.

It starts as caffeine, $C_8H_{10}N_4O_2$, the stimulant in coffee and tea; the Toy tab has sugars, medicines, vitamins and DNA too. The atoms always jiggle a little on their bonds, as real ones do; tap it to heat it up and they shake hard, the light hydrogens furthest, each bond stretching and squeezing with its atoms, then it cools. In the Toy tab, pick another molecule, or type a name, a formula or a SMILES string (a way of writing a molecule on one line) to build your own.

HOW TO PLAY

- **Tap:** Bonds vibrate and stretch (like heat), then calm.
- **Sound:** Soft little ticks of the atoms jostling as it heats, thinning out as it settles; no wind, no building blocks.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

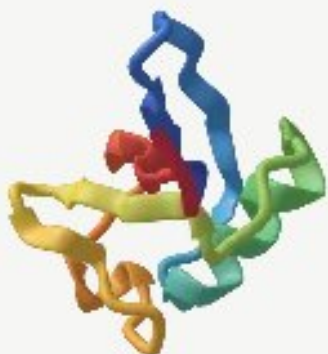
- **Molecule:** Water, Carbon dioxide, Methane, Ammonia, Ethanol, Benzene, Caffeine, Buckyball (C_{60}) and 23 more

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/molecule/

CREDITS

- **Molecule gallery:** [PubChem 3D conformers \(glucose, aspirin, ATP and the rest\)](#) by NCBI PubChem, Public domain.
- **DNA:** [RCSB Protein Data Bank entry 1BNA \(B-DNA dodecamer\)](#), [hydrogens added](#) by H. R. Drew, R. M. Wing, T. Takano, C. Broka, S. Tanaka, K. Itakura, R. E. Dickerson, CC0 1.0.



Protein

Tap it to pull it apart. Pick a protein, or open your own file, in the Toy tab.



Scan to play with it.

Learn more: [Protein on Wikipedia](#)

Proteins are the tiny machines of living things. Each is a long chain of building blocks called amino acids that folds up into its own shape: coils called helices, flat strands and loops. Hemoglobin carries oxygen in the blood, insulin helps control sugar, and a jellyfish protein, GFP, glows green.

These are real shapes from the Protein Data Bank, a free library of well over 200,000 structures that scientists share. Tap it to pull it apart into its pieces, and they come back by themselves; tap again while it is apart and they come back at once. In GFP the glowing part lights up while it is open. You can open any PDB or mmCIF file from the library in the Toy tab.

HOW TO PLAY

- **Tap:** The protein pulls apart into its helices, strands and loops, then locks back together.
- **Sound:** A real tearing apart as it separates, with a whoosh, then a pressing-together and a soft thud as it reassembles (no building blocks).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

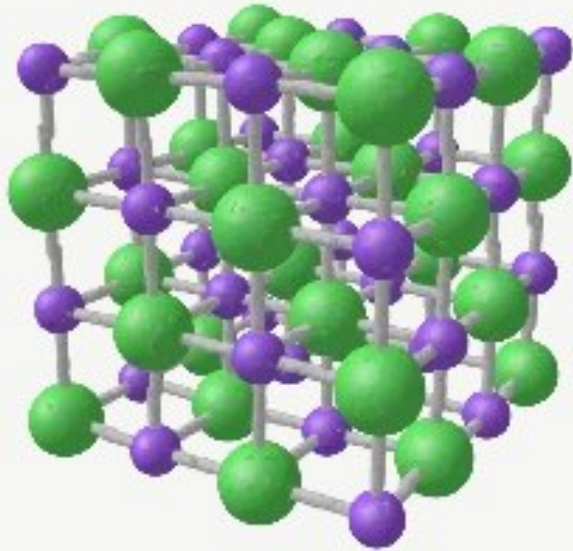
- **Protein:** Ubiquitin (1UBQ), Insulin (4INS), Green fluorescent protein (1EMA), Hemoglobin (4HHB), Your own (below)
- **Colour:** Rainbow, start to end, By chain, Helices, strands, loops

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/protein/

CREDITS

- **Protein structures:** [RCSB Protein Data Bank entries 1UBQ, 4INS, 1EMA and 4HHB](#) by the wwPDB and the structures' authors, CC0 1.0.
- **Sound:** [Velcro](#) by paulocorona, CC0 1.0.



Crystal lattice

Tap to send a wave through it. Pick one of eleven crystals in the Toy tab.



Scan to play with it.

Learn more: [Crystal structure on Wikipedia](#)

In a crystal, atoms are lined up in a pattern that repeats over and over, like tiles on a floor. In table salt, sodium and chlorine take turns in rows of little cubes. Diamond and graphite are both pure carbon: in diamond each atom holds on to four others, which makes it very hard, while graphite's flat sheets slide apart, which is why a pencil writes.

Tap it to send a wave through it: a ripple runs across, each slice of atoms rising and falling in turn, the bonds between them stretching and bending. Sound and heat travel through solids as waves like this. In ice, the water molecules spread out into an open pattern, so ice floats on water. Metals, quartz and graphene are in the Toy tab too.

HOW TO PLAY

- **Tap:** A wave of vibration travels through the lattice.
- **Sound:** One cleaner, softer metallic chime with a faint glassy ring, no rattle after.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

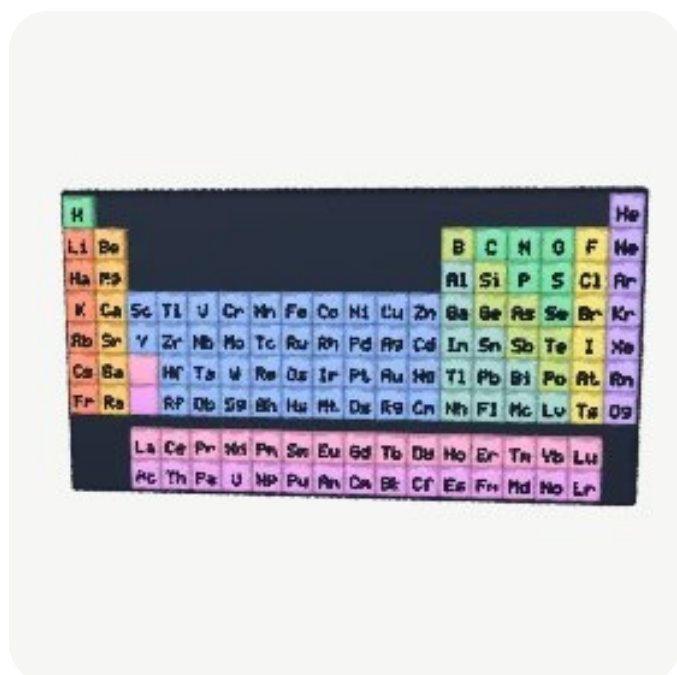
- **Crystal:** Salt (NaCl), Diamond, Graphite, Ice, Iron (body-centered cubic), Copper (face-centered cubic), Cesium chloride (CsCl), Fluorite (CaF₂) and 3 more

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/crystal-lattice/

CREDITS

- **Sound:** [Steel chain soft drop](#) by ani_music, CC0 1.0.



Periodic table

Tap a tile to raise its atom, again to lower it. Tap the board for a tour, 57-71 for its row.



Scan to play with it.

Learn more: [Periodic table on Wikipedia](#)

The periodic table lists all 118 elements in order of their protons, in rows that repeat their chemistry: each column's elements behave alike. The colors are the families, from the alkali metals on the left to the noble gases on the right.

Tap an element and its atom rises and builds itself: the protons (red) and neutrons (gray) of a typical isotope, then the electrons, filling their shells in the real order. Tap the atom and an electron jumps up a shell and falls back, giving off light the color of the element's brightest visible line: red for hydrogen, yellow for sodium, green for copper. Elements with no visible line measured flash white. Tap the board for a tour. Tap 57-71 or 89-103 to light that row; Wide table shows it in place. Numbers from NIST, PubChem and IUPAC.

HOW TO PLAY

- **Tap:** Tap an element's tile and its atom rises out of the table and builds itself: every proton and neutron of its commonest isotope, then the electrons filling their shells in order.
- **Sound:** The atom's whoosh much quieter, and a faint, soft tick for each proton and neutron as it packs into the nucleus, in sync.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

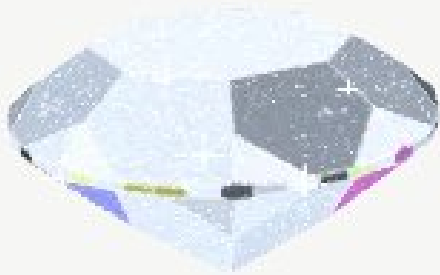
- **Element:** 1 Hydrogen, 2 Helium, 3 Lithium, 4 Beryllium, 5 Boron, 6 Carbon, 7 Nitrogen, 8 Oxygen and 110 more
- **Wide table:** switch
- **Tour order:** By atomic number, Shuffled
- **Excite an electron:** button
- **Tour:** button
- **Show the row:** button

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/periodic-table/

CREDITS

- **Element data:** [Atomic Weights and Isotopic Compositions](#); [Atomic Spectra Database ground levels](#); [Handbook of Basic Atomic Spectroscopic Data \(strong lines\)](#) by NIST Physical Measurement Laboratory, Public domain (U.S. government data).
- **Element families and the newest elements:** [PubChem Periodic Table of Elements](#); [IUPAC Periodic Table of the Elements \(2022\)](#) by NCBI PubChem; IUPAC, Public domain (facts).



Diamond

Tap it to turn it in the light and flash its fire.



Scan to play with it.

Learn more:
[Diamond on Wikipedia](#)

A diamond is pure carbon, squeezed into a crystal deep underground by great heat and pressure, and carried up to the surface by volcanic eruptions long ago. It is the hardest natural material. This one has a brilliant cut, with dozens of small flat faces, called facets, to catch the light.

Tap it and it turns one way and the other, the way you would tilt a ring: facet after facet flashes with rainbow sparks. Jewelers call these colors fire. They come from the stone splitting white light into its colors, as a prism does.

HOW TO PLAY

- **Tap:** Fire: rainbow sparkles flash off the facets as it turns.
- **Sound:** A real bright glassy tap, and smaller tinkles as the facets flash.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Tint:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/diamond/

CREDITS

- **Sound:** [Glass Tap.wav](#) by Unicornaphobist, CC0 1.0.



Ruby

Tap it to make it glow deep red.



Scan to play with it.

Learn more: [Ruby](#)
on [Wikipedia](#)

A ruby is a red form of the mineral corundum, which is made of aluminum and oxygen. A little chromium in the crystal gives it its red color. Corundum is very hard, second only to diamond among gems, and in any other color it is called sapphire.

The chromium also makes rubies glow red under ultraviolet light. Tap it and the stone lights up deep red from its heart outward, throbs twice and fades. In 1960 the first laser was made with a rod of ruby.

HOW TO PLAY

- **Tap:** Glows deep red from inside.
- **Sound:** A lower real glassy tap and a warm low glow as it lights red.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

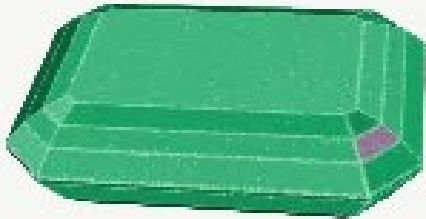
- **Colour:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/ruby/

CREDITS

- **Sound:** [Glass Tap.wav](#) by Unicornaphobist, CC0 1.0.



Emerald

Tap it to send green light spiraling around its steps.



Scan to play with it.

Learn more:
[Emerald on Wikipedia](#)

An emerald is a green form of the mineral beryl, colored by tiny amounts of chromium or vanadium. Most emeralds have little specks and cracks inside them, and jewelers often cut them in rows of long, flat steps. That shape is so common for emeralds that it is called the emerald cut.

A step cut works like a hall of mirrors. Tap it and green light races around the edges of the steps, each one a moment behind the one outside it, so the light spirals in to the flat top, the table. Some of the finest emeralds come from Colombia.

HOW TO PLAY

- **Tap:** Green light runs along the step cuts.
- **Sound:** The crisp gem tap alone, without the whoosh.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colour:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/emerald/

CREDITS

- **Sound:** [Glass Tap.wav](#) by Unicornaphobist, CC0 1.0.



Amethyst geode

Tap to close the geode; tap again to open it.



Scan to play with it.

Learn more:
[Amethyst on Wikipedia](#)

A geode is a rock with a hollow inside, lined with crystals. Over a very long time, water full of dissolved minerals seeped into the hollow, and the crystals slowly grew inward from the walls. Amethyst is purple quartz; its color comes from a little iron in the crystal.

The geode starts open. Tap it to close it; tap again and it opens, its halves swing wider, and the crystals glow violet with twinkling tips, then settle. Some amethyst geodes from Brazil and Uruguay are taller than a person.

HOW TO PLAY

- **Tap:** The geode opens wider and the crystals glow violet.
- **Sound:** Stone crack and a chime.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

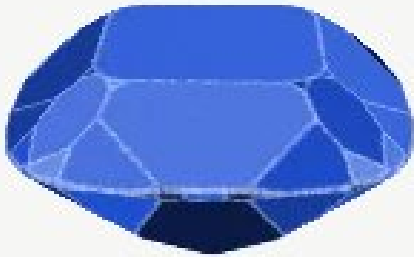
- **Crystals:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/amethyst-geode/

CREDITS

- Built from a recipe in the browser; no outside files.



Sapphire

Tap it to catch the six-rayed star gliding across it.



Scan to play with it.

Learn more:
[Sapphire on Wikipedia](#)

A sapphire is the mineral corundum, the same as ruby, but colored blue by iron and titanium. Sapphires come in many colors, from yellow to pink, but blue is the best known. They are very hard, so they last a long time in rings.

Some sapphires show a star of six rays, because tiny needles of another mineral inside reflect the light. The star is called an asterism, and it moves as the stone or the light moves. Tap it to catch the star: its rays spread out from a bright center, it glides across the top, and the rays draw back in at the far side.

HOW TO PLAY

- **Tap:** A star shimmer (asterism) glides across it.
- **Sound:** The crisp gem tap alone, without the wind and sandy shuffle.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colour:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/sapphire/

CREDITS

- **Sound:** [Glass Tap.wav](#) by Unicornaphobist, CC0 1.0.



Quartz cluster

Tap it to light the crystal points one by one.



Scan to play with it.

Learn more: [Quartz on Wikipedia](#)

Quartz is one of the most common minerals in the Earth's crust, made of silicon and oxygen. Its crystals grow as six-sided columns with pointed tips, often many together in a cluster like this one. Clear quartz is also called rock crystal.

Tap it and the ten points light up one by one, from left to right, each glowing from inside with a glint at its tip and a rising chime. A tiny sliver of quartz shakes at a very steady beat when electricity runs through it, so it keeps time in many watches and clocks.

HOW TO PLAY

- **Tap:** Points light up one by one.
- **Sound:** Tinkling chimes.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Tint:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/quartz-cluster/

CREDITS

- Built from a recipe in the browser; no outside files.



Opal

Tap it to tilt it and roll flashes of color across it. Pick an opal in the Toy tab.



Scan to play with it.

Learn more: [Opal on Wikipedia](#)

Opal is made of tiny balls of silica, the stuff of quartz and sand, with a little water between them. In precious opal the balls are packed in neat rows, and light passing through them splits into flashes of color that change as the stone moves. This is called play of color. Fire opal is named for its orange body. Opal is cut into a smooth dome called a cabochon, not into facets.

Tap it and it rocks in the light while patches of color roll across it, changing as they go. In the Toy tab, pick a white, black or fire opal. Most of the world's opal comes from Australia.

HOW TO PLAY

- **Tap:** Colour play: flashes of colour roll across it.
- **Sound:** A softer, duller real tap and the stone rocking gently on the table (no synth shimmer).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Opal:** White opal, Black opal, Fire opal

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/opal/

CREDITS

- **Sound:** [Glass Tap.wav](#) by Unicornaphobist, CC0 1.0.



Pearl

Tap to close the oyster; tap again to open it.



Scan to play with it.

Learn more: [Pearl on Wikipedia](#)

A pearl is a gem made by a living animal. When an irritant, usually a scrap of the animal's own tissue or a bead put in by a pearl farmer, ends up inside an oyster or mussel, the animal coats it with layer after layer of nacre, or mother-of-pearl, the same shiny stuff that lines its shell. The thin layers give a pearl its soft glow, called luster.

This oyster starts open, with its pearl inside. Tap it to close the shell; tap again to open it. A pearl can take years to grow, and most pearls sold today are grown on pearl farms.

HOW TO PLAY

- **Tap:** Open or close
- **Sound:** Soft round clink.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

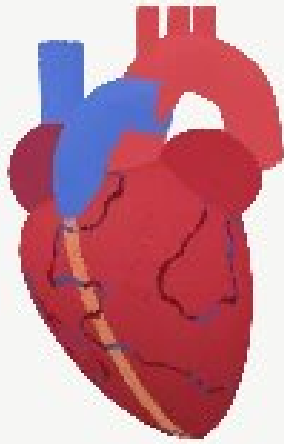
- **Pearl:** pick a color
- **Oyster shell:** switch

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/pearl/

CREDITS

- Built from a recipe in the browser; no outside files.



Beating heart

Tap it to set it racing. Set its beat, or pick a love heart, in the Toy tab.



Scan to play with it.

Learn more: [Heart on Wikipedia](#)

The heart is a muscle that pumps blood around the body. It has four chambers: the two atria on top fill with blood, then the two ventricles below squeeze it out, the right side to the lungs and the left side to the rest of the body.

This heart beats by itself, the atria first and then the ventricles. Tap it and it races, as it does when you run, then calms down again. A resting adult heart beats about 60 to 100 times a minute, which is about 100,000 beats a day. Style switches between an anatomical heart and a love heart; Color changes its tint.

HOW TO PLAY

- **Tap:** Racing heartbeat: faster beats with a glow, then calms.
- **Sound:** The same real heartbeats at the same tempo, about 60 percent louder.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

VIEWS

- **Style:** Anatomy, Love heart

SETTINGS

- **Colour:** pick a color
- **Beat:** slider

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/heart/

CREDITS

- **Sound:** [Human Heartbeat \(60 BPM\)](#) by FenrirFangs, CC0 1.0.



Brain

Tap it to think. Try the Sparks slider in the Toy tab.



Scan to play with it.

Learn more: [Human brain on Wikipedia](#)

The brain is the body's control center. Its wrinkled outer layer, the cortex, is folded so that a lot of it fits inside the skull, and it is split into lobes that do different jobs: the back handles seeing, the sides hearing and memory, the top touch, and the front moving, planning and deciding. It works by sending tiny electrical signals between about 86 billion nerve cells.

Tap it to spark a thought: light races along the folds of the lobe you tapped, then through the others, and every lobe flashes at once. The Sparks slider sets how many sparks twinkle over it, and the Toy tab can color it by lobe or plain pink.

HOW TO PLAY

- **Tap:** Sparks of light race across the folds (a thought).
- **Sound:** A subtle electric crackle as the sparks race round (softer and lower than the neuron's), and a small snap as the side lights up; no wind.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colours:** Lobes, Plain pink
- **Sparks:** slider

In the Toy tab of the app.

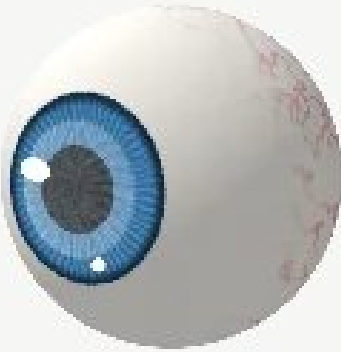
Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/brain/

CREDITS

- **Sound:** [Muffled Distant Explosion](#) by NenadSimic, CC0 1.0.

Eye

Tap it to blink. Try the Pupil slider in the Toy tab.



Scan to play with it.

Learn more: [Human eye on Wikipedia](#)

The eye is how we see. Light goes in through the pupil, the black hole in the middle, and a lens focuses it onto the retina at the back, which sends a picture to the brain. The colored ring around the pupil is the iris: its muscles make the pupil smaller in bright light and bigger in the dark.

This eye glances around by itself. Tap it and it blinks, turns to look at you and snaps its pupil small. The Pupil slider sets how wide the pupil is, and you can pick the iris color. People blink about 15 to 20 times a minute, which keeps the eye wet and clean.

HOW TO PLAY

- **Tap:** Idle: the eye looks around.
- **Sound:** A soft, small click as the lids close, and a fainter one as they open.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

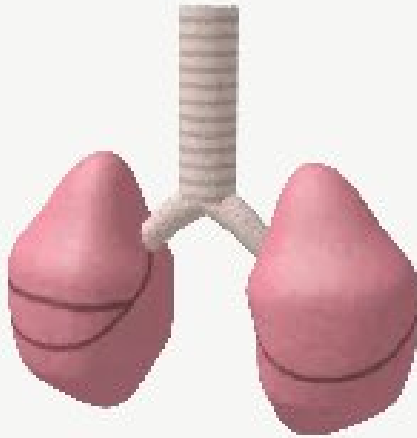
- **Iris:** pick a color
- **Pupil:** slider

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/eye/

CREDITS

- Built from a recipe in the browser; no outside files.



Lungs

Tap for a deep breath. Try the Breath slider in the Toy tab.



Scan to play with it.

Learn more: [Lung](#) on Wikipedia

The lungs are two spongy organs in the chest that bring air into the body. Air comes down the windpipe, splits into two tubes and branches into smaller and smaller airways that end in millions of tiny air sacs, where oxygen passes into the blood and carbon dioxide passes out.

These lungs breathe gently by themselves. Tap for a deep breath: both lungs swell out big, hold, then empty further than usual and settle back into their rhythm. The Breath slider sets how deeply they breathe at rest. An adult at rest takes about 12 to 20 breaths a minute.

HOW TO PLAY

- **Tap:** A deep breath in and out.
- **Sound:** A real person's slow deep breath in, then out.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colour:** pick a color
- **Breath:** slider

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/lungs/

CREDITS

- **Sound:** - [Deep Breath](#) by rrehl, CC0 1.0.



Tooth

Tap it to polish it until it sparkles.



Scan to play with it.

Learn more: [Tooth on Wikipedia](#)

A tooth is made in layers. On the outside is enamel, the hardest material in the whole body. Under it is a softer, yellowish layer called dentin, and in the middle is the pulp, with the nerves and blood vessels. Roots hold the tooth in the jawbone.

This is a molar, one of the wide back teeth that grind food. Tap it to polish it: a bright sheen wipes across, star sparkles pop where it passes and twinkle while it gleams. Pick the enamel's color in the Toy tab. Children have 20 baby teeth, and adults usually have 32.

HOW TO PLAY

- **Tap:** A sparkle wipe cleans it bright white.
- **Sound:** A soft polishing swish instead of the squeak, then the same short gleam and sparkles.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

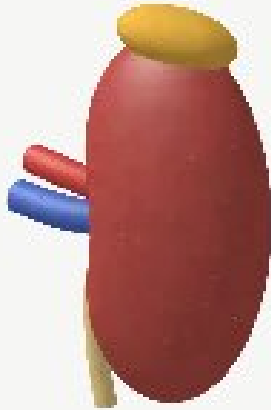
- **Enamel:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/tooth/

CREDITS

- Built from a recipe in the browser; no outside files.



Kidney

Tap it to pump blood through it.



Scan to play with it.

Learn more: [Kidney on Wikipedia](#)

The kidneys are two bean-shaped organs, each about the size of a fist, tucked in the back just below the ribs. They clean the blood: they filter out waste and extra water, which leave the body as urine, and send the clean blood back.

Tap it to pump three pulses of blood through: each comes in along the artery as light, spreads through the kidney as it swells a little, leaves along the vein, and a drop runs down the ureter, the tube to the bladder. Together, the kidneys filter all the body's blood many times a day.

HOW TO PLAY

- **Tap:** Fluid flows through in pulses along the vessels.
- **Sound:** Gentle trickle.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colour:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/kidney/

CREDITS

- Built from a recipe in the browser; no outside files.



Oak tree

Tap it to shake down its leaves. Pick a season in the Toy tab.



Scan to play with it.

Learn more: [Oak on Wikipedia](#)

The oak is a big, broad tree with a thick trunk and wavy-edged leaves, and its seeds are acorns. Oaks grow slowly and can live for hundreds of years. A single old oak is home to hundreds of kinds of insects, birds and other animals.

Tap it and the crown rocks on its trunk, and leaves come loose one by one and flutter down to the grass; then fresh leaves open in their place. In the Toy tab, pick spring, summer, autumn or winter, or a Seed for a different tree. In winter some oaks keep a few dead brown leaves on their twigs, and those are the ones that fall.

HOW TO PLAY

- **Tap:** Shakes and leaves flutter down.
- **Sound:** A leafy rustle as the crown rocks, then soft rustles as the loose leaves land (synthesized; no flutter or patter); real CC0 recordings now (oak-leaves.mp3), with the synthesized sound as a fallback.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Season:** Spring, Summer, Autumn, Winter
- **Seed:** 1, 2, 3, 4

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/oak/

CREDITS

- **Sound:** [FallingLeaves](#) by falcospizaetus, CC0 1.0.



Pine tree

Tap it to shake the snow off.
Turn on Snow in the Toy tab
for a snowy day.



Scan to play with it.

Learn more: [Pine on Wikipedia](#)

A pine is an evergreen tree: it keeps its long, thin needles all year, and its seeds grow inside woody cones. Its sloping branches help heavy snow slide off instead of breaking them, so pines grow well in cold, snowy places.

Tap it and the tree rocks, and the snow on its boughs drops off from the top down in puffs of powder, lies on the ground and melts. On a day without snow, a quick shower dusts the branches first. With Snow on in the Toy tab, fresh snow settles on the boughs again afterward.

HOW TO PLAY

- **Tap:** Shakes off a dusting of snow (approved instead of lights, which the decorated tree already has).
- **Sound:** The needles rustle as it rocks, with a soft jingle of sleigh bells; the snow's hiss and thump stay (no chime).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Snow:** switch

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/pine/

CREDITS

- Built from a recipe in the browser; no outside files.



Palm tree

Tap it to shake down the coconuts.



Scan to play with it.

Learn more:
[Areaceae on Wikipedia](#)

The coconut palm grows on warm, sandy coasts all around the tropics. It has no branches: its tall, bendy trunk ends in a crown of giant feathery leaves, called fronds, with the coconuts hanging in bunches beneath them.

Tap it and the crown shakes, and the five coconuts drop one after another, thump into the sand, bounce and roll to a stop; then new green coconuts swell in the crown. A coconut can float, so it can drift across the sea for weeks and sprout on a faraway beach.

HOW TO PLAY

- **Tap:** The coconuts fall and bounce, then regrow.
- **Sound:** Two coconut bonks.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/palm/

CREDITS

- Built from a recipe in the browser; no outside files.



Cherry blossom

Tap it to shake down the blossom and watch it bloom again.



Scan to play with it.

Learn more: [Cherry blossom on Wikipedia](#)

Cherry trees flower in spring, covering their bare branches in clouds of pale pink blossom before the leaves come out. In Japan, where the blossom is called sakura, people gather under the trees for picnics to enjoy it, a custom called hanami, which means "flower viewing."

Tap it and every blossom drops in a flurry of petals that settle on the grass, then the bare branches bloom again. Pick a Seed in the Toy tab for a differently shaped tree. The blossom only lasts a week or two, which is part of why people treasure it.

HOW TO PLAY

- **Tap:** All the petals fall in a flurry, then it blooms again.
- **Sound:** The opening of the traditional "Sakura Sakura" on a plucked koto-like string, over a faint rustle of petals (no wind).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Seed:** 1, 2, 3, 4

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/cherry-blossom/

CREDITS

- Built from a recipe in the browser; no outside files.



Maple tree

Tap it to send a gust whirling through. Pick the leaf color in the Toy tab.



Scan to play with it.

Learn more: [Maple on Wikipedia](#)

The maple is a tree known for its leaves with pointed lobes, which turn fiery red, orange and gold in autumn. Leaves are green in summer because of chlorophyll, the stuff that catches sunlight; in autumn the tree stops making it, so yellow and orange pigments show through, and many maples also make new red ones. Its seeds have wings and spin down like little helicopters.

Tap it and a whirling gust tears leaves off and carries them around the tree in a widening spiral before they settle in a ring on the grass; then fresh leaves open. Maple syrup is made by boiling down maple sap: about 40 liters of sap make 1 liter of syrup.

HOW TO PLAY

- **Tap:** Leaves swirl down in a spiral gust (different from the cherry's gentle fall).
- **Sound:** A soft gust and the leaves rustling as they swirl and land (no crunching clicks, much less wind).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Leaves:** Fiery red, Orange, Gold, Summer green
- **Seed:** 1, 2, 3, 4

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/maple/

CREDITS

- Built from a recipe in the browser; no outside files.



Bonsai

Tap it to grow a new branch, then trim it.



Scan to play with it.

Learn more: [Bonsai on Wikipedia](#)

A bonsai is a real tree kept small by careful growing in a shallow pot. Its keeper trims the branches and roots and shapes the trunk, sometimes for many years, so it looks like a tiny old tree from the wild. The art grew in China and was refined in Japan.

Tap it and a new branch grows out of the trunk with a pad of leaves on its end; then a pair of bonsai scissors slides in and snips it off, the cut piece drops onto the moss and is cleared away, and the stub heals over. Pick the pot's color in the Toy tab. Some bonsai trees are hundreds of years old.

HOW TO PLAY

- **Tap:** Grows a new branch, then the scissors snip it back.
- **Sound:** The branch creaks and its leaves rustle as it grows, the scissors open and snip, and the cut piece drops onto the moss; real CC0 recordings now (bonsai-snip.mp3), with the synthesized sound as a fallback.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Pot:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/bonsai/

CREDITS

- **Sound:** [scissors cut.wav](#) by mywhats, CC0 1.0.



Weeping willow

Tap it to send a breeze through the hanging branches.



Scan to play with it.

Learn more: [Salix babylonica on Wikipedia](#)

The weeping willow is a tree whose long, thin branches hang down almost to the ground like a curtain. It loves water and often grows on the banks of rivers and ponds. Its branches are so bendy that people have long woven them into baskets.

The strands sway gently by themselves. Tap it to send a breeze across from the left: the hanging strands swing away with it a curtain at a time, bending most at their tips, then swing back and settle while the crown leans a little. Pick a Seed in the Toy tab for a different tree.

HOW TO PLAY

- **Tap:** The long branches sway in a wave like a breeze passing.
- **Sound:** The same breath of wind, quieter, with the strands rustling.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Seed:** 1, 2, 3, 4

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/willow/

CREDITS

- Built from a recipe in the browser; no outside files.



Sunflower

Tap it to bring out the sun and watch the flower turn to face it.



Scan to play with it.

Learn more: [Common sunflower on Wikipedia](#)

A sunflower's big head is not one flower but many: the dark middle is packed with hundreds of tiny flowers that each become a seed, and the yellow petals around the edge belong to flowers of their own. The seeds grow in spirals that curve both ways.

The head nods gently at rest. Tap it to bring out the sun at the top left: the head turns up to face it and its petals spread wide open; then the sun goes in, the head turns back and the petals lift again. Young sunflowers really do follow the Sun, turning from east to west during the day.

HOW TO PLAY

- **Tap:** The head turns to face the sun and petals open wide.
- **Sound:** A warm low swell as the sun comes out (no bouncing tone) and the petals rustling open.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/sunflower/

CREDITS

- Built from a recipe in the browser; no outside files.



Rose

Tap it to open the bloom.
Pick its color in the Toy tab.



Scan to play with it.

Learn more: [Rose on Wikipedia](#)

The rose is one of the best-loved flowers in the world, grown in gardens for thousands of years for its soft, layered petals and its scent. There are thousands of kinds, in almost every color. The sharp points on its stems, often called thorns, help it climb and keep animals from eating it.

Tap it and the bloom opens further, every petal turning outward, and one outer petal comes loose and flutters down to the grass. Then the bloom closes again and a new petal fills the gap. After the flowers fade, many roses grow round red fruits called hips.

HOW TO PLAY

- **Tap:** Blooms open further, a petal drops.
- **Sound:** As before, plus the daisy's soft tap when the loose petal lands on the grass.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colour:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/rose/

CREDITS

- Built from a recipe in the browser; no outside files.

Dandelion

Tap it to blow the seeds away.



Scan to play with it.

Learn more:
[Taraxacum on Wikipedia](#)

A dandelion starts as a bright yellow flower. When it is done flowering, it turns into a round, fluffy seed head called a clock. Each seed hangs from its own tiny parachute of soft white hairs, so the wind can carry it far away to grow a new plant.

Tap it to blow: a gust bends the stalk, the seeds fly off and drift away, and then the seed head grows back. A single clock holds about 150 to 200 seeds. The name comes from the French for "lion's tooth," after the jagged edges of its leaves.

HOW TO PLAY

- **Tap:** Blow the seeds
- **Sound:** Puff of breath.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/dandelion/

CREDITS

- Built from a recipe in the browser; no outside files.



Tulips

Tap it to open the tulips to the sun. Pick their color in the Toy tab.



Scan to play with it.

Learn more: [Tulip on Wikipedia](#)

Tulips are spring flowers that grow from bulbs, each stem holding one cup-shaped flower. They first grew wild in the mountains of Central Asia. In the Netherlands in the 1600s, tulip bulbs became so prized that a single rare bulb could cost as much as a house.

Tap it and the three tulips open wide to the sun one after another, each petal turning out to show the dark stamens and the pale pistil inside, then they close up again. Real tulips open like this in warm sunshine and close again in the cool of the evening.

HOW TO PLAY

- **Tap:** Petals open wide to the sun, then close.
- **Sound:** Gentle pluck.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colour:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/tulip/

CREDITS

- Built from a recipe in the browser; no outside files.

Daisies

Tap it to spin the big daisy:
loves me, loves me not.



Scan to play with it.

Learn more: [Bellis perennis](#) on Wikipedia



A daisy looks like one flower, but its yellow middle is made of many tiny flowers, and each white petal around it is a flower too. The name comes from the old words "day's eye," because many daisies open in the morning and close at night.

Tap it and the big daisy spins like a pinwheel, twice around, flinging off eight petals one after another while the small ones bob; then new petals fill the gaps. It plays the old game of pulling off petals one by one: "loves me, loves me not," and the last petal gives the answer.

HOW TO PLAY

- **Tap:** Petals spin like a pinwheel (loves me, loves me not).
- **Sound:** Each petal's soft tap now lands with it: scheduled for the landing itself, so it stays in sync at any frame rate.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/daisy/

CREDITS

- Built from a recipe in the browser; no outside files.



Lotus

Tap it to lift the flower out of the water and open it.



Scan to play with it.

Learn more:
[Nelumbo nucifera](#)
on [Wikipedia](#)

The lotus is a water plant that roots in the mud at the bottom of ponds and holds its big round leaves and flowers above the water. Its leaves are covered in tiny waxy bumps, so water rolls off in beads and carries the dirt away with it. Lotus seeds can last a very long time: one more than 1,000 years old has sprouted.

Tap it and the flower folds into a bud, rises out of the water on its stalk as a ring of ripples spreads, and opens wide in the air, the outer petals first. Then it sinks back onto its leaf, still open.

HOW TO PLAY

- **Tap:** The flower rises and opens on the water with ripples.
- **Sound:** Water drop and a bell.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/lotus/

CREDITS

- Built from a recipe in the browser; no outside files.



Toadstool

Tap the big cap to puff out a cloud of spores.



Scan to play with it.

Learn more:
[Mushroom on Wikipedia](#)

These red toadstools with white spots are fly agarics, perhaps the most famous mushrooms of fairy tales. A mushroom is the part of a fungus that pops up above ground; most of the fungus is a web of fine threads hidden in the soil. Fly agarics are poisonous, so they are for looking at, not eating.

Tap it and the big cap dips and springs back, and a cloud of glowing spores puffs out from the gills underneath and drifts away. Spores are how a fungus spreads, like tiny seeds, and one mushroom can let out billions of them. Pick the cap's color in the Toy tab.

HOW TO PLAY

- **Tap:** Puffs a cloud of glowing spores.
- **Sound:** Soft puff.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Cap:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/mushroom/

CREDITS

- Built from a recipe in the browser; no outside files.

Fern

Tap it to unroll the fiddleheads into new fronds.



Scan to play with it.

Learn more: [Fern on Wikipedia](#)

Ferns are some of the oldest plants on Earth: they grew long before the dinosaurs, and they still grow in shady woods today. They have no flowers or seeds. Instead they spread by spores, tiny specks that grow in brown dots under their leaves, which are called fronds.

Each new frond starts as a tight coil called a fiddlehead, because it looks like the curled end of a violin. Tap it and the two fiddleheads in the middle unroll into new fronds, joint by joint from the base to the tip, then roll back up. Pick a Seed in the Toy tab for a different fern.

HOW TO PLAY

- **Tap:** The fronds unroll from fiddleheads.
- **Sound:** A soft leafy rustle and a gentle stem creak as each frond unrolls, and again as they roll back up (no sandy flutter).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Seed:** 1, 2, 3, 4

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/fern/

CREDITS

- Built from a recipe in the browser; no outside files.



Saguaro cactus

Tap for the spines, then a look inside. Set its arms and flowers in the Toy tab.



Scan to play with it.

Learn more: [Saguaro on Wikipedia](#)

The saguaro is a giant cactus of the Sonoran Desert in Arizona and Mexico. It grows very slowly but can reach about 12 meters tall and live for 150 years or more, and it may not grow its first arm until it is at least 50 years old. Its white flowers bloom on the tips in late spring.

Taps take turns. The first shoots all the spines out and draws them back; the next slides a wedge out of the trunk to show the inside: wet, pale green flesh around a ring of woody ribs. Its pleats spread like an accordion to store water after rain. Set the number of arms and the flowers in the Toy tab.

HOW TO PLAY

- **Tap:** The spines shoot out and back; a second tap shows a cut-away with the watery inside.
- **Sound:** Prickly zing.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

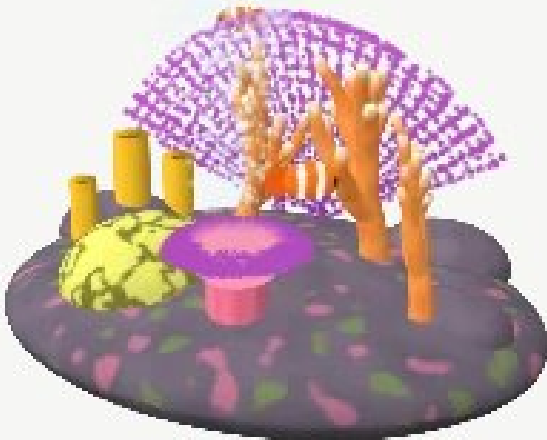
- **Arms:** slider
- **Flowers:** switch
- **Seed:** 1, 2, 3, 4

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/saguaro/

CREDITS

- Built from a recipe in the browser; no outside files.



Coral reef

Tap it to open the polyps and let the fish out.



Scan to play with it.

Learn more: [Coral reef on Wikipedia](#)

Corals may look like rocks or plants, but they are animals. This reef-building coral is a colony of tiny polyps with tentacles that build a calcium carbonate skeleton; other corals can be soft or solitary. Over thousands of years, corals build reefs. Reefs cover a tiny part of the ocean floor but are home to about a quarter of all the kinds of sea life.

Tap it and the polyps open all over this staghorn coral, like tiny tentacled stars, and six little fish dart out of the reef, hover, turn and dart back in; then the polyps close. Pick a Seed in the Toy tab for a different reef.

HOW TO PLAY

- **Tap:** Polyps open and little fish dart out and back.
- **Sound:** Bubbles.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Seed:** 1, 2, 3, 4

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/coral/

CREDITS

- Built from a recipe in the browser; no outside files.



Pinecone

Tap it to open the scales and drop them, then put it back together.



Scan to play with it.

Learn more: [Cone \(botany\) on Wikipedia](#)

A pinecone is the part of a pine tree that holds its seeds. Its woody scales are arranged in spirals, and under each one sit seeds with thin wings. In dry weather the scales open so the seeds can fall out and spin away on the wind; in wet weather they close up tight again.

Tap it and the cone opens as it does on a dry day, and winged seeds spin down like little propellers. Then the scales facing you break off one after another from the bottom up and tumble into a pile, showing the core. Then they fly back up into place and the cone closes.

HOW TO PLAY

- **Tap:** The scales open wide (as in dry weather) and seeds drift out.
- **Sound:** Woody creak.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/pinecone/

CREDITS

- Built from a recipe in the browser; no outside files.



Acorns

Tap it to pop the caps and sprout the acorns.



Scan to play with it.

Learn more: [Acorn](#) on Wikipedia

An acorn is the nut of an oak tree, with a little scaly cap on top. Inside is a seed that can grow into a whole new oak. Squirrels and jays bury acorns to eat later and forget some of them, and those can sprout, so the animals help plant new oak forests.

Tap it and the caps pop off, flip through the air and land upside down on the leaf, and a green sprout pokes out of each acorn and opens two tiny leaves. Then the sprouts draw back in and the caps hop home. Pick an autumn or a green leaf in the Toy tab.

HOW TO PLAY

- **Tap:** The caps pop off and a sprout pokes out.
- **Sound:** Tiny pop.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Leaf:** Autumn, Green

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/acorn/

CREDITS

- Built from a recipe in the browser; no outside files.



Succulent

Tap it to open the rosette and send up a flower.



Scan to play with it.

Learn more:

[Succulent plant on Wikipedia](#)

Succulents are plants that store water in fleshy leaves, stems or roots, so they can live through long dry spells. This one grows in a rosette: a tight circle of leaves, like the petals of a rose, with new leaves coming from the middle.

Tap it and the rosette opens, every leaf tipping outward, and a flower stalk rises from the middle and arches over with little coral-colored bells; then it draws back and the rosette closes. Pick the color of the leaf tips in the Toy tab. Many succulents grow new plants from a single fallen leaf.

HOW TO PLAY

- **Tap:** The rosette opens and a small flower stalk rises.
- **Sound:** Soft rising pluck.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Tips:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/succulent/

CREDITS

- Built from a recipe in the browser; no outside files.



Bamboo

Tap it to make the new shoots grow up tall.



Scan to play with it.

Learn more:
[Bamboo on Wikipedia](#)

Bamboo is a giant grass with tall, hollow stems split into sections by rings called nodes. It is one of the fastest-growing plants in the world: some kinds can grow almost a meter in a single day. People use its strong, light stems to build houses, scaffolding and furniture.

Three young shoots sit on the ground. Tap it and they shoot up a section at a time, each new section sliding up out of the one below with a hollow knock, and a tuft of leaves opens at the top; later they sink back down.

HOW TO PLAY

- **Tap:** Shoots grow tall very fast, segment by segment.
- **Sound:** Hollow wooden knocks.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/bamboo/

CREDITS

- Built from a recipe in the browser; no outside files.



Pebbles

Drag a pebble to pick it up and set it on another to stack them. Tap to tumble them.



Scan to play with it.

Learn more: [Pebble on Wikipedia](#)

Pebbles are small stones rounded smooth by water. As rivers and waves roll them over and over, knocking against sand and other stones, their sharp corners wear away, a process that can take hundreds or thousands of years. A cairn is a pile of stones stacked up by people, often to mark a trail.

Drag a pebble to lift it and set it on the sand or on another stone. A careful stack stands; one set on the edge topples. The 🖐️ button turns this off, and reset puts every stone back. Tap it and five pebbles hop one at a time onto a cairn, biggest at the bottom, then all hop back. Pick a pile or a cairn in the Toy tab.

HOW TO PLAY

- **Tap:** The pebbles drop and scatter, then stack into a cairn.
- **Sound:** Real stone-on-stone knocks as the pile shifts and tumbles home; the stacking knocks stay; real CC0 recordings now (rocks-pebbles.mp3), with the synthesized sound as a fallback.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

VIEWS

- **Style:** Pebble pile, Cairn

SETTINGS

- **Seed:** 1, 2, 3, 4

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/rocks/

CREDITS

- **Sound:** [Slow Pebble Tumble.wav](#) by bbrocer, CC0 1.0.



Kelp

Tap it to bring fish in to nibble the kelp.



Scan to play with it.

Learn more: [Kelp on Wikipedia](#)

Kelp is a giant seaweed, a kind of brown algae that grows in cool, shallow seas. It holds on to the rocks with rootlike grips, and little bulbs full of gas keep its long blades floating up toward the light. Kelp can grow so thick that it makes underwater forests, full of fish, snails, sea urchins and sea otters.

The kelp sways gently by itself. Tap it and three little fish swim in, each noses up to a blade for a few nibbles, and the kelp sways away from them, then swings back as they swim off. Giant kelp can grow more than half a meter in a day.

HOW TO PLAY

- **Tap:** Fish swim in and nibble; the kelp sways away.
- **Sound:** Underwater bubbles.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/kelp/

CREDITS

- Built from a recipe in the browser; no outside files.



Campfire

Tap it to stoke the fire. Set the fire's size in the Toy tab.



Scan to play with it.

Learn more:
[Campfire on Wikipedia](#)

A fire needs three things: fuel to burn, heat to start it, and oxygen from the air. Take any one away and it goes out. The yellow and orange glow of the flames comes from tiny bits of soot, heated until they shine.

This campfire has logs leaning together in a ring of stones, on a bed of glowing embers. Tap it to stoke the fire and the flames leap up with more sparks, then settle down again. Set how big the fire is in the Toy tab. A ring of stones helps keep a real campfire from spreading.

HOW TO PLAY

- **Tap:** Stoke the fire
- **Sound:** The volcano's kind of fire, a low flickering rush with a few soft pops of wood (no clicks).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Fire size:** slider

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/campfire/

CREDITS

- Built from a recipe in the browser; no outside files.

Storm cloud

Tap for thunder: lightning strikes under your finger. Try the Rain slider in the Toy tab.



Scan to play with it.

Learn more:

[Cumulonimbus cloud on Wikipedia](#)

A thunderstorm cloud can tower more than 10 kilometers into the sky. Inside it, bits of ice and water crash together and build up electric charge, until a giant spark, lightning, jumps across. Lightning heats the air around it to about 30,000 degrees Celsius, hotter than the surface of the Sun, and the air bursts outward with the boom we call thunder.

This cloud rains and flashes with lightning by itself. Tap it for a big strike and a clap of thunder: the bolt comes straight down from the cloud where you tapped. Light travels much faster than sound, so count the seconds between flash and thunder: every 3 seconds is about 1 kilometer away.

HOW TO PLAY

- **Tap:** Thunder
- **Sound:** Thunder rumble.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Rain:** slider

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/storm-cloud/

CREDITS

- Built from a recipe in the browser; no outside files.



Lava lamp

Tap to heat it up. Pick colors, blobs and flow in the Toy tab.



Scan to play with it.

Learn more: [Lava lamp on Wikipedia](#)

A lava lamp is a glass bottle of liquid with blobs of colored wax inside, and a lamp in its base. When the wax is cool it is a little denser than the liquid and rests at the bottom. The lamp warms it, the wax spreads out and gets less dense, and a blob rises; at the top it cools and sinks again.

Tap it to heat it up: the blobs move much faster, the wax glows and shifts color, and the liquid brightens, then it all eases back. In the Toy tab, pick a color set or your own colors, how many blobs, their size and shape, how fast they flow, and a glow from within.

HOW TO PLAY

- **Tap:** Blobs speed up, glow and shift colour for a few seconds.
- **Sound:** Only the gloops and bubbles (the rising hum is gone).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colour set:** Pick below, Ocean, Violet, Lime, Sunset, Midnight
- **Wax:** pick a color
- **Liquid:** pick a color
- **Base:** pick a color
- **Blobs:** slider
- **Blob size:** slider
- **Blob shape:** Mixed, Round, Tall
- **Flow:** slider
- **Glow:** slider

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/lava-lamp/

CREDITS

- Built from a recipe in the browser; no outside files.

Snow globe

Tap it to swirl the snow.
Hands-on: pick it up and shake it hard.



Scan to play with it.

Learn more: [Snow globe on Wikipedia](#)

A snow globe is a glass ball filled with water, with a little winter scene inside and white flakes that settle on the ground. Shake it and the flakes swirl up, then drift down slowly, because the water holds them back, like snow on a calm day.

This one holds a snowman with a hat, a scarf and a carrot nose, next to a snowy fir tree, and gentle snow falls all the time. Tap it to shake the globe: it wobbles, the snow swirls up and around, then settles. You can pick the color of the base in the Toy tab.

HOW TO PLAY

- **Tap:** Shake the globe
- **Sound:** The water sloshing as it is shaken, then the twinkle (no rattle of clicks).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

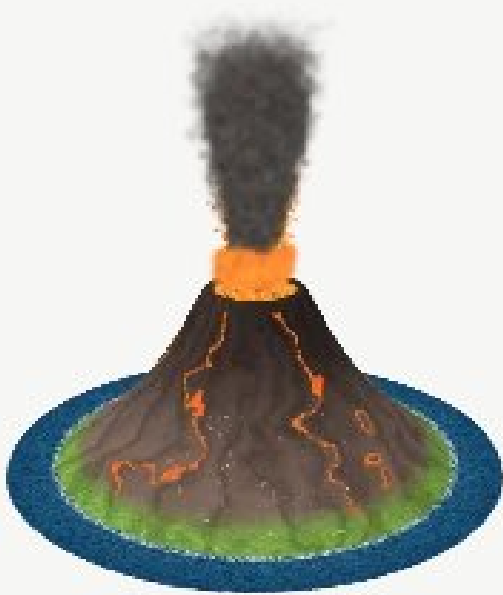
- **Base:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/snow-globe/

CREDITS

- Built from a recipe in the browser; no outside files.



Volcano

Tap it to make it erupt.



Scan to play with it.

Learn more:
[Volcano on Wikipedia](#)

A volcano is an opening in the Earth's crust where melted rock, called magma, comes up from deep below. Once it reaches the surface it is called lava. Gas trapped in the magma can make an eruption explosive, blasting out ash and rock.

Tap it to make it erupt. Most volcanoes sit where the great plates of the Earth's crust meet, and about 1,350 on land may still erupt; many more lie hidden under the sea.

HOW TO PLAY

- **Tap:** Erupt
- **Sound:** Deep rumble and boom.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/volcano/

CREDITS

- Built from a recipe in the browser; no outside files.



Ice swan

Tap to melt the swan and refreeze it. Try the Temperature slider in the Toy tab.



Scan to play with it.

Learn more: [Ice sculpture on Wikipedia](#)

An ice sculpture is carved from a big block of clear ice with saws, chisels and even small power tools. Clear ice is made by freezing water slowly, so the air bubbles can escape instead of turning it white. Swans are a favorite shape for parties, and whole towns of ice are built at winter festivals.

Tap it and the swan melts faster and faster: water drips into a spreading puddle, the beak wears away, the wing tips and the head crack off and fall in, and the body wears down to a lump. Then it freezes back, piece by piece, with a sweep of frost. The Temperature slider melts it partway and back.

HOW TO PLAY

- **Tap:** Specks drift; tap melts the swan into drips, then it refreezes.
- **Sound:** The drips and the refreezing; the wind at the start is gone.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Temperature:** slider

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/ice-statue/

CREDITS

- Built from a recipe in the browser; no outside files.



Candle

Tap to blow out the candle;
tap again to light it.



Scan to play with it.

Learn more: [Candle](#)
on [Wikipedia](#)

A candle is a stick of wax with a string, the wick, down the middle. The flame's heat melts the wax, the melted wax soaks up the wick, and near the flame it turns into a gas, which is what really burns. The blue part at the bottom burns with plenty of air; the yellow above is tiny bits of soot glowing hot.

This candle starts lit. Tap it to blow it out, and a thin trail of smoke rises from the wick; tap again to light it. A flame is shaped like a teardrop because its hot gases rise; in space, where nothing rises, a candle flame is a round, blue ball. You can pick the wax color in the Toy tab.

HOW TO PLAY

- **Tap:** Blow out or light
- **Sound:** Soft blow and a match strike.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Wax:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/candle/

CREDITS

- Built from a recipe in the browser; no outside files.

Tornado

Tap to spin it up. Set its power in the Toy tab.



Scan to play with it.

Learn more:
[Tornado on Wikipedia](#)

A tornado is a spinning column of air that reaches down from a thunderstorm to the ground. It is the most violent kind of storm on Earth: its winds can blow faster than 400 kilometers an hour, though most tornadoes are much weaker and last only a few minutes. The United States has more tornadoes than any other country, about 1,200 a year.

Tap it to spin it up: the funnel widens and whirls faster, dust boils up at its foot, and the planks, clods and leaves on the field are lifted one by one, spiral up around it, then fall back where they lay as it weakens. The Power slider in the Toy tab sets how hard it swirls.

HOW TO PLAY

- **Tap:** Spins up bigger and faster, pulling debris, then weakens.
- **Sound:** Rising roar of wind.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Power:** slider

In the Toy tab of the app.

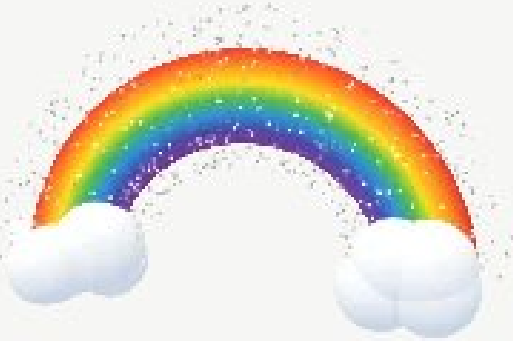
Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/tornado/

CREDITS

- Built from a recipe in the browser; no outside files.

Rainbow

Tap it to draw the rainbow again, color by color.



Scan to play with it.

Learn more:
[Rainbow on Wikipedia](#)

A rainbow appears when sunlight shines into falling raindrops. Each drop bends the light, splits it into its colors and bounces it back out, so you see a rainbow when the Sun is behind you and rain is in front. Its colors always go in the same order: red on the outside, then orange, yellow, green, blue and violet on the inside.

Tap it and the arc blinks away, then draws itself again color by color, red first, each band sweeping from one cloud to the other behind a bright little pen. Then sparkles burst from both clouds. Seen from a plane, a rainbow can be a whole circle.

HOW TO PLAY

- **Tap:** The arc draws itself colour by colour and sparkles at the ends.
- **Sound:** Rising harp glissando.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/rainbow/

CREDITS

- Built from a recipe in the browser; no outside files.



Iceberg

Tap it to break off a chunk into the sea.



Scan to play with it.

Learn more: [Iceberg on Wikipedia](#)

An iceberg is a huge piece of ice that has broken off a glacier or an ice shelf and floats out to sea. It is made of fresh water that fell as snow long ago. Ice is a little lighter than seawater, so an iceberg floats, but only about one tenth of it shows above the water.

When a chunk breaks off, it is called calving. Tap it and a chunk cracks off, showing fresh pale ice, and tips into the sea with a splash of spray and rings of ripples. The chunk bobs, and the iceberg, now lighter, bobs and rocks. Then the chunk melts away and the toy grows it back.

HOW TO PLAY

- **Tap:** A chunk calves off and splashes; the berg bobs.
- **Sound:** Real ice breaking, a deep crack with its groan and splinters (no crackle of clicks), then the splash; real CC0 recordings now (iceberg-crack.mp3), with the synthesized sound as a fallback.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/iceberg/

CREDITS

- **Sound:** [Ice Crack 1](#) by j_p_higgins, CC0 1.0.



Waterfall

Tap it to send a surge of water over the falls.



Scan to play with it.

Learn more:
[Waterfall on Wikipedia](#)

A waterfall forms where a river runs over a steep drop, often where hard rock sits on top of softer rock. The falling water wears away the softer rock below, so over thousands of years many waterfalls slowly move back upstream. The tallest on Earth, Angel Falls in Venezuela, drops nearly 1 kilometer.

Tap it to send a surge: a front of white water runs along the river and over the edge, a thicker sheet of white water pours down the falls, foam spreads over the pool and a cloud of mist billows up, then it all calms again.

HOW TO PLAY

- **Tap:** A surge: the flow doubles and mist billows.
- **Sound:** Rushing water swell.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/waterfall/

CREDITS

- Built from a recipe in the browser; no outside files.

Ocean wave

Tap it to break the wave; a new one curls up behind it.



Scan to play with it.

Learn more: [Wind wave on Wikipedia](#)

Most ocean waves are made by wind blowing across the water, sometimes far out at sea. The wave's energy travels on, but the water itself mostly just rises and falls. Near the shore, the shallow bottom slows the wave, so it grows taller and steeper until its top curls over and breaks.

Tap it to break it: the top of the wave throws forward and crashes into the water in front, white water and spray burst up and foam spreads out as the wave flattens. Then a new swell rises behind, steepens and curls over. Surfers call the tube under a curling wave the barrel.

HOW TO PLAY

- **Tap:** The wave curls and breaks into foam.
- **Sound:** Wave crash.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/ocean-wave/

CREDITS

- Built from a recipe in the browser; no outside files.

Geyser

Tap it to make it erupt.



Scan to play with it.

Learn more: [Geyser on Wikipedia](#)

A geyser is a hot spring that now and then shoots water and steam high into the air. Deep underground, hot rock heats water trapped in narrow cracks until it boils and bursts up out of the hole. The word comes from Geysir, a famous geyser in Iceland.

This geyser bubbles by itself, with a bigger eruption every so often. Tap it to make it erupt, shooting up a plume of water in a cloud of steam. The bright colors around its pool are mats of tiny living things that love hot water. More than half of the world's geysers are in Yellowstone National Park.

HOW TO PLAY

- **Tap:** Erupt
- **Sound:** Hiss and a whoosh of steam.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/geyser/

CREDITS

- Built from a recipe in the browser; no outside files.



Ice cream

Tap to melt it and refreeze it. Pick scoops and flavors in the Toy tab.



Scan to play with it.

Learn more: [Ice cream on Wikipedia](#)

Ice cream is a frozen mix of milk or cream, sugar and flavors. It is stirred while it freezes, so the ice crystals stay tiny and air is whipped in: up to half of a scoop of soft ice cream can be air, which is why it feels light and smooth.

Tap it and it melts: the scoops slump and drips run down the cone, then it freezes firm again. The Warmth slider in the Toy tab melts it slowly by hand. You can also pick one, two or three scoops and their flavors, and add sprinkles and a cherry.

HOW TO PLAY

- **Tap:** Tap melts it: drips run down the cone, then it refreezes.
- **Sound:** A slow, thick melt with heavy drops plopping down (no bubbles or drips).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Scoops:** slider
- **First scoop:** Strawberry, Vanilla, Chocolate, Mint choc chip, Pistachio, Blueberry, Lemon, Cookies & cream
- **Second scoop:** Strawberry, Vanilla, Chocolate, Mint choc chip, Pistachio, Blueberry, Lemon, Cookies & cream
- **Third scoop:** Strawberry, Vanilla, Chocolate, Mint choc chip, Pistachio, Blueberry, Lemon, Cookies & cream
- **Sprinkles:** switch
- **Cherry:** switch
- **Warmth:** slider

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/ice-cream/

CREDITS

- Built from a recipe in the browser; no outside files.



Watermelon

Tap it to chop it into slices that fan open.



Scan to play with it.

Learn more:
[Watermelon on Wikipedia](#)

A watermelon is a big fruit that grows on a vine along the ground. Under its hard green rind is sweet red flesh dotted with black seeds, and it is about 92 percent water, which makes it a favorite on hot days. Watermelons were first grown in Africa thousands of years ago.

Tap it and a big knife chops the whole melon five times, then the six slices fan open like an accordion, showing the red flesh, the pale rind and the seeds, and fold shut again. Pick a whole melon, a wedge, or both in the Toy tab.

HOW TO PLAY

- **Tap:** Cuts into slices that fan out.
- **Sound:** No whoosh at the start, only a soft settle; the chops and the fan are unchanged.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

VIEWS

- **Style:** Melon and wedge, Whole, Wedge

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/watermelon/

CREDITS

- Built from a recipe in the browser; no outside files.



Birthday cake

Tap to blow out the candles; tap again to light them. Choose how many in the Toy tab.



Scan to play with it.

Learn more: [Birthday cake on Wikipedia](#)

A birthday cake is a sweet, frosted cake with candles on top, often one for each year. People sing to the birthday person, who makes a wish and tries to blow out all the candles in one breath.

This cake starts with its candles lit. Tap to blow them out, with a puff of smoke from each wick, and tap again to light them. In the Toy tab, choose from one to nine candles, the colors of the frosting and the drip, and a vanilla, chocolate, red velvet or strawberry sponge. A candle flame needs air: blowing hard pushes the hot, burning gas away from the wick, and the flame goes out.

HOW TO PLAY

- **Tap:** Blow out
- **Sound:** Blown out with a breath, then the last line of "Happy Birthday to You" on the piano with its chords (no voices); relighting keeps the match.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Frosting:** pick a color
- **Drip:** pick a color
- **Sponge:** Vanilla, Chocolate, Red velvet, Strawberry
- **Candles:** slider

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/birthday-cake/

CREDITS

- Built from a recipe in the browser; no outside files.

Popcorn

Tap it to pop more kernels up out of the bucket.



Scan to play with it.

Learn more: [Popcorn on Wikipedia](#)

Popcorn is a kind of corn whose kernels puff up when they are heated. Each kernel has a hard shell with a little water inside. When it gets hot, the water turns to steam, the pressure builds until the shell bursts with a pop, and the soft starch inside puffs out into a white, crunchy foam.

Tap it and fresh kernels pop up out of the striped bucket, tumble and land on the heap. People in the Americas were popping corn thousands of years ago, long before movie theaters.

HOW TO PLAY

- **Tap:** Pop!
- **Sound:** A clean, single dry pop for each kernel as it jumps (14 of them, at the toy's own times, from ten different real pops with the pot's rumble and the oil's hiss cut out); no sizzle.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/popcorn/

CREDITS

- **Sound:** [Popcorn](#) by elricadavis, CC0 1.0.



Jelly

Drag it to stretch it; let go and it wobbles back. Tap it to poke it.



Scan to play with it.

Learn more:

[Gelatine dessert on Wikipedia](#)

A jelly, or gelatin dessert, is fruit juice or sweet flavored water set with gelatin and turned out of a mold. Gelatin makes a fine, stretchy mesh that traps the water, so the jelly holds its shape but wobbles when you touch it.

This one always jiggles a little. Drag any part of it and it stretches toward your finger; let go and it springs back and wobbles on its plate. Tap it to poke it: it squashes and wobbles hard, then settles down. In the Toy tab, pick strawberry, lime, orange, blueberry, grape or rainbow, and choose whether it has fruit set inside.

HOW TO PLAY

- **Tap:** Poke
- **Sound:** A wet, heavy wobble instead of a cartoon boing.
- **Drag:** stretches it; let go and it springs back.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Flavour:** Strawberry, Lime, Orange, Blueberry, Grape, Rainbow
- **Fruit inside:** switch

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/jelly/

CREDITS

- Built from a recipe in the browser; no outside files.



Pancakes

Tap to flip the top pancake.
Set the stack and the syrup
in the Toy tab.



Scan to play with it.

Learn more:
[Pancake on
Wikipedia](#)

Pancakes are flat, round cakes made from a runny batter of flour, eggs and milk, cooked on a hot pan. Baking powder in the batter makes bubbles of gas, so they puff up soft and fluffy. When bubbles pop on the top, it is time to flip.

Tap it to flip the top pancake: it hops up, turns a full somersault in the air with its syrup and butter, and lands back on the stack. In the Toy tab, set how many pancakes are in the stack, from two to seven, and try the Syrup slider.

HOW TO PLAY

- **Tap:** Flip the top one
- **Sound:** A smooth sizzle in the pan, the flip and the soft slap as it lands (no crackling ticks).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Pancakes:** slider
- **Syrup:** slider

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/pancakes/

CREDITS

- Built from a recipe in the browser; no outside files.



Cupcake

Tap it to flick the cherry up;
it plops back down.



Scan to play with it.

Learn more:
[Cupcake on Wikipedia](#)

A cupcake is a small cake baked in a paper cup, called a liner, and topped with swirls of frosting. Because it is small, it bakes quickly, and each person gets a whole cake of their own.

Tap it and the springy frosting flicks the cherry up: it tumbles and drops back with a plop, the frosting squashes and wobbles, and the sprinkles jump off and rain back down. In the Toy tab, pick the colors of the frosting and the liner, a vanilla, chocolate or red velvet cake, and the topping.

HOW TO PLAY

- **Tap:** A cherry drops on top and sprinkles rain down.
- **Sound:** Plop and sprinkle patter.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Frosting:** pick a color
- **Liner:** pick a color
- **Cake:** Vanilla, Chocolate, Red velvet
- **Topping:** Sprinkles and cherry, Sprinkles, Cherry, Plain

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/cupcake/

CREDITS

- Built from a recipe in the browser; no outside files.



Lollipop

Tap it to spin it fast; the swirl seems to pour inward.



Scan to play with it.

Learn more: [Lollipop on Wikipedia](#)

A lollipop is a hard candy on a stick. Swirl lollipops are made by twisting long ropes of soft, warm candy in different colors together and coiling them into a flat spiral before the candy cools and sets hard.

It turns slowly by itself. Tap it and the swirl whirls up to nearly three turns a second, so the spiral seems to pour inward toward the middle, then slows again. That is a trick of the eye: a turning spiral looks as if it is moving in or out. Pick its colors in the Toy tab.

HOW TO PLAY

- **Tap:** Spins fast and the swirl hypnotises.
- **Sound:** Only the wind as it spins (the rising "vroom" hum is gone).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colours:** Rainbow, Strawberry, Blueberry, Citrus

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/lollipop/

CREDITS

- Built from a recipe in the browser; no outside files.

Candy cane

Tap it to twist it until it snaps. With 🖐️ on, pull the hook until it snaps off.



Scan to play with it.

Learn more: [Candy cane on Wikipedia](#)

A candy cane is a stick of hard peppermint candy bent into a hook, with red and white stripes, and it is often hung on trees in the Christmas season. The stripes are made by twisting ropes of red and white candy together while they are warm and soft.

Tap it and each cane twists, the hook turning and the stripes winding tighter, until it snaps with a crack. The top half springs clear, sugar chips fly, then the halves come back together and mend with a glint. Pick a pair with a bow or a single cane, and the stripe color, in the Toy tab. With 🖐️ Hands-on on, pull the hook: the cane bends a little, then snaps in two with a crack, and you can pick up the broken piece; reset mends it.

HOW TO PLAY

- **Tap:** The stripes wind around like a barber pole, then it snaps in two and rejoins.
- **Sound:** The sugar strains as it twists (no whistle); the snaps and the mends stay.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

VIEWS

- **Style:** Pair with a bow, Single

SETTINGS

- **Stripes:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/candy-cane/

CREDITS

- Built from a recipe in the browser; no outside files.



Macarons

Drag a macaron to pick it up and stack it. Tap and the two in front hop up onto the stack.



Scan to play with it.

Learn more:
[Macaron on Wikipedia](#)

A macaron is a small French sweet: two light, round cookies made of ground almonds, egg whites and sugar, stuck together with a creamy filling. The tops are smooth and domed, and each cookie has a frilly edge at the bottom, called the foot, that rises as it bakes.

Drag any macaron to pick it up and set it down on the plate or on top of the stack; set one down crooked and it slides off. Tap it and the two macarons in front hop, one after the other, up onto the stack and land with a soft tap. The tower of five sways, then they hop back down. Pick a pastel mix or one flavor in the Toy tab.

HOW TO PLAY

- **Tap:** They stack themselves into a tower.
- **Sound:** Soft taps.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Flavours:** Pastel mix, Raspberry, Pistachio, Chocolate, Lemon

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/macarons/

CREDITS

- Built from a recipe in the browser; no outside files.



Donut

Tap it to break it apart and put it back together.



Scan to play with it.

Learn more: [Doughnut on Wikipedia](#)

A frosted donut with sprinkles, built by the computer from a simple ring shape. To a mathematician, a ring like this is a torus. In topology, the math of shapes that can stretch but not tear, a donut and a coffee mug count as the same shape, because each has exactly one hole.

Tap it and it snaps into chunks that fly apart and tumble, showing the dough inside, while the sprinkles spray off. Then it all flies back together. The Make tab builds shapes like this from a shape, a palette and a seed.

HOW TO PLAY

- **Tap:** Sprinkles jump off and rain back down.
- **Sound:** A soft, dry tear as it pulls apart (a real bread tear) and a soft pat as the halves meet; nothing wet.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/donut/

CREDITS

- **Sound:** [Squish impact](#) by Bertsz, CC0 1.0.
- **Sound:** [bread slice serrated knife cut and rip](#) by spanrucker, CC0 1.0.



Gummy bear

Drag the bear to stretch it; let go and it springs back. Tap to squish it.



Scan to play with it.

Learn more: [Gummy bear on Wikipedia](#)

A gummy bear is a small, chewy candy shaped like a bear, made of sugar, fruit flavors and gelatin. Gelatin is what makes it stretchy and springy: it forms a mesh that bends and then pulls back. Gummy bears were first made in Germany in the 1920s.

Drag any part of the bear and it stretches, up to about a body length, then springs back with a few wobbles when you let go. A drag beside it turns the view instead. Tap it for a jelly squish. Pick its flavor in the Toy tab, from cherry red to a clear pineapple.

HOW TO PLAY

- **Tap:** Drag to stretch (like the Magnet, with springy release); tap does a jelly jiggle.
- **Sound:** Only the bounce (the whistle is gone).
- **Drag:** stretches it; let go and it springs back.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Flavour:** Cherry red, Orange, Lemon, Apple green, Pineapple (clear), Raspberry pink

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/gummy-bear/

CREDITS

- Built from a recipe in the browser; no outside files.



Pretzel

Tap it to twist it like a knot and let it spring back.



Scan to play with it.

Learn more: [Pretzel on Wikipedia](#)

A pretzel is a baked bread shaped from a long rope of dough into a loop with a twist in the middle. Before baking, it is dipped in a special bath of water and an alkali, such as lye or baking soda, which gives it its shiny, dark brown crust. Then it is sprinkled with coarse salt.

Tap it and it twists like a knot, all in one piece: its two sides wring opposite ways and its loops fold a little toward you. Then it springs a little past its shape into the opposite twist and wobbles to a stop.

HOW TO PLAY

- **Tap:** Untwists into a rope and twists back.
- **Sound:** Soft dough squished and pulled, with a wet stretching sound.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/pretzel/

CREDITS

- Built from a recipe in the browser; no outside files.

Croissant

Tap it to slice it open and melt a pat of butter inside.



Scan to play with it.

Learn more: [Croissant on Wikipedia](#)

A croissant is a flaky, crescent-shaped pastry; its name is the French word for crescent. The dough is folded around a slab of butter again and again, making dozens of thin layers. In the hot oven, the water in the butter turns to steam and puffs the layers apart, so the inside is soft and full of holes and the outside is crisp.

Tap it and it is sliced open along its middle. The top lifts and tips back like a lid, showing the soft layers inside, where a pat of butter melts and spreads. Then the top settles back down.

HOW TO PLAY

- **Tap:** Puffs up in an oven glow (rising), then flakes fly off.
- **Sound:** The flaky layers shattering softly under the knife, and a smooth butter sizzle (no clicks).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/croissant/

CREDITS

- Built from a recipe in the browser; no outside files.



Pizza

Tap to take a slice; tap again to go back.



Scan to play with it.

Learn more: [Pizza on Wikipedia](#)

Pizza is a flat, round bread topped with tomato sauce and cheese and baked in a very hot oven. It comes from Naples, in Italy, and is now eaten all over the world. The margherita pizza, with red tomato, white mozzarella and green basil, shows the colors of the Italian flag.

Tap it to take a slice: it slides out with strings of melted cheese stretching behind it. Warm mozzarella stretches because the proteins in it line up in long, stringy strands. Tap again to put the slice back. Pick pepperoni, margherita, veggie or cheese in the Toy tab. With 🖐️ Hands-on on, drag the slice away yourself: the cheese strings sag, stretch and snap, and you can set the slice back.

HOW TO PLAY

- **Tap:** Take a slice
- **Sound:** The bite and a softer, shorter cheese stretch; not windy.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Topping:** Pepperoni, Margherita, Veggie, Cheese

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/pizza/

CREDITS

- Built from a recipe in the browser; no outside files.



Burger

Tap to spread out the layers; tap again to stack them up.



Scan to play with it.

Learn more:
[Hamburger on Wikipedia](#)

A burger is a round patty of ground meat, or of vegetables, cooked and served in a sliced bun with toppings. The name comes from the city of Hamburg, in Germany. This one has a sesame seed bun, a patty, a slice of cheese, a leaf of lettuce and tomato.

Tap it to spread out the layers in the air, one above the other, so every part shows, and tap again to stack them back up. Engineers call a picture like this an exploded view: it shows how the parts of a machine fit together without taking the real thing apart.

HOW TO PLAY

- **Tap:** Explode view
- **Sound:** The patty's smooth sizzle, no clicking.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/burger/

CREDITS

- Built from a recipe in the browser; no outside files.

Sushi

Tap a piece: the chopsticks pick that one up, dip it in soy sauce and put it back.



Scan to play with it.

Learn more: [Sushi on Wikipedia](#)

Sushi is a Japanese dish of rice mixed with a little vinegar and sugar, served with fish, vegetables or egg. This board has nigiri, a pillow of rice with a slice of fish draped on top, and maki, rice and a filling rolled up in a sheet of dried seaweed called nori and cut into rounds.

Tap a piece and the chopsticks lift off the board as if held by an invisible hand, pinch that piece with a click, carry it to a little dish of soy sauce and dip it twice, then set it back and lie down again.

HOW TO PLAY

- **Tap:** Chopsticks pick up a roll, dip it and put it back.
- **Sound:** Wooden chopstick clicks.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/sushi/

CREDITS

- Built from a recipe in the browser; no outside files.

Taco

Tap it to break the shell in half; then it closes up again.



Scan to play with it.

Learn more: [Taco on Wikipedia](#)

A taco is a Mexican dish: a tortilla, a thin, round flatbread of corn or wheat, folded around a filling such as meat, beans, cheese, lettuce and salsa. People in Mexico have made corn tortillas for thousands of years. This one has a crunchy, fried corn shell.

Tap it and the shell snaps across the middle with a crunch. The halves pull apart and swing open like a book, showing the filling in each break, and bits drop out of the break onto the plate, bounce a little and slide to a stop. Then they hop back in and the halves close.

HOW TO PLAY

- **Tap:** Shell cracks in half and the filling spills out, then it reassembles.
- **Sound:** A hard shell cracking open; the filling dropping out is unchanged; real CC0 recordings now (taco-crack.mp3), with the synthesized sound as a fallback.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/taco/

CREDITS

- **Sound:** [chips crunch sound](#) by TomatoHater, CC0 1.0.

Boiled egg

Tap to crack it open; tap again to go back.



Scan to play with it.

Learn more: [Boiled egg on Wikipedia](#)

A soft-boiled egg is cooked in its shell in boiling water for only a few minutes, so the white sets firm but the yolk stays runny. It is served in an egg cup, with strips of toast for dipping into the yolk.

Tap it to crack it open: the top of the shell pops off, and the runny yolk shows and drips down the side. Tap again to put the top back. In the Toy tab, pick a brown or a white shell and the color of the egg cup. The color of a shell depends on the kind of hen that laid it; the egg inside is the same.

HOW TO PLAY

- **Tap:** Crack
- **Sound:** Shell tap.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Shell:** Brown, White
- **Egg cup:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/egg/

CREDITS

- Built from a recipe in the browser; no outside files.

Coffee

Tap it to stir. Pick the latte art and the steam in the Toy tab.



Scan to play with it.

Learn more: [Coffee on Wikipedia](#)

Coffee is made from the roasted seeds, called beans, of the coffee plant, which first grew wild in Ethiopia. A latte is a strong shot of coffee topped with a lot of steamed milk. Pouring the milk foam carefully draws a picture on top, called latte art: a heart, a leafy rosetta or a tulip.

Tap it to stir: the latte art twists into a swirl, the middle turning further than the edge, then relaxes back, with a thick curl of steam. In the Toy tab, pick the latte art and the color of the cup, and try the Steam slider.

HOW TO PLAY

- **Tap:** Latte art swirl spins; steam curls up.
- **Sound:** A simple "tap tap" of a spoon on the cup.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Latte art:** Heart, Rosetta, Tulip
- **Cup:** pick a color
- **Steam:** slider

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/coffee/

CREDITS

- Built from a recipe in the browser; no outside files.



Apple

Tap to take a bite; tap again and a worm peeks out before it's whole again.



Scan to play with it.

Learn more: [Apple on Wikipedia](#)

Apples grow on trees and come in thousands of kinds, from sharp green ones to sweet red and golden ones. The wild apple that most of them come from still grows in the mountains of Central Asia.

Tap it to take a bite: a chunk comes away, leaving a scalloped bite of pale flesh, and the apple stays bitten. Tap again and a little worm pokes out of the bite, looks about, ducks back in, and the apple grows whole again. Pick a red, green or golden apple in the Toy tab. The flesh of a cut apple turns brown in the air, much as a sliced potato does.

HOW TO PLAY

- **Tap:** A bite appears (crunch); a second tap, a worm peeks out.
- **Sound:** A crisp bite into an apple (a crunch of tiny fractures and a juicy snap), not a zipper; real CC0 recordings now (apple-bite.mp3), with the synthesized sound as a fallback.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Apple:** Red, Green, Golden

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/apple/

CREDITS

- **Sound:** [Bite \(Apple\)](#) by wadaltmon, CC0 1.0.



Bananas

Tap it to pull the bananas apart and peel all three.



Scan to play with it.

Learn more: [Banana on Wikipedia](#)

Bananas grow in big hanging bunches on giant plants that look like trees but are not: the trunk is made of tightly rolled leaves. A banana is picked green and turns yellow as it ripens, and then brown spots appear as its starch turns to sugar and it gets sweeter.

Tap it and the three bananas pull apart. Each is peeled from its tip: its skin splits into three strips that curl back, showing the pale fruit. Then the strips fold back up and the bunch comes together. Pick green, just right or spotty bananas in the Toy tab.

HOW TO PLAY

- **Tap:** One banana peels open.
- **Sound:** A soft rustle as the bunch comes apart, and a quiet real banana peel as each one peels (no zipper).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Ripeness:** Green, Just right, Spotty

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/banana/

CREDITS

- **Sound:** [Banana Peel](#) by spanrucker, CC0 1.0.



Orange

Tap it to open it into wedges. With 🖐️ on, pull a wedge out; bring it back and it clicks in.



Scan to play with it.

Learn more: [Orange \(fruit\) on Wikipedia](#)

An orange is a citrus fruit. Its bright peel is dotted with tiny pockets of fragrant oil, and inside, the fruit is divided into segments packed with little sacs of juice. Oranges are full of vitamin C, and they were first grown in southern China and Southeast Asia.

Tap it and the whole orange opens like a flower: eight wedges fall open outward, one just after another, showing their juicy faces, while juice squirts up out of the middle. Then it closes up again. Pick a whole orange, a half, or both in the Toy tab. With 🖐️ Hands-on on, pull the wedges out one by one and set them down; bring a wedge back over its place and it clicks home.

HOW TO PLAY

- **Tap:** Splits into segments that fan out.
- **Sound:** Juicy squirt.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

VIEWS

- **Style:** Whole and half, Whole, Half

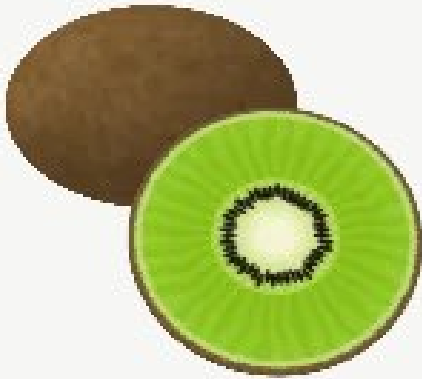
Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/orange/

CREDITS

- Built from a recipe in the browser; no outside files.

Kiwi

Tap it to cut it open and show the green inside.



Scan to play with it.

Learn more:
[Kiwifruit on Wikipedia](#)

A kiwifruit is a small fruit with thin, fuzzy brown skin and bright green flesh around a pale core ringed with tiny black seeds. It first grew wild in China. Farmers in New Zealand made it popular and named it after the kiwi, their fuzzy brown national bird.

Tap it and the whole kiwi is cut across the middle; the halves slide apart and turn to show their green faces and the ring of seeds, then close up again. Pick a whole kiwi, a half, or both in the Toy tab.

HOW TO PLAY

- **Tap:** Halves open to show the green inside and seeds.
- **Sound:** Soft slice.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

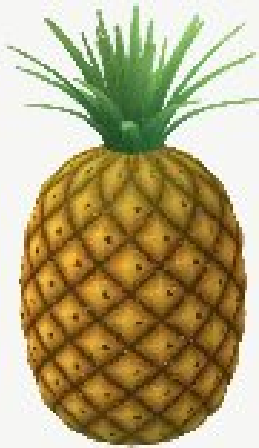
VIEWS

- **Style:** Whole and half, Whole, Half

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/kiwi/

CREDITS

- Built from a recipe in the browser; no outside files.



Pineapple

Tap it to slice it into rings.



Scan to play with it.

Learn more:
[Pineapple on Wikipedia](#)

A pineapple is a tropical fruit with a tough, scaly skin and a spiky crown of leaves. It is really many small fruits grown together: each scale on the skin comes from a single flower. Pineapples first grew in South America, and a crown twisted off and planted can grow a whole new plant.

Tap it for four chops, top first. With each chop, everything above lifts and leans toward you, until five rings stand apart in a leaning stack, showing their golden flesh and pale cores. Then they drop back onto each other.

HOW TO PLAY

- **Tap:** Cut into rings that stack.
- **Sound:** Chop.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/pineapple/

CREDITS

- Built from a recipe in the browser; no outside files.



Cherries

Pull a cherry out on its stem and let go; it swings back and knocks the other. Tap to swing them.



Scan to play with it.

Learn more: [Cherry on Wikipedia](#)

Cherries are small, round fruits that grow on trees, often in pairs whose long stems are joined at the top. Each cherry has one hard stone in the middle with the seed inside, so it is called a stone fruit, like a plum or a peach.

Pull a cherry out on its stem and let go: it swings back and knocks the other one, which swings out in turn, until they settle. Tap it and they swing like a small Newton's cradle, at their real speed: one cherry is flicked out, swings back and knocks the other out, which swings back and knocks the first, with a plink each time, until the swing dies away. A cherry on a stem about 4.5 cm long swings to and fro in under half a second.

HOW TO PLAY

- **Tap:** The pair swings and bounces on its stems.
- **Sound:** No string note, only the stems' soft swish; the knocking is unchanged.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/cherries/

CREDITS

- Built from a recipe in the browser; no outside files.



Grapes

Tap it: grapes drop off the bunch, bounce and hop back.



Scan to play with it.

Learn more: [Grape on Wikipedia](#)

Grapes grow in bunches on woody vines, from a few dozen to a few hundred on a bunch. They come in purple, green and red, and people eat them fresh, dry them into raisins and press them for juice.

Tap it and ten grapes on the front come off one after another, drop and bounce on the table and roll a little, each on its own path, then hop back up to their places one by one. Pick purple, green or red grapes in the Toy tab. The Photoreal shelf has a real grape that peels.

HOW TO PLAY

- **Tap:** Grapes drop off the stem and bounce, then return.
- **Sound:** Plop-plop cascade.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

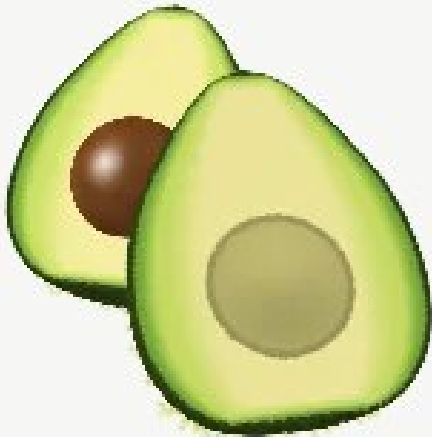
- **Grapes:** Purple, Green, Red

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/grapes/

CREDITS

- Built from a recipe in the browser; no outside files.



Avocado

Tap it to pop the stone into the other half and back.



Scan to play with it.

Learn more:
[Avocado on Wikipedia](#)

An avocado is a fruit with a bumpy green skin, soft, creamy green flesh and one big, round stone in the middle, which is its seed. It first grew in Mexico and Central America. Avocados are unusual: they do not ripen on the tree, only after they are picked.

Tap it and the stone pops out of its half, flies over and drops into the empty half with a thock, rocking it, then pops back home and the first half rocks. Pick both halves, or just the half with the stone, in the Toy tab.

HOW TO PLAY

- **Tap:** Splits open and the stone pops out and back.
- **Sound:** Thock.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

VIEWS

- **Style:** Both halves, Half with the stone

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/avocado/

CREDITS

- Built from a recipe in the browser; no outside files.

Building bricks

Drag a brick onto another to stack it; it clicks onto the studs. Tap to build a model.



Scan to play with it.

Learn more: [Toy block on Wikipedia](#)

Building bricks are small plastic blocks with round studs on top that press into tubes underneath the next brick, so they hold together firmly and can be pulled apart again. With enough of them you can build almost anything, and even six ordinary bricks can be stacked together in hundreds of millions of different ways.

Here eighteen bricks lie spread out on the table. Drag one to pick it up and hold it over another: it turns square and lines up with the studs, and clicks on when you let go, so you can build your own. Each tap pops the last model apart and builds a new one in the middle, brick by brick from the bottom up, each clicking into place: a tower, a bridge, stairs, a dog or a tree. Pick a set of colors in the Toy tab.

HOW TO PLAY

- **Tap:** Bricks rearrange into a random small build (tower, bridge, animal), different each tap.
- **Sound:** A lower plastic clack, for the tap and each brick landing.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colours:** Classic, Pastel, Rainbow, Ocean

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/bricks/

CREDITS

- Built from a recipe in the browser; no outside files.



Rubber duck

Tap it to squeeze it: it squeaks, hops and bobs.



Scan to play with it.

Learn more: [Rubber duck on Wikipedia](#)

A rubber duck is a little yellow bath toy that floats. The first ones were made of solid rubber; today most are hollow and made of soft plastic, with a small hole underneath. Squeeze one and the air rushes out through the hole with a squeak, and when you let go it sucks air back in.

Tap it to squeeze it: it squashes, squeaks and hops, then bobs and rocks as if it were floating in the bath, and settles down. Pick its color in the Toy tab.

HOW TO PLAY

- **Tap:** Keep the effect.
- **Sound:** Rubber duck quack.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colour:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/rubber-duck/

CREDITS

- Built from a recipe in the browser; no outside files.

Spinning top

Tap it to spin it faster; it wobbles more, then steadies.



Scan to play with it.

Learn more: [Spinning top on Wikipedia](#)

A spinning top is one of the oldest toys in the world, played with in many lands for thousands of years. While it spins fast, it balances on its tiny point, because a spinning thing holds the direction of its spin. As it slows, it starts to wobble, and its handle traces a slow circle; scientists call that circling precession.

This top is always spinning. Tap it to spin it much faster: it tilts and circles more widely for a few seconds, then settles back to a steady spin. Pick one of three sets of colors in the Toy tab.

HOW TO PLAY

- **Tap:** Spin it
- **Sound:** A soft flick as it starts; its spin is a quiet real hum that follows its speed (louder and a little higher when fast, gone as it slows), also when a drag spins it.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colours:** Circus, Ocean, Candy

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/spinning-top/

CREDITS

- **Sound:** [Metallic Spinning Top \(Dry Sound\).wav](#) by Uzbazur, CC0 1.0.



Dice

Tap to roll. Pick two six-sided dice or a d20 in the Toy tab.



Scan to play with it.

Learn more: [Dice on Wikipedia](#)

Dice are small shapes with a number on each face, rolled to get a number by chance. A fair die lands on each face just as often as any other. On an ordinary six-sided die the opposite faces always add up to seven: 1 and 6, 2 and 5, 3 and 4.

Tap to roll: the dice tumble and land on new faces, so you can use them for a real game. In the Toy tab, pick two six-sided dice or a d20, a die with 20 triangle faces used in many tabletop games, and pick their color.

HOW TO PLAY

- **Tap:** Roll
- **Sound:** Two dice thrown on a wooden table, bouncing with quicker, quieter knocks and settling; real CC0 recordings now (dice-throw.mp3), with the synthesized sound as a fallback.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

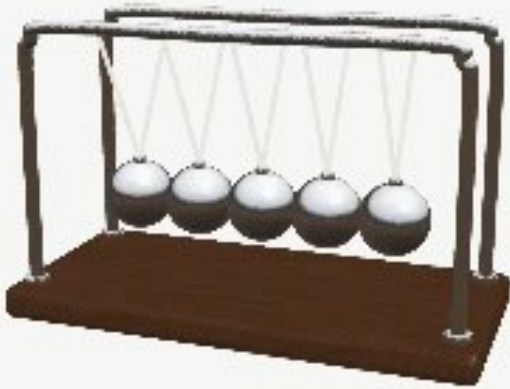
- **Dice:** Two d6, d20
- **Colour:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/dice/

CREDITS

- **Sound:** [Casino Audio 1.1: dice-throw-1.ogg and dice-throw-3.ogg](#) by Kenney (kenney.nl), CC0 1.0.



Newton's cradle

Drag a ball out to the side and let go. Or tap to lift the end ball.



Scan to play with it.

Learn more: [Newton's cradle on Wikipedia](#)

Newton's cradle is a row of steel balls, each hung on two strings so it can only swing in one line. Lift the end ball and let go: it strikes the row, the ball at the far end flies out, and the balls in between barely move. Let go of two and two fly out.

It shows two rules of physics at once: in each knock both the momentum and the energy carry through the row, so the same number of balls leaves as arrived. Each clack turns a little energy into sound and heat, so the swings slowly die away. Drag any ball out (the balls beside it come too) and let go, or tap to lift the end ball.

HOW TO PLAY

- **Tap:** Tap lifts and drops an end ball hard so the clacks are obvious; later, drag a ball back and let go.
- **Sound:** Crisp steel clacks.
- **Drag:** plays with the toy itself; a drag beside it turns the view.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/newtons-cradle/

CREDITS

- Built from a recipe in the browser; no outside files.



Teddy bear

Tap it and it waves hello.



Scan to play with it.

Learn more: [Teddy bear on Wikipedia](#)

A teddy bear is a soft, stuffed toy bear. It is named after Theodore "Teddy" Roosevelt, a president of the United States: in 1902 a newspaper cartoon showed him sparing a young bear on a hunting trip, and soon toymakers in the United States and Germany were making stuffed bears.

This bear sways its head and arms a little. Tap it and it lifts one arm and waves hello, tipping its head, then puts its arm back down. Pick the color of its fur in the Toy tab.

HOW TO PLAY

- **Tap:** Wave hello
- **Sound:** The soft rustle of a plush bear's fabric and stuffing; no creak.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Fur:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/teddy-bear/

CREDITS

- **Sound:** [08. Relleno de oso.wav](#) by lemigoga, CC0 1.0.



Yo-yo

Tap it to throw it down; it spins down the string and climbs back up.



Scan to play with it.

Learn more: [Yo-yo on Wikipedia](#)

A yo-yo is two disks joined by a short axle, with a string tied around the axle. Thrown down, it unwinds and spins fast; the spin keeps it steady, and a little tug on the string makes it wind itself back up into your hand. Children in ancient Greece played with yo-yos of wood, metal and clay.

This one bobs gently on its string. Tap it to throw it: it drops, spinning as the string unwinds, and then climbs back up the string to where it started. Pick its color in the Toy tab. With 🖐 Hands-on on, pull the yo-yo down and let go: it drops to the end of the string, spins there and climbs back.

HOW TO PLAY

- **Tap:** Throw
- **Sound:** The string unwinding, the whirr at the bottom and the smack back into the hand.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colour:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/yo-yo/

CREDITS

- Built from a recipe in the browser; no outside files.



Puzzle cube

Drag across a face to turn a row or column. Tap to scramble or solve it; tap again to pause.



Scan to play with it.

Learn more: [Combination puzzle on Wikipedia](#)

A twisting puzzle cube: 26 small cubes around a hidden core, with one color on each of its six faces. Each turn moves a whole row or column of nine cubes, and the puzzle is to bring every face back to one color.

Here every little cube is its own piece, and the cube keeps track of each turn, so you can really solve it: drag across a face to turn that row or column. A tap scrambles a solved cube, or turns a scrambled one back to solved, one layer at a time. Tap while it turns and it pauses once the turn in progress lands, so you can take over by hand: your own move ends the automatic one, and the next tap starts again from there. Solving it by hand earns a hop and a chime. The cube has about 43 quintillion arrangements, yet any of them can be solved in 20 moves or fewer when a half-turn counts as one move.

HOW TO PLAY

- **Tap:** Tap scrambles with several random layer turns.
- **Sound:** Each turn a plastic slide over the ridges and a soft clack as the layer seats; real CC0 recordings now (puzzle-cube-turn.mp3), with the synthesized sound as a fallback.
- **Drag:** plays with the toy itself; a drag beside it turns the view.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

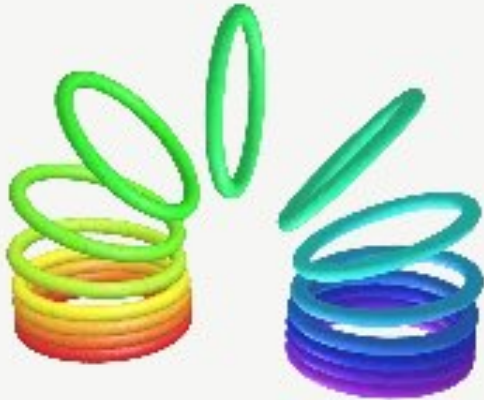
Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/puzzle-cube/

CREDITS

- **Sound:** [Cube Turn - 9 \(a puzzle cube turning\)](#) by SpaceJoe, CC0 1.0.

Spring toy

Pull its top coils up or sideways; let go and it springs back. Tap to hurry it along.



Scan to play with it.

A spring toy is a long, loose coil of metal or plastic that can walk down stairs by itself. Set one end on a lower step and it flips over, end over end, as each coil tips across in turn and the weight moves from one end to the other. It was invented in the 1940s by an engineer who saw a spring fall off a shelf and keep moving.

This one walks by itself, its coils flipping over from one side to the other. Pull its top coils up or to the side and it gathers into a stack and stretches out, coil by coil; let go and it springs back down and wobbles, then walks on. Tap it to hurry it along. Pick a rainbow, metal or pastel coil in the Toy tab.

HOW TO PLAY

- **Tap:** Make it walk
- **Sound:** A real metal Slinky's shimmering coils as it walks.
- **Drag:** plays with the toy itself; a drag beside it turns the view.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

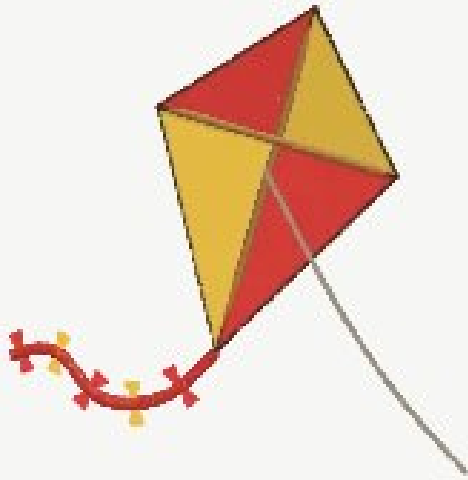
VIEWS

- **Style:** Rainbow, Metal, Pastel

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/spring-toy/

CREDITS

- **Sound:** [slinky Copy.wav](#) by foxraid, CC0 1.0.



Kite

Tap it for a gust of wind: the kite loops and the tail whips.



Scan to play with it.

Learn more: [Kite on Wikipedia](#)

A kite is a light frame covered with paper or cloth that flies on the wind at the end of a long line. The wind pushing under the tilted kite lifts it up, and a tail helps keep it pointing the right way. Kites were first flown in China more than 2,000 years ago.

Tap it for a big gust: the kite climbs around a loop, turning once while it rises, and its tail whips behind it. The line stays tied on the whole time. Pick its two colors in the Toy tab. With 🖐 Hands-on on, drag the kite or its line: the kite swoops back up into the wind, its tail flapping.

HOW TO PLAY

- **Tap:** A big gust: the kite loops the loop and the tail whips.
- **Sound:** The same, with the wind dialed back.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colour 1:** pick a color
- **Colour 2:** pick a color

In the Toy tab of the app.

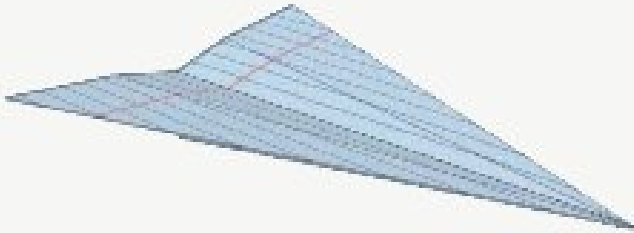
Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/kite/

CREDITS

- Built from a recipe in the browser; no outside files.

Paper plane

Tap it to do a barrel roll.



Scan to play with it.

Learn more: [Paper plane on Wikipedia](#)

A paper plane is a sheet of paper folded into a glider. It has no engine: once thrown, it slowly trades height for speed, and air flowing over its wings holds it up, while the weight at the pointed nose keeps it flying straight. Small folds at the back of the wings can make it climb, dive or turn.

This plane glides and bobs gently in place. Tap it and it does a barrel roll, one full turn around its long middle, while it lifts a little. In the Toy tab, pick the color of the paper, and choose lined paper or plain.

HOW TO PLAY

- **Tap:** Barrel roll
- **Sound:** The same fold and glide, with the air lower.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Paper:** pick a color
- **Lined paper:** switch

In the Toy tab of the app.

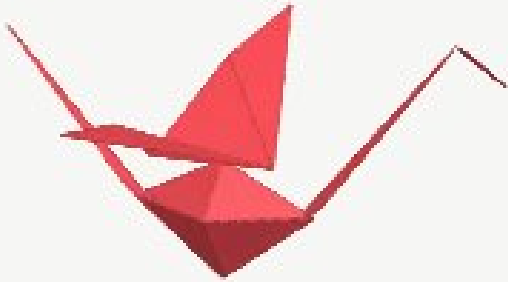
Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/paper-plane/

CREDITS

- Built from a recipe in the browser; no outside files.

Origami crane

Tap it to flap its wings.



Scan to play with it.

Learn more: [Orizuru](#)
on [Wikipedia](#)

Origami is the Japanese art of folding a single square of paper into a shape, without cutting or gluing. The crane is its most famous model, folded with a long neck, a pointed tail and wide wings. In Japan, a string of a thousand folded cranes is a traditional wish for good health and for peace.

Tap it to flap its wings: they beat up and down a few times, as a real crane's do. Some paper cranes are folded so that the wings really flap when you pull the tail. Pick the color of the paper in the Toy tab.

HOW TO PLAY

- **Tap:** Flap the wings
- **Sound:** Paper flutter.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

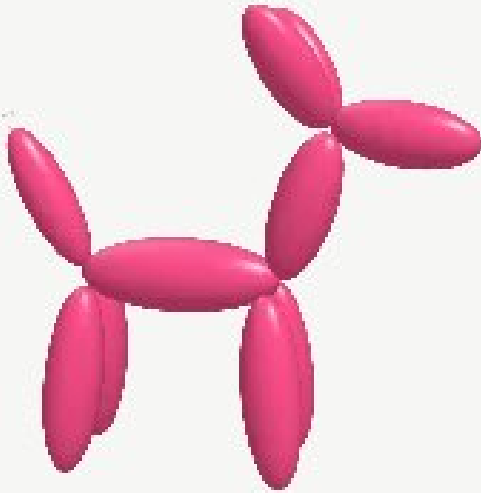
- **Paper:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/origami-crane/

CREDITS

- Built from a recipe in the browser; no outside files.



Balloon dog

Tap it to pop it; then it blows back up.



Scan to play with it.

Learn more: [Balloon modelling on Wikipedia](#)

A balloon dog is made from one long, thin balloon, twisted into a chain of bubbles and folded into a head, ears, a body, four legs and a tail. Each twist traps the air in its own bubble, so the shape holds.

Tap it and it pops: bits of rubber fly out and fall away. Then the dog blows back up, part by part, until it stands whole again. Pick the color of the balloon in the Toy tab.

HOW TO PLAY

- **Tap:** Pop
- **Sound:** A real balloon bursting (no whistle first); real CC0 recordings now (balloon-dog-pop.mp3), with the synthesized sound as a fallback.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Balloon:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/balloon-dog/

CREDITS

- **Sound:** [Balloon-Burst-07.wav](#) by Gniffelbaf, CC0 1.0.

Soap bubbles

Tap it to blow a stream of bubbles.



Scan to play with it.

Learn more: [Soap bubble on Wikipedia](#)

A soap bubble is a very thin skin of soapy water wrapped around some air. The skin pulls itself as small as it can, and the smallest skin that can hold the air is a sphere, so bubbles are round. The swirling colors come from light bouncing off the inside and outside of the skin, which is thinner than a hair.

Bubbles drift up slowly from the wand by themselves. Tap it to blow, and many more stream out faster, float away and fade. Pick the color of the wand in the Toy tab.

HOW TO PLAY

- **Tap:** Blow bubbles
- **Sound:** A soft breath through the wand, and a soft real pop as each bubble bursts while it's blown.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Wand:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/soap-bubbles/

CREDITS

- **Sound:** [Blow air short](#) by yehdawgo, CC0 1.0.
- **Sound:** [multiple bubbles bursting](#) by florianreichelt, CC0 1.0.



Wind-up robot

Tap it to wind it up: its key whirs and its arms swing wide.



Scan to play with it.

Learn more: [Wind-up toy on Wikipedia](#)

A wind-up robot is a tin toy with a spring inside. Turning the key on its back winds the spring tight, and as the spring slowly unwinds, it turns little gears that swing the arms and move the legs. Wind-up tin robots were a favorite toy in the middle of the 1900s.

This robot always swings its arms a little, and its antenna light glows. Tap it to wind it up: the key spins fast, the arms swing wide and it rocks and bobs from side to side, then it slowly runs down again. Pick its color in the Toy tab.

HOW TO PLAY

- **Tap:** Wind it up
- **Sound:** A few clicks of the winding key, then clockwork whirring as it unwinds and its tin feet clanking along; real CC0 recordings now (robot-wind.mp3), with the synthesized sound as a fallback.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colour:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/robot/

CREDITS

- **Sound:** [Wind-up sound](#) by Breviceps, CC0 1.0.



Chess set

Tap a piece, then a square, to move it. Or press play in the bar to watch a game.



Scan to play with it.

Learn more: [Chess on Wikipedia](#)

Chess is a game for two players on a board of 64 squares. Each side starts with 16 pieces (a king, a queen, two rooks, two bishops, two knights and eight pawns), and the goal is to checkmate the other king: to attack it so that it has no way out.

The board is set up for the Opera Game, played in Paris in 1858 at the opera house, where Paul Morphy beat the Duke of Brunswick and Count Isouard in 17 moves. Press play in the bar under the board to watch it, or tap a piece and then a square to play your own moves. In the Toy tab you can open or paste any game written in PGN, the usual way of writing down chess games.

HOW TO PLAY

- **Tap:** Plays a famous public-domain game move by move; later, tap a piece then a square to move it.
- **Sound:** Pieces set down on a felted wooden board, a woody thock (a sharper one for captures), like a good online-chess theme; real CC0 recordings now (chess-set-move.mp3), with the synthesized sound as a fallback.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Restart:** button
- **Move a piece:** button
- **Move speed:** slider

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/chess-set/

CREDITS

- **3D model:** [Chess Set](#) by Riley Queen, CC0 1.0.
- **Sound:** [Piece Placement.mp3](#) by el_boss, CC0 1.0.



Treasure chest

Tap to open or close the lid. With 🖐️ on, lift the lid by hand: let go high and it stays open.



Scan to play with it.

Learn more: [Chest \(furniture\) on Wikipedia](#)

A treasure chest is a strong wooden box with metal bands, a heavy lid and a lock, made to keep coins, jewels and other precious things safe. Long ago, before banks were common, people kept their valuables in chests like this, and ships carried money and goods in them.

Tap it to open the lid and see the heap of gold coins and bright jewels inside, and tap again to close it. Pick the color of the wood in the Toy tab. With 🖐️ Hands-on on, lift the lid on its hinge yourself: let go past upright and it stays open; lower, it drops shut with a thud.

HOW TO PLAY

- **Tap:** Open or close
- **Sound:** Creak and a treasure chime.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Wood:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/chest/

CREDITS

- Built from a recipe in the browser; no outside files.

Storybook

Tap to close the book; tap again to open it.



Scan to play with it.

Learn more: [Picture book on Wikipedia](#)

A storybook is a book of stories, with pages of words and often pictures. A book like this, with pages sewn together along one edge between two hard covers, is a way of keeping writing that people have used for about two thousand years. Printing with movable metal type, which made books much cheaper, began in Europe in the 1450s.

The book starts open. Tap to close it: the pages turn over one by one and the cover shuts. Tap again to open it. Pick the color of the cover in the Toy tab.

HOW TO PLAY

- **Tap:** Open or close
- **Sound:** Real paper pages turning (no wind); real CC0 recordings now (book-page.mp3), with the synthesized sound as a fallback.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Cover:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/book/

CREDITS

- **Sound:** [Book Turn Page 2.wav](#) by esperri, CC0 1.0.



Laptop

Tap the keys, or type on your keyboard. Drag on the trackpad. Tap elsewhere to close it.



Scan to play with it.

Learn more: [Laptop on Wikipedia](#)

A laptop is a computer small enough to carry and use on your lap. Its screen is in the lid, and the base holds the keyboard, a touch pad called a trackpad, and a battery. The order of the letters on the keyboard, starting Q, W, E, R, T, Y, comes from typewriters of the 1870s.

Tap any key, or type on your own keyboard, and the key goes down and the letter appears on the screen. Tap or drag on the trackpad to move the pointer and click. Tap anywhere else on the laptop to close the lid, and again to open it. Pick the color of the case in the Toy tab.

HOW TO PLAY

- **Tap:** Open or close
- **Sound:** Opening keeps only the screen-on sound (no twinkle), closing has no thunk, and the keys sound like a real keyboard (a tick and a thock).
- **Drag:** plays with the toy itself; a drag beside it turns the view.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Case:** pick a color
- **Type:** button
- **Trackpad:** button

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/laptop/

CREDITS

- Built from a recipe in the browser; no outside files.



Music box

Tap to close or open the lid. With 🖐️ on, turn the crank round to play the tune note by note.



Scan to play with it.

Learn more: [Music box on Wikipedia](#)

A music box plays a tune all by itself. Inside, a spring turns a metal cylinder covered in tiny pins. As it turns, the pins pluck the teeth of a steel comb, and each tooth rings with its own note: the longer the tooth, the lower the note.

This music box starts open, with a little dancer twirling on its stand while the tune plays and music notes float up. Tap to close the lid and stop the music, and tap again to open it. Pick the color of the wood in the Toy tab. With 🖐️ Hands-on on, lift or close the lid on its hinge yourself, and turn the crank by drawing circles round it: each click plays the tune's next note and the dancer turns.

HOW TO PLAY

- **Tap:** Open or close
- **Sound:** A music-box melody (plucked comb tones) while open.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Wood:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/music-box/

CREDITS

- Built from a recipe in the browser; no outside files.



Alarm clock

Tap it to ring the alarm.



Scan to play with it.

Learn more: [Alarm clock on Wikipedia](#)

A wind-up alarm clock has two metal bells on top and a little hammer between them. At the set time a spring inside lets the hammer go, and it strikes the two bells back and forth very fast, loud enough to wake a sleeper. Before alarm clocks were common, some people were paid to go around town tapping on windows to wake others up.

The hands on this clock show the real time where you are, or in another time zone you pick in the Toy tab. Tap it to ring the alarm: the hammer rattles between the bells and the whole clock shakes and hops. Pick its color in the Toy tab.

HOW TO PLAY

- **Tap:** Ring the bell
- **Sound:** A real wind-up alarm clock, the hammer rattling between its two bells, in two rings.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Color:** pick a color
- **Time zone:** This device, UTC, Los Angeles, Denver, Chicago, New York, São Paulo, London and 8 more

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/clock/

CREDITS

- Built from a recipe in the browser; no outside files.



Gift box

Tap to open the present; tap again to close it.



Scan to play with it.

Learn more: [Gift wrapping on Wikipedia](#)

A gift box is a present wrapped in bright paper and tied with a ribbon and a bow. People give wrapped presents for birthdays, holidays and other happy days, and the fun is not knowing what is inside until it is opened.

Tap it to open the present: the lid with its bow pops up and tips back, a golden star rises out of the box, and a burst of confetti fills the air. Tap again to close it. Pick the colors of the paper and the ribbon in the Toy tab.

HOW TO PLAY

- **Tap:** Open the present
- **Sound:** Ribbon rustle and a ta-da.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Paper:** pick a color
- **Ribbon:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/gift-box/

CREDITS

- Built from a recipe in the browser; no outside files.



Umbrella

Tap to close the umbrella;
tap again to open it.



Scan to play with it.

Learn more:
[Umbrella on Wikipedia](#)

An umbrella is a folding shade of cloth stretched over thin metal ribs, held up on a stick. When it opens, a sliding ring pushes little struts that spread the ribs out, and the cloth pulls tight to keep off the rain or the sun. People in ancient Egypt, China and Greece used umbrellas as sunshades thousands of years ago.

The umbrella starts open. Tap to close it, folding the ribs down along the stick, and tap again to open it. Pick a rainbow, striped or plain canopy and its color in the Toy tab.

HOW TO PLAY

- **Tap:** Open or close
- **Sound:** Fwump of the canopy opening.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Canopy:** Rainbow, Stripes, Plain
- **Colour:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/umbrella/

CREDITS

- Built from a recipe in the browser; no outside files.



Desk fan

Tap to switch it off or on.
Turn its swing on or off in
the Toy tab.



Scan to play with it.

Learn more: [Fan \(machine\) on Wikipedia](#)

A desk fan cools you by moving air: its angled blades push air forward as they spin, and moving air carries heat and sweat away from your skin faster. A fan does not make the air colder; it only makes you feel cooler.

This fan starts on, spinning and swinging slowly from side to side. Tap it to switch it off, and the blades slow to a stop; tap again to switch it on. In the Toy tab, turn the swing on or off and pick its color.

HOW TO PLAY

- **Tap:** Switch on or off
- **Sound:** The blades' soft thrum of air swelling in as it starts and fading as it stops (no rising motor hum).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colour:** pick a color
- **Swing:** on or off

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/desk-fan/

CREDITS

- Built from a recipe in the browser; no outside files.

Desk lamp

Tap to switch the light off or on.



Scan to play with it.

A desk lamp has a jointed arm that bends, so you can point its shade right where you need light, over a book or a drawing. The shade stops the light from shining in your eyes and sends it down onto the desk.

The lamp starts on, with a warm glow under the shade. Tap it to switch the light off, and tap again to switch it on. Pick its color in the Toy tab.

HOW TO PLAY

- **Tap:** Switch the light
- **Sound:** Switch click.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colour:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/lamp/

CREDITS

- Built from a recipe in the browser; no outside files.

Potion bottle

Tap it to pop the cork.



Scan to play with it.

A potion is a magic drink from fairy tales and wizard stories, said to make you fly, shrink or fall asleep. This one sits in a round glass flask with a long neck and a cork, the kind that chemists really use: the round bottom is strong and heats evenly.

Bubbles rise through the potion all the time. Tap it to pop the cork: it shoots up with a puff of glittering sparkles, then drops back into the neck. Pick the color of the potion in the Toy tab.

HOW TO PLAY

- **Tap:** Pop the cork
- **Sound:** Cork pop and fizz.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Potion:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/potion-bottle/

CREDITS

- Built from a recipe in the browser; no outside files.

Telescope

Tap to collapse it; tap again to pull it out.



Scan to play with it.

Learn more:
[Telescope on Wikipedia](#)

A telescope uses lenses or mirrors to make faraway things look nearer and bigger. This is a sliding spyglass, the kind sailors carried: its tubes slide inside one another, so it folds up short and pulls out long, with a big lens at the far end that gathers light. Galileo Galilei used a small telescope in 1610 to see the moons of Jupiter.

The telescope starts pulled out, on a three-legged stand. Tap to collapse it, sliding the tubes together, and tap again to pull it out.

HOW TO PLAY

- **Tap:** Extend or collapse
- **Sound:** Brass slide and a twinkle.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/telescope/

CREDITS

- Built from a recipe in the browser; no outside files.

Fountain pen

Tap it to uncap the pen and write a swirl in wet blue ink.



Scan to play with it.

Learn more: [Fountain pen on Wikipedia](#)

A fountain pen carries its own ink inside. The ink runs from the barrel through a thin channel called the feed to a split metal nib, and the slit draws it down to the paper by capillary action, the same pull that lets a paper towel soak up water.

This one is made from a detailed 3D model of a real green pen. Tap it and the cap slides off and clicks onto the back end, the nib writes a looping swirl that shines while it is wet and dries darker, and the cap goes back on.

HOW TO PLAY

- **Tap:** The cap (cut out of the scan as a solid piece) slides off and clicks onto the end, and the nib writes a swirl in wet blue ink that glistens, then dries darker; the cap goes back on (4 s).
- **Sound:** A cap click, a smooth nib scratch, another click.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/fountain-pen/

CREDITS

- **Fountain pen:** [Fountain pen in translucent green \(maker's marks painted out, cap moved\)](#) by chemicalIX, CC BY 4.0.



Running shoe

Tap it to untie the laces and tie them again, then watch it tap its toe.



Scan to play with it.

Learn more: [Sneakers on Wikipedia](#)

A running shoe is laced through rows of eyelets, so the laces pull the shoe snug around the foot. The usual bow is a reef knot with two loops: if the second half is tied the wrong way round, it becomes a granny knot, which slips and comes undone much more easily.

This shoe is a photo scan of a real trail shoe. Tap it and its laces come undone, cross over and tie themselves into a bow again, and the shoe taps its toe twice. Pick a flag in the top bar and the shoe wears its colors.

HOW TO PLAY

- **Tap:** Its laces (kit-built jointed cords over the scan's own laces) come undone, cross back through the eyelets and tie themselves into a bow, and the shoe taps its toe twice (4 s).
- **Sound:** Laces zipping through the eyelets, a soft tug, two toe taps.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/running-shoe/

CREDITS

- **Running shoe:** [PB158 Sneaker Low](#) (photogrammetry scan; its bow left out) by SCANIMAT, CC BY 4.0.



Hoodie

Tap it to flip the hood and cross the sleeves.



Scan to play with it.

Learn more: [Hoodie on Wikipedia](#)

A hoodie is a sweatshirt with a hood, first made in the 1930s for people working in cold places. Its soft fleece is usually knitted cotton or a cotton blend, brushed on the inside, and a drawstring runs through the edge of the hood to pull it snug. This one has raglan sleeves, whose seams run from the armpit up to the collar.

It is made from a detailed 3D model of a real hoodie. Tap it and the hood flops forward and flips back up as the drawstrings swing, then the sleeves swing up, cross in front and settle back. Pick a flag in the top bar and the hoodie wears its colors. With 🖐 Hands-on on, pull the hood: it flops like cloth and springs back up.

HOW TO PLAY

- **Tap:** The hood (cut out of the scan as a solid piece) flips up and its drawstrings swing, then the sleeves (cut at the shoulders) swing in and cross, and it all settles back (4.5 s).
- **Sound:** The fabric's soft brushes and the zip, with much less wind.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/hoodie/

CREDITS

- **Hoodie:** [Hoodie \(drawstrings and inside label left out\)](#) by Virtual Pandora, CC BY 4.0.

Sunglasses

Tap them (the frame or a lens) to fold the arms, flip the glasses round and darken the lenses.



Scan to play with it.

Learn more: [Sunglasses on Wikipedia](#)

These are round glasses with light-changing lenses. Special molecules in the lenses change shape in the sun's ultraviolet light and start to absorb light, so the lenses turn dark outdoors and clear again a few minutes after you go inside.

The frame comes from a detailed 3D model of real vintage spectacles. Tap them and the arms fold in on their hinges one after the other, the glasses flip over to face you and the lenses darken from clear to deep gray, then everything unfolds and clears. Pick a flag in the top bar and the frame wears its colors.

HOW TO PLAY

- **Tap:** The arms fold in on their hinges one after the other, the glasses flip to face you and the lenses darken from clear to deep gray like light-changing lenses; then they unfold (3.5 s).
- **Sound:** Each fold is a plastic hinge sliding shut and seating; the lenses darken silently (no electronic clicks or shimmer).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/sunglasses/

CREDITS

- **Sunglasses:** [Round Spectacles \(lenses left out\)](#) by Sean Buckley, CC0 1.0.



Baseball cap

Tap it to flip the cap off its stand and spin it like a flying disc.



Scan to play with it.

Learn more: [Baseball cap on Wikipedia](#)

A baseball cap is a round crown sewn from six panels, with a button on top and a stiff brim that shades the eyes. Baseball players wore caps like it in the 1800s, and now people wear them everywhere.

This gray cap is made from a detailed 3D model of a real one, resting on a wooden stand. Tap it and it flips up off the stand, spins flat like a flying disc and lands brim backward, then a second flip turns it round the right way. A spinning disc stays level because its spin resists being tipped over. Pick a flag in the top bar and the cap wears its colors.

HOW TO PLAY

- **Tap:** It flips up off its stand, spins flat like a flying disc and lands brim backward; then a second flip turns it round the right way (3 s).
- **Sound:** Fabric rustles instead of whooshes as it flips; the flick and the soft landings are unchanged.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/baseball-cap/

CREDITS

- **Baseball cap:** [Baseball Cap](#) by Scott VanArsdale, CC BY 4.0.



Sword in the stone

Tap to pull the sword from the stone. With 🖐️ on, drag it up out of the stone and back in.



Scan to play with it.

Learn more:
[Excalibur on Wikipedia](#)

The sword in the stone comes from the old legends of King Arthur of Britain. A sword was stuck fast in a great stone, and only the true king could pull it out. Many strong knights tried and failed, until the young Arthur drew it out easily and was made king.

Tap it to try: the sword sticks at first and wiggles, then slides free and rises in a shower of golden sparkles. Tap again to put it back in the stone. The Arthur stories have been told and retold for more than 800 years. With 🖐️ Hands-on on, pull the sword up by hand: it sticks and wiggles at first, then slides free and stays where you let go; push it back in with a clank.

HOW TO PLAY

- **Tap:** Pull
- **Sound:** Metal ring.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/sword-in-stone/

CREDITS

- Built from a recipe in the browser; no outside files.



Heraldic shield

Tap it and it blocks a blow where you tap: sparks fly and the emblem gleams.



Scan to play with it.

Learn more:

[Escutcheon \(heraldry\) on Wikipedia](#)

A heraldic shield carries a coat of arms, a painted design that showed who a knight was, even when a helmet hid the face. Heraldry has its own words: the background is the field, and a shape on it is a charge. A chevron is an upside-down V, a bend is a stripe from corner to corner, a saltire is an X-shaped cross, and per pale means split down the middle.

Tap it and it takes an unseen knock right where you tapped: it jolts and rocks away from the blow, sparks spray off the spot, and a gleam sweeps across the emblem. Pick one of seven designs and the colors of the field and the charge in the Toy tab.

HOW TO PLAY

- **Tap:** Blocks an invisible hit: it jolts, sparks and the emblem gleams.
- **Sound:** The clang and the gleam's shimmer, no crackling clicks.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Design:** Quarterly, Chevron, Cross, Saltire, Bend, Per pale, Stars
- **Field:** pick a color
- **Charge:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/shield/

CREDITS

- Built from a recipe in the browser; no outside files.



Bow and target

Pull the arrow back on the string and let go to shoot it. Tap to shoot too.



Scan to play with it.

Learn more: [Bow and arrow on Wikipedia](#)

Archery is the sport of shooting arrows with a bow at a target. Drawing back the string bends the bow's springy limbs, which store the energy of your pull; when the string is let go, the limbs spring back and send the arrow flying. A target has rings of color, with gold in the middle, and the closer to the center, the more points.

Pull the arrow back on the string, up or down a little to aim, and let go: the further you pull, the faster it flies, and it sticks where it lands, in the target or the grass. Tap to shoot too: the string draws back, the arrow flies across and lands in the target, and a fresh arrow waits on the string. Tap again to go back. Archery was first part of the Olympic Games in 1900.

HOW TO PLAY

- **Tap:** Shoot
- **Sound:** A real bowstring's thump and the arrow's hiss, not a guitar-like note; the hit is unchanged; real CC0 recordings now (bow-and-target-release.mp3), with the synthesized sound as a fallback.
- **Drag:** plays with the toy itself; a drag beside it turns the view.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/bow-and-target/

CREDITS

- **Sound:** [Bow Release \(Bow and Arrow\) 3](#) by Ali_6868, CC0 1.0.



Trebuchet

Tap it to launch: the weight drops and the long arm flings a stone.



Scan to play with it.

Learn more:
[Trebuchet on Wikipedia](#)

A trebuchet is a giant throwing machine from the Middle Ages. A long wooden arm swings on an axle, with a heavy weight on its short end and a sling on its long end. When the weight falls, the long end whips up and over, the sling swings out and lets go, and the stone flies a long way. Carpenters built them from great timbers, and the biggest could throw a stone as heavy as a person.

Tap it to launch: the weight drops, the arm swings up and the stone sails off, then the arm comes back down, ready for the next one. Today people build trebuchets for fun, to throw pumpkins in contests.

HOW TO PLAY

- **Tap:** Launch
- **Sound:** The timber beam creaks and strains, then a softer swing (much less wind).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/trebuchet/

CREDITS

- Built from a recipe in the browser; no outside files.



Crossbow

Tap it to shoot a bolt; the string snaps and a new bolt loads.



Scan to play with it.

Learn more:
[Crossbow on Wikipedia](#)

A crossbow is a short, strong bow fixed crosswise on a wooden stock. The drawn string is held back by a catch, so it can wait until the trigger lets it go. It shoots short, thick arrows called bolts. Crossbows were used in China more than 2,000 years ago and later all over medieval Europe, and today people shoot them at targets as a sport.

Tap it: the trigger lets the string go, the string snaps forward with a buzz and the bolt flies off along its groove. Then the string is drawn back and a new bolt appears.

HOW TO PLAY

- **Tap:** Shoot
- **Sound:** No click; the string's real thump, then the bolt hitting as before; real CC0 recordings now (bow-and-target-release.mp3), with the synthesized sound as a fallback.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/crossbow/

CREDITS

- **Sound:** [Bow Release \(Bow and Arrow\) 3](#) by Ali_6868, CC0 1.0.



Knight's helmet

Tap to open the visor; tap again to close it.



Scan to play with it.

Learn more: [Great helm on Wikipedia](#)

A knight's helmet was made of steel, hammered into shape by a skilled craftsman called an armorer. The front piece, the visor, has narrow slits and holes to see and breathe through, and it swings up on pivots at the sides so the knight could show their face. Plumes of colored feathers on top made the knight easy to spot at a tournament.

Tap it to open the visor, and tap again to close it. Pick the color of the plume in the Toy tab. A full suit of steel plate armor weighed about 20 to 25 kilograms (45 to 55 pounds).

HOW TO PLAY

- **Tap:** Open the visor
- **Sound:** Visor clank.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Plume:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/knights-helmet/

CREDITS

- Built from a recipe in the browser; no outside files.

Crown

Tap it: it rises and its jewels light up one by one.



Scan to play with it.

Learn more: [Crown on Wikipedia](#)

A crown is a ring of gold or silver worn on the head by a king or queen, a sign of their rule. Many crowns are set with jewels and lined with a soft velvet cap. Royal crowns are often kept with other treasures called the crown jewels, and some are hundreds of years old.

Tap it and the crown lifts and hovers. Its eight jewels light up one after another around the band, each flashing white, while golden sparkles drift up around it; then the jewels go out in turn and it settles back down. Pick the color of the velvet in the Toy tab.

HOW TO PLAY

- **Tap:** Jewels light up in turn and it rises with a glow.
- **Sound:** The fanfare on a natural trumpet (a lip scoop, breath and late vibrato) instead of synthetic brass; the jewels' pings stay.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Velvet:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/crown/

CREDITS

- Built from a recipe in the browser; no outside files.



Dragon egg

Tap to hatch the egg; tap again to go back.



Scan to play with it.

Dragons are creatures of legend, told of all over the world. In many European tales they are winged, fire-breathing beasts that guard treasure; in Chinese stories they are wise, long, snake-like beings that bring rain and good luck. In stories, a dragon hatches from an egg, just as lizards and snakes do.

Tap it to hatch the egg: cracks run across the scaly shell, it breaks open and a baby dragon peeks out in a burst of golden sparkles. Tap again to go back. Pick an emerald, ruby, sapphire or gold egg in the Toy tab.

HOW TO PLAY

- **Tap:** Hatch
- **Sound:** Crack and a tiny roar.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Egg:** Emerald, Ruby, Sapphire, Gold

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/dragon-egg/

CREDITS

- Built from a recipe in the browser; no outside files.



Wizard's orb

Tap it to cast a spell: sparks spiral out and runes circle the orb.



Scan to play with it.

A wizard's orb is a glass ball from stories of magic, where a wizard gazes into it to see faraway places or the future. This one sits on a stand, with glowing magic swirling inside. Runes, the letters of old alphabets once used in northern Europe, are often carved on magic things in stories.

Tap it to cast a spell: a flash of light fills the glass, the swirl spins up, sparks spiral out in five arms, and a ring of runes rises and circles the orb before it fades. Pick the color of the magic in the Toy tab.

HOW TO PLAY

- **Tap:** A bigger spell: runes orbit, light bursts and sparks spiral out.
- **Sound:** A deep, slow hum of power swelling in the glass (no wind or twinkles).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Magic:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/wizards-orb/

CREDITS

- Built from a recipe in the browser; no outside files.



Crystal ball

Tap it to swirl the mist and raise a glowing sign.



Scan to play with it.

Learn more: [Crystal ball on Wikipedia](#)

A crystal ball is a ball of clear glass or quartz. In old stories, fortune tellers gaze into one to see visions of what is to come. A real clear ball also works like a lens: look through it and you see the room behind it, small and upside down.

This one is filled with glowing mist. Tap it and the mist whips around and thins, and a glowing sign rises out of it, turns once and fades: a star, a moon or a heart in turn. Pick the color of the mist in the Toy tab. Sparkle adjusts the sparkling effect in the ball.

HOW TO PLAY

- **Tap:** Mist swirls hard and a glowing shape appears inside, then fades.
- **Sound:** A soft misty whoosh and a real crystal glass ringing as the sign rises (no theremin).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Mist:** pick a color
- **Sparkle:** slider

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/crystal-ball/

CREDITS

- **Sound:** [Wine glass clink deeper.wav](#) by lmr9, CC0 1.0.



Jellyfish

Tap it to make it swim: one strong stroke jets it upward.



Scan to play with it.

Learn more:
[Jellyfish on Wikipedia](#)

A jellyfish is a soft sea animal with no brain, no heart and no bones; its body is mostly water. It swims by squeezing its bell-shaped body, which pushes water out behind it, and it trails long tentacles armed with tiny stingers to catch its food. Jellyfish have drifted in the oceans for more than 500 million years.

Tap it and one strong stroke squeezes the bell and jets it up, trailing its glowing tentacles, then it drifts slowly back down. Pick a moon jelly, a sea nettle or a blue one in the Toy tab.

HOW TO PLAY

- **Tap:** A strong pulse that jets it upward, trailing glowing tentacles.
- **Sound:** The same two swimming pulses, a little louder.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colour:** Moon jelly, Sea nettle, Blue

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/jellyfish/

CREDITS

- Built from a recipe in the browser; no outside files.



School of fish

Tap it for a bait ball. Hands-on: drag through the school and the fish dart away.



Scan to play with it.

Learn more:
[Shoaling and schooling on Wikipedia](#)

A school is a big group of fish swimming together, all turning at once. Each fish keeps pace with its neighbors by watching them and by feeling the water move along a line of special sense organs down its sides. Being one of many makes it much harder for a hunter to pick out any single fish.

These 48 fish each swim on their own. Tap it and the school tightens into a spinning bait ball, the shape small fish make when a hunter comes near, then bursts outward in every direction and swims back into place. Pick silver or tropical fish in the Toy tab.

HOW TO PLAY

- **Tap:** The school swirls into a bait ball and scatters.
- **Sound:** Water swishes and bursts, with bubbles.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Fish:** Silver, Tropical

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/fish-school/

CREDITS

- Built from a recipe in the browser; no outside files.



Butterfly

Tap it to make it flutter faster.



Scan to play with it.

Learn more:
[Butterfly on Wikipedia](#)

A butterfly is an insect with four large wings covered in tiny, overlapping scales, and the scales make its colors and patterns. Every butterfly starts life as a caterpillar, which sheds its skin to become a pupa, called a chrysalis, before emerging as a butterfly. Some colors, like the shining blue of the blue morpho, come from the shape of the scales, not from any paint-like color in them.

It flaps its wings slowly all the time. Tap it to flutter, beating its wings faster and wider for a moment. Pick a monarch, a blue morpho, a swallowtail or a rose butterfly in the Toy tab.

HOW TO PLAY

- **Tap:** Flutter
- **Sound:** Delicate flutter.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Wings:** Monarch, Blue morpho, Swallowtail, Rose

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/butterfly/

CREDITS

- Built from a recipe in the browser; no outside files.



Pufferfish

Tap it to poke it. Try the Puff slider in the Toy tab.



Scan to play with it.

Learn more:

[Tetraodontidae on Wikipedia](#)

A pufferfish is a slow swimmer with a clever defense: when it is frightened, it gulps water and swells up into a ball several times its normal size, and in many kinds, spines stand out all over it. That makes it very hard for a hungry fish to swallow. Many pufferfish are also poisonous to eat.

This one rests slim. Tap it to poke it: it puffs up into a big, spiky ball, holds it, then lets the water out with a sputter and shrinks back. Try the Puff slider in the Toy tab too.

HOW TO PLAY

- **Tap:** Inflates to a big spiky ball, then deflates.
- **Sound:** Balloon inflate and a sputter.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Puff:** slider

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/pufferfish/

CREDITS

- Built from a recipe in the browser; no outside files.



Nautilus

Tap it to startle it: it hides its tentacles, then peeks out again.



Scan to play with it.

Learn more:
[Nautilus on Wikipedia](#)

A nautilus is a sea animal related to the octopus and the squid, but it lives in a coiled, striped shell. Inside, the shell is split into chambers: the nautilus lives in the biggest, newest one and fills the older ones with gas and a little water to float up or sink down. It has up to 90 small tentacles and swims by squirting water. Animals like it have lived in the sea for about 500 million years.

Tap it and, startled, it jets back a little and pulls its tentacles in behind its hood. It waits, peeks out halfway, then slowly reaches out again.

HOW TO PLAY

- **Tap:** Tentacles pull into the shell, then peek back out.
- **Sound:** A jet of water, the hollow shell, a knock and bubbles.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/nautilus/

CREDITS

- Built from a recipe in the browser; no outside files.



Ladybug

Tap to open the wings; tap again to close them.



Scan to play with it.

Learn more:
[Coccinellidae on Wikipedia](#)

A ladybug is a small, round beetle. Its red, spotted back is really a pair of hard wing cases, and under them, thin flying wings lie folded up. Its bright colors warn birds that it tastes bad. Gardeners love ladybugs, because they eat the tiny aphids that harm plants.

Tap it to open the wings: the red wing cases lift and spread, and the thin wings unfold beneath them, ready to fly. Tap again to fold them away. Counting its spots will not tell you how old a ladybug is: the number depends on the kind of ladybug.

HOW TO PLAY

- **Tap:** Open the wings
- **Sound:** The same wing whirr, about 5 dB quieter.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/ladybug/

CREDITS

- Built from a recipe in the browser; no outside files.

Snail

Tap to make it hide in its shell; tap again to bring it out.



Scan to play with it.

Learn more: [Snail on Wikipedia](#)

A snail carries its home on its back: a coiled shell it can pull its whole body into. It glides along on one long, muscular foot over a thin layer of slime, and its eyes sit at the tips of its two long upper tentacles. In dry weather, a snail can seal its shell with a layer of dried slime and wait for rain.

Tap it and it hides, as a real snail does: the eye stalks roll in first, then the head and the foot are drawn in through the shell's opening, and the shell settles on the ground. Tap again and it slides back out, the eye stalks unrolling last.

HOW TO PLAY

- **Tap:** Keep hiding in the shell, but keep the shell clearly visible and ease in more gently.
- **Sound:** Wetter and slimier, a squish with a sticky stretch.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/snail/

CREDITS

- Built from a recipe in the browser; no outside files.



Octopus

Tap it to squirt ink.



Scan to play with it.

Learn more: [Octopus on Wikipedia](#)

An octopus is a soft-bodied sea animal with eight arms lined with suckers. It has three hearts and blue blood, no bones at all, and it can change the color of its skin in a split second to hide or to signal.

When something scares it, an octopus squirts a cloud of dark ink and jets away behind it. Tap this one and it does just that: the ink billows out while it shoots up and away with its arms streaming, then it drifts back as the ink thins. You can pick its color in the Toy tab. With 🖐 Hands-on on, drag its body and the arms trail behind and curl back; pull an arm and let it go.

HOW TO PLAY

- **Tap:** A big ink cloud billows out while it jets away and back.
- **Sound:** A squirt, then the ink bubbling out as a liquid (no wind).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

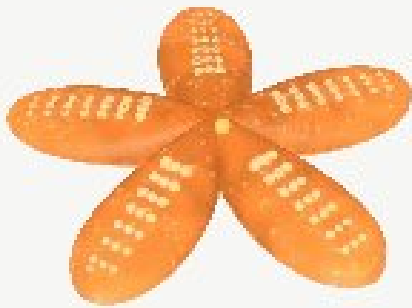
- **Colour:** Coral, Purple, Orange, Teal

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/octopus/

CREDITS

- Built from a recipe in the browser; no outside files.



Starfish

Tap it to wave its arms.



Scan to play with it.

Learn more: [Starfish on Wikipedia](#)

A starfish, or sea star, is not a fish at all but a relative of the sea urchin. It has no brain. Seawater pumped through canals in its body works the hundreds of tiny tube feet under each arm, which let it creep along and grip rocks, and at the tip of each arm is a simple eye. Many kinds can regrow a lost arm.

Tap it and its five arms lift and curl in turn, like a slow wave, then settle back. Pick an orange, red, purple or blue starfish in the Toy tab.

HOW TO PLAY

- **Tap:** Wave the arms
- **Sound:** The pop is kept, and each arm makes a soft wet lift as it rises and a soft pat as it settles, one after another.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

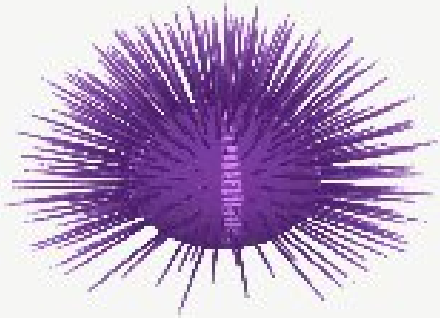
- **Colour:** Orange, Red, Purple, Blue

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/starfish/

CREDITS

- Built from a recipe in the browser; no outside files.



Sea urchin

Tap it: its spines sweep in waves and it creeps along.



Scan to play with it.

Learn more: [Sea urchin on Wikipedia](#)

A sea urchin is a round, spiny sea animal, a relative of the starfish. Its spines protect it, and between them it has long, thin tube feet with suckers that it uses to walk and to hold on. Its mouth is underneath, with five strong teeth for scraping seaweed off rocks.

Tap it and its spines sweep around it in waves, each tilting on its base, while pink tube feet reach out, and it creeps a little way to the side and back. Pick a purple, black, red or green urchin in the Toy tab.

HOW TO PLAY

- **Tap:** Spines ripple in waves and it creeps along.
- **Sound:** A prickly rustle, wave after wave.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colour:** Purple, Black, Red, Green

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/sea-urchin/

CREDITS

- Built from a recipe in the browser; no outside files.



Frog

Tap it: a fly buzzes in and the frog catches it with its tongue.



Scan to play with it.

Learn more: [Frog on Wikipedia](#)

A frog is an amphibian: many species start as tadpoles in water, then grow legs and lungs, though others develop directly into small frogs. It catches insects with a long, sticky tongue that flips out of its mouth in a flash. To swallow, a frog pulls its big eyes down into its head, and they help push the food down its throat.

Tap it: a fly buzzes in and hovers, the frog's tongue shoots out, catches it and snaps back, its eyes sink to swallow, and it croaks twice with a puff of its throat. Pick a green, red, blue or yellow frog in the Toy tab.

HOW TO PLAY

- **Tap:** Tongue shoots out to catch a fly.
- **Sound:** A real fly's wandering buzz, and real croaks after it eats the fly; real CC0 recordings now (frog-fly.mp3, frog-croak.mp3), with the synthesized sound as a fallback.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colour:** Green, Red, Blue, Yellow

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/frog/

CREDITS

- **Sound:** [Frog croaking sound effect](#) by betterchinese, CC0 1.0.
- **Sound:** [Fly_Buzzing_Edited.wav](#) by AmberdeMeillon, CC0 1.0.



Penguin

Tap it to flap its flippers.



Scan to play with it.

Learn more:
[Penguin on Wikipedia](#)

A penguin is a bird that cannot fly, but it is a superb swimmer: its wings are stiff, strong flippers that it uses to fly through the water. Almost all penguins live in the southern half of the world. The largest, the emperor penguin, stands about 1.1 meters (3.7 feet) tall.

Tap it and it flaps its flippers up and down. Its colors help it hide in the sea: from above, its black back blends with the dark water below, and from below, its white front blends with the bright surface.

HOW TO PLAY

- **Tap:** Flap
- **Sound:** Squawk.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/penguin/

CREDITS

- Built from a recipe in the browser; no outside files.



Owl

Tap it and it turns its head to look behind.



Scan to play with it.

Learn more: [Owl on Wikipedia](#)

Owls are birds that mostly hunt at night. Their big eyes cannot move in their sockets, so an owl turns its whole head instead, as far as about 270 degrees, three-quarters of the way around. Soft, fringed edges on its feathers let it fly almost without a sound.

This owl sits on a branch. Tap it and it turns its head far around to look behind it, holds there a moment, then turns back to face you.

HOW TO PLAY

- **Tap:** Turn the head
- **Sound:** A short, natural owl hoot (h'HOO-oo and a soft hoo), about 1 s.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

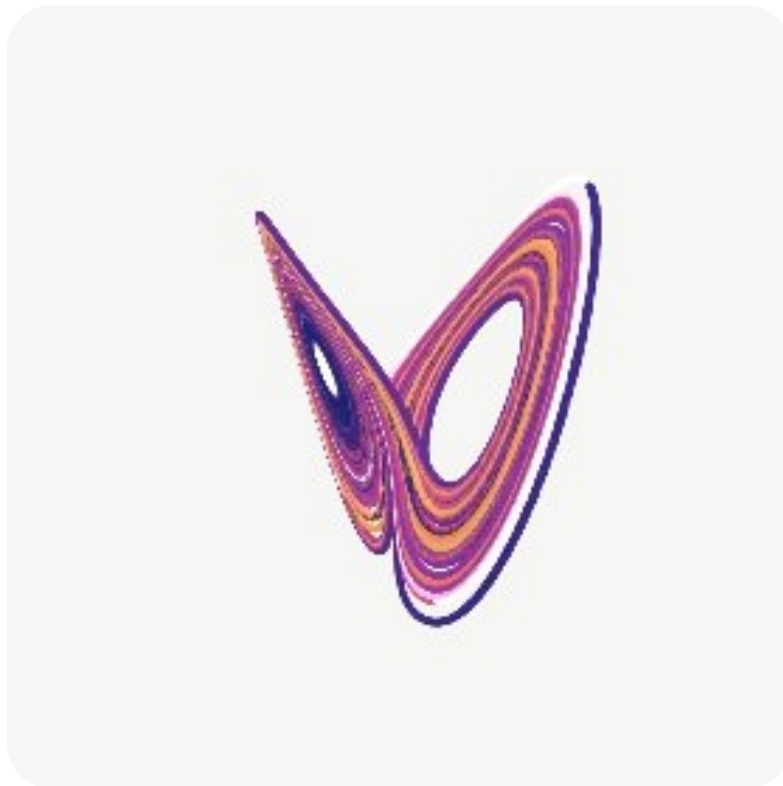
SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/owl/

CREDITS

- **Sound:** [owl_hooting_000102_0145S3 002-070 000-002 068-074.wav](#) by Gerent, CC0 1.0.



Lorenz attractor

Tap to race along the path.
Set the glow in the Toy tab.



Scan to play with it.

Learn more: [Lorenz system on Wikipedia](#)

In 1963 the scientist Edward Lorenz made a very simple model of moving air: three short equations. He found that its path loops around two centers forever, never repeating and never crossing itself, in a shape like a pair of butterfly wings. This shape is called the Lorenz attractor.

Tap it and a bright spark races along the whole path, drawing it again in light that then fades. Two starting points that are almost the same soon follow very different paths. This is called chaos, or the butterfly effect, and it is why weather is so hard to forecast far ahead.

HOW TO PLAY

- **Tap:** A glowing point races along the trail, drawing a new path.
- **Sound:** Something epic, a deep hit, a dark slow rush and a wide low chord (no whistling howl).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Glow:** slider

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/lorenz/

CREDITS

- Built from a recipe in the browser; no outside files.



Möbius strip

Tap to send the rider round.
Pick a rider in the Toy tab.



Scan to play with it.

Learn more: [Möbius strip on Wikipedia](#)

A Möbius strip is a loop with a half twist in it. It has only one side and only one edge: a line drawn along its middle comes back to where it started without ever crossing an edge. It is named after August Möbius, who wrote about it in the 1850s. You can make one from a strip of paper and some tape.

Tap it and the rider goes along the middle of the band. After one lap it is underneath where it started, and after two it is back on top. Pick a race car, a beach ball, a duck on a bike or an ant in the Toy tab. Pick its colors and glow in the Toy tab too.

HOW TO PLAY

- **Tap:** An ant-like dot walks the single side; the band twists.
- **Sound:** No step tune; each rider has its own sound on the ride: a race car's engine, a rolling beach ball, a bicycle with one soft quack, the ant's tiny feet.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colours:** Sunset, Ocean, Jewel
- **Rider:** Race car, Beach ball, Duck on a bike, Ant
- **Glow:** slider

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/mobius/

CREDITS

- Built from a recipe in the browser; no outside files.



Klein bottle

Tap to send water through. Set the glow in the Toy tab.



Scan to play with it.

Learn more: [Klein bottle on Wikipedia](#)

A Klein bottle is a surface with no inside and no outside. Its neck bends around and passes through its own side to join the bottom from within, so an ant crawling on it could reach every part of it without ever crossing an edge. Felix Klein described it in 1882. A true Klein bottle needs four dimensions; in our world its neck has to cut through the wall.

Tap it and a surge of glowing water pours in at the base, runs up the body, through the neck and around into the bottom, then fades. Set the glow, or pick the color of the glass, in the Toy tab.

HOW TO PLAY

- **Tap:** A clear surge of water flows through the neck and round.
- **Sound:** Water moving in a bottle, sloshing and glugging (no bubbles or waterfall); real CC0 recordings now (klein-bottle-slosh.mp3), with the synthesized sound as a fallback.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Glass:** pick a color
- **Glow:** slider

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/klein-bottle/

CREDITS

- **Sound:** [Bottle Slosh 2](#) by alegemaate, CC0 1.0.



Menger sponge

Tap to close and carve the holes. Pick the level in the Toy tab.



Scan to play with it.

Learn more: [Menger sponge on Wikipedia](#)

A Menger sponge is a fractal. Start with a cube and cut it into 27 smaller cubes, like a puzzle cube. Take out the one in the middle and the six at the middle of each face, leaving 20. Then do the same to each of those 20 cubes, and again, forever. Karl Menger described it in 1926.

Tap it and every hole is plugged, the smallest first, until it is a plain cube. Then it is carved again: the big cubes slide out of the faces, then the next size down, then the smallest. Pick level 2 or 3 in the Toy tab: level 2 keeps 400 little cubes, and level 3 keeps 8,000.

HOW TO PLAY

- **Tap:** Zooms into itself: the holes open to show the next level down.
- **Sound:** Descending recursive blips.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

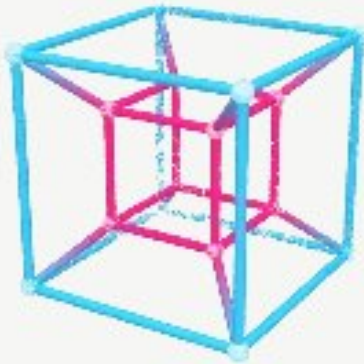
- **Level:** Level 2, Level 3

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/menger-sponge/

CREDITS

- Built from a recipe in the browser; no outside files.



Hypercube

Tap to turn it inside out. Try the 4D turn slider in the Toy tab.



Scan to play with it.

Learn more: [Hypercube on Wikipedia](#)

A hypercube, or tesseract, is a cube in four dimensions. A square has 4 corners and a cube has 8; a tesseract has 16 corners and 32 edges, and its sides are 8 cubes. We can't see four dimensions, so this toy shows its shadow in our three: a cube inside a cube, with their corners joined.

At rest it rocks gently in 4D. Tap it to turn it once around through the fourth dimension: the pink inner cube swells out to become the outer one while the blue one folds inside, then it turns on back to where it started. Every edge stays perfectly straight the whole time. The 4D turn slider changes the strength of its gentle resting rotation.

HOW TO PLAY

- **Tap:** Rotates through the fourth dimension, turning inside out.
- **Sound:** No "vroom"; a great slow rush as it turns inside out and again as it turns home, over a low chord.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **4D turn:** slider

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/hypercube/

CREDITS

- Built from a recipe in the browser; no outside files.



Torus knot

Tap it: the knot pulls loose and springs back. Drag to turn it. Pick a knot in the Toy tab.



Scan to play with it.

Learn more: [Torus knot on Wikipedia](#)

A torus knot is a knot that winds around the surface of a donut shape, called a torus. It goes around the hole a few times one way while it loops through it a few times the other way. The simplest is the trefoil, which winds 2 and 3 times; it is the simplest true knot, one that can never be untied without cutting it.

Tap it and the knot is pulled loose, its loops stretching and swirling; then it is let go and springs back past its rest shape and wobbles to a stop, like a spring. Choose winding counts (2, 3), (2, 5), (3, 4), (3, 5) or (2, 7) in the Toy tab: the pair counts turns around the torus's two directions. Colors changes its palette and Glow its brightness.

HOW TO PLAY

- **Tap:** Contorts through different knot shapes.
- **Sound:** It strains as it is pulled and lets go with a real thump and swish (no banjo or guitar notes).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

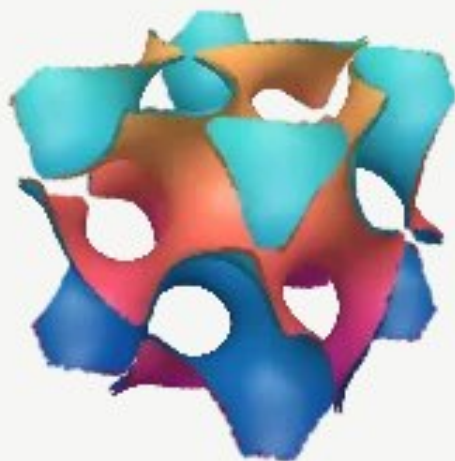
- **Knot:** Trefoil (2, 3), Cinquefoil (2, 5), (3, 4), (3, 5), (2, 7)
- **Colours:** Sunset, Ocean, Jewel, Candy
- **Glow:** slider

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/torus-knot/

CREDITS

- Built from a recipe in the browser; no outside files.



Gyroid

Tap to make it breathe. Pick a cube or a ball in the Toy tab.



Scan to play with it.

Learn more: [Gyroid on Wikipedia](#)

A gyroid is a curving surface that splits space into two tangled mazes of tunnels that never meet. It has no straight lines and no flat parts, and it repeats forever in every direction. Alan Schoen, a scientist working for NASA, found it in 1970. Nature makes it too: tiny gyroid crystals in some butterfly wings give them their shiny green color.

Tap it and it breathes: the surface slides along itself, so one set of tunnels swells while the other narrows, then the other way, and it settles. In the Toy tab, cut it into a cube or a ball.

HOW TO PLAY

- **Tap:** The surface flows and shifts phase like a breathing sponge.
- **Sound:** Slow rushes one way and back over a low chord (no rising hum, no bubbles).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

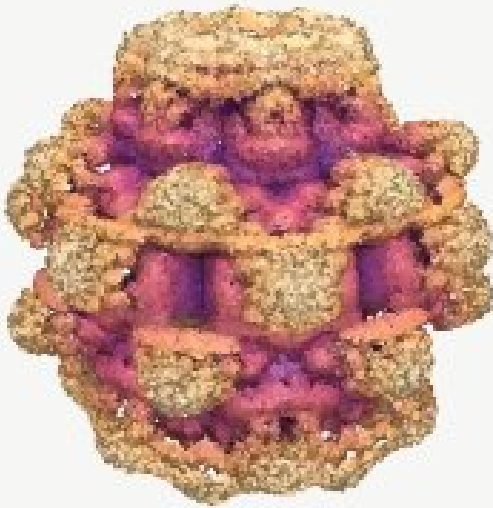
- **Shape:** Cube, Ball

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/gyroid/

CREDITS

- Built from a recipe in the browser; no outside files.



Mandelbulb

Tap it to turn its discs like the dials of a lock.



Scan to play with it.

Learn more:
[Mandelbulb on Wikipedia](#)

The Mandelbulb is a 3D fractal, a cousin of the famous flat Mandelbrot set. It is made by repeating one short formula over and over for each point in space and keeping only the points that never fly away. The mathematical fractal has detail on many scales; this toy has a fixed, finite set of points. It was found in 2009 by Daniel White and Paul Nylander.

Tap it and its seven slices turn like the dials of a combination lock, neighbors in opposite directions, then back again. Each turns one seventh of a turn, because the bulb looks the same after a seventh of a turn, so it lands on the same picture.

HOW TO PLAY

- **Tap:** Morphs through powers (bulb shapes twist and bloom).
- **Sound:** Deep morphing tone.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

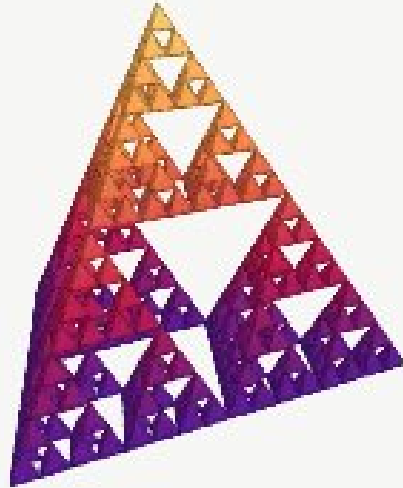
SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/mandelbulb/

CREDITS

- Built from a recipe in the browser; no outside files.



Sierpinski tetrahedron

Tap to explode it; tap again to put it back. Pick the level in the Toy tab.



Scan to play with it.

Learn more:
[Sierpiński triangle](#)
on [Wikipedia](#)

A Sierpinski tetrahedron is a fractal pyramid. It is made of four smaller copies of itself, each half as tall, and each of those is made of four smaller ones, and so on. It is the 3D cousin of the Sierpinski triangle, named after the mathematician Waclaw Sierpiński, who described the triangle in 1915.

Tap it to explode it into its pieces; tap again to put it back together. Pick the level in the Toy tab: each level has four times as many little pyramids, so levels 3, 4 and 5 have 64, 256 and 1,024 respectively.

HOW TO PLAY

- **Tap:** Explode
- **Sound:** Stacked triangle chimes.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Level:** Level 3, Level 4, Level 5

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/sierpinski/

CREDITS

- Built from a recipe in the browser; no outside files.



Platonic solids

Tap to explode it; tap again to put it back. Pick one of the five solids in the Toy tab.



Scan to play with it.

Learn more:
[Platonic solid on Wikipedia](#)

The Platonic solids are the only five convex polyhedra whose faces are all the same regular shape, meeting in the same way at every corner: the tetrahedron (4 triangles), the cube (6 squares), the octahedron (8 triangles), the dodecahedron (12 pentagons) and the icosahedron (20 triangles). They are named after the Greek thinker Plato, and Euclid proved there can be no others.

It starts as a dodecahedron. Tap it to explode its faces apart; tap again to put them back. Pick any of the five solids, and its colors, in the Toy tab. Many game dice come in these shapes.

HOW TO PLAY

- **Tap:** Explode
- **Sound:** Five-note chime.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Solid:** Tetrahedron, Cube, Octahedron, Dodecahedron, Icosahedron
- **Colours:** Ocean, Sunset, Jewel, Candy

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/platonic/

CREDITS

- Built from a recipe in the browser; no outside files.



Seashell spiral

Tap it to hear the sea in the shell.



Scan to play with it.

Learn more:
[Logarithmic spiral](#)
[on Wikipedia](#)

Many seashells grow in a spiral. As the animal inside grows, it adds new shell at the opening, each turn wider than the one before but the same shape, so the shell gets bigger without changing its form. Mathematicians call this a logarithmic spiral.

People say you can hear the sea in a shell. Tap it and three soft swells of sea-blue light run down the spiral to the opening, and with each one a ripple rolls out of the mouth, widening and fading like a wave. The sound in a real shell is the noise around you, echoing inside it.

HOW TO PLAY

- **Tap:** Glows softly and waves ripple out of the opening.
- **Sound:** Ocean hush.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

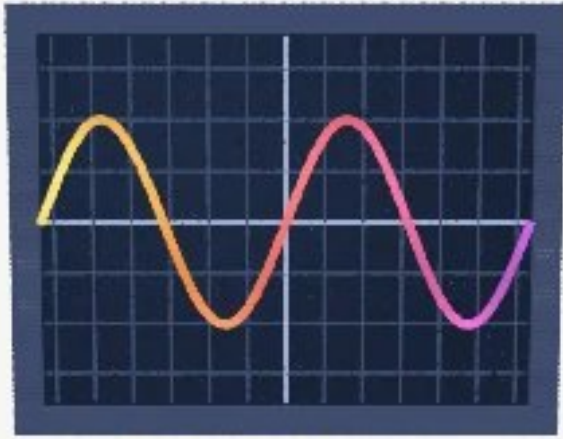
SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/seashell-spiral/

CREDITS

- Built from a recipe in the browser; no outside files.



Graph plotter

Tap to draw the curve. Pick a curve, or type your own, in the Toy tab.



Scan to play with it.

Learn more: [Graph of a function on Wikipedia](#)

A graph turns an equation into a picture. For $y = x^2$, every x along the bottom gets a height y , and joining them draws a curve. René Descartes wrote about this kind of grid in 1637. Curves can be drawn other ways too: by distance and angle (polar), or by following a moving point over time.

Tap it and a pen draws the curve across the grid, humming a note that follows its height; then the a slider sweeps and the curve bends with it. Pick one of 42 famous curves in the Toy tab, or type your own: $y = \dots$ with x , $r = \dots$ with θ , or $x = \dots$, $y = \dots$ with t . Put a in it to see it bend.

HOW TO PLAY

- **Tap:** A pen traces the curve across a lit grid, then the a slider sweeps and the curve bends in real time before settling back.
- **Sound:** A tone that follows the curve's height as the pen draws it.
- **Drag:** plays with the toy itself; a drag beside it turns the view.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Curve:** Sine wave, Bell curve, Parabola, Cubic, Double well, Tangent, Hyperbola, Exponential and 35 more
- **b (your own curve):** slider
- **a:** slider

In the Toy tab of the app.

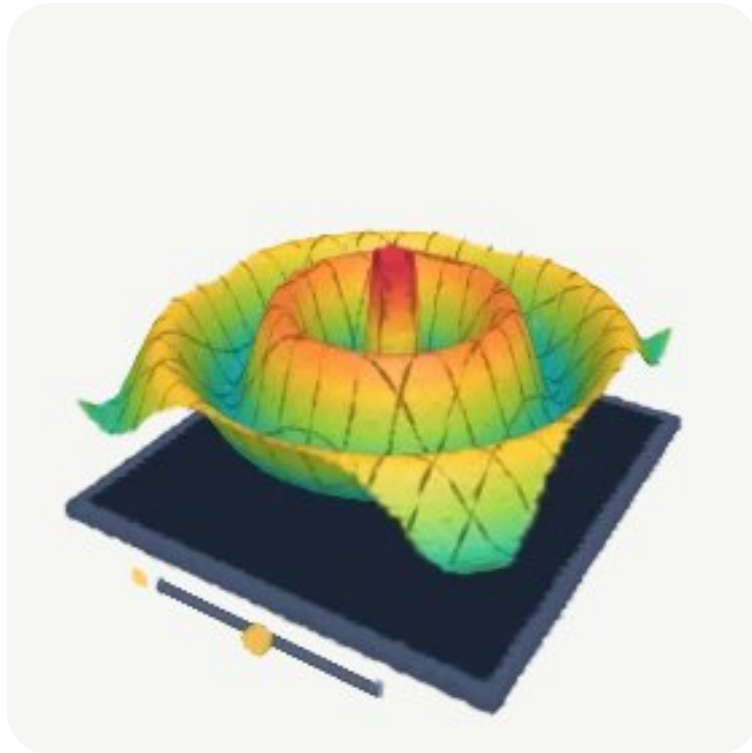
Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/graph-plotter/

CREDITS

- Built from a recipe in the browser; no outside files.

Surface plotter

Tap to raise the surface. Pick one, or type your own, in the Toy tab.



Scan to play with it.

A surface plot is a graph in 3D. For each point on a flat floor, with its x and y , the equation gives a height z , so the graph becomes a landscape of hills, valleys and saddles. Colors show how high each part is.

Tap it and the surface lies flat, then rises out of the sheet and ripples, twists or breathes as a number called a changes. Pick one of 16 famous surfaces in the Toy tab, such as the sombrero, the saddle or Rosenbrock's banana valley, which is used to test computer methods that hunt for the lowest point. Or type your own: $z = \dots$ with x and y , or with r and θ .

HOW TO PLAY

- **Tap:** The surface rises out of a flat sheet, colored by height, then its parameter plays so it ripples, twists or breathes, and it settles.
- **Sound:** Subtle, a soft low chord swelling as the surface rises (no wave, no "vroom").
- **Drag:** plays with the toy itself; a drag beside it turns the view.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Surface:** Saddle, Monkey saddle, Sombrero, Egg crate, Gaussian hill, Rosenbrock's banana valley, Paraboloid (bowl), Ocean waves and 9 more
- **a :** slider

In the Toy tab of the app.

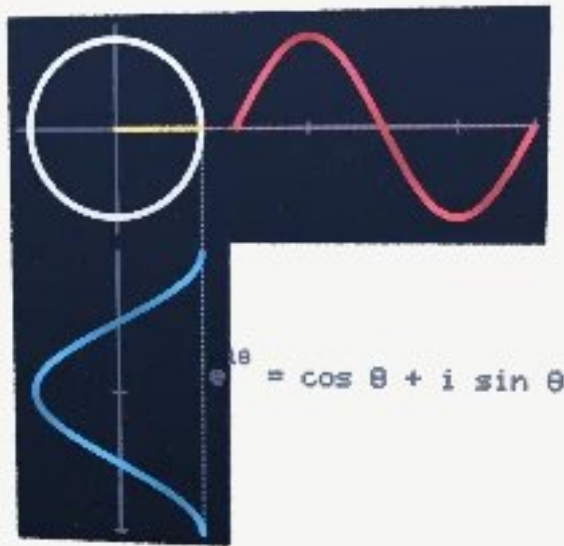
Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/surface-plotter/

CREDITS

- Built from a recipe in the browser; no outside files.

Circle and waves

Tap to send the point round. Try 3D, or type your own path, in the Toy tab.



Scan to play with it.

Learn more: [Unit circle on Wikipedia](#)

The unit circle is a circle with a radius of 1. As a point goes around it, turning through an angle called θ , its height is the sine of θ and its left-right place is the cosine. Traced out over time, they make two waves, the same shapes as sound and light waves.

Tap it and the point goes once around while its height draws the sine wave and its left-right place draws the cosine wave. For the circle, Euler's formula, $e^{i\theta} = \cos \theta + i \sin \theta$, lights up piece by piece. In the Toy tab, try a 3D helix, another path, up to three turns, or type your own: $x = \dots$, $y = \dots$ with t , or $r = \dots$ with θ .

HOW TO PLAY

- **Tap:** A point runs round the unit circle while its shadows unroll as a sine wave and a cosine wave on two walls, and Euler's formula lights up term by term.
- **Sound:** Warmer, a soft low chord that swells with the turn and a gentle rush a quarter turn later (one option to try).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

VIEWS

- **View:** 2D diagram, 3D helix

SETTINGS

- **Path:** Circle, Ellipse, Figure eight, Cardioid, Rose, Your own (below)
- **Turns:** slider

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/unit-circle/

CREDITS

- Built from a recipe in the browser; no outside files.



Fourier circles

Tap to spin the circles. Type a word or a curve in the Toy tab.



Scan to play with it.

Learn more: [Fourier series on Wikipedia](#)

Circles turning on circles can draw almost any closed shape, given enough of them. Each circle spins a whole number of times per loop while riding on the rim of the one before, and the tip of the last one traces the drawing. This is a Fourier series, named after Joseph Fourier, who showed in the early 1800s that repeating patterns can be built by adding up simple waves.

Tap to spin them and watch the shape drawn again. Few circles draw a wobbly shape and many draw a crisp one: set how many in the Toy tab, pick a heart, a star or a square wave, stack them in 3D, or type a word and each letter gets its own chain of circles.

HOW TO PLAY

- **Tap:** A chain of spinning circles, each riding on the last, draws a shape with its tip; a Circles option sets how many.
- **Sound:** Each circle hums its own frequency, building into a chord.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

VIEWS

- **View:** 2D, 3D (stacked)

SETTINGS

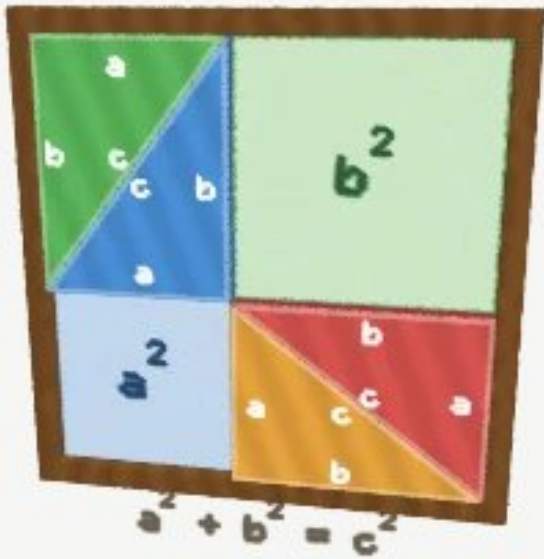
- **Shape:** Heart, Star, Square wave, Your words (below), Your own curve (below)
- **Circles:** slider

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/fourier-circles/

CREDITS

- Built from a recipe in the browser; no outside files.



Pythagoras proof

Tap it to slide the triangles and show that $a^2 + b^2 = c^2$.



Scan to play with it.

Learn more:

[Pythagorean theorem on Wikipedia](#)

In a right triangle, the two short sides a and b and the long side c always fit a rule: $a^2 + b^2 = c^2$. So a square drawn on the long side has the same area as the squares on the two short sides put together. It is named after Pythagoras, a Greek thinker of about 2,500 years ago, though people in Babylon knew it even earlier.

This is a rearrangement proof. Four copies of the triangle fill a big square, leaving two empty squares, a^2 and b^2 . Tap it and the triangles slide into the corners, and the empty space becomes one tilted square, c^2 , so $a^2 + b^2$ must equal c^2 . Then they slide back.

HOW TO PLAY

- **Tap:** Four copies of a right triangle slide as solid pieces inside a big square, so the empty space changes from $a^2 + b^2$ into c^2 ; then they slide back.
- **Sound:** Wooden slides, and a click as each piece lands.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

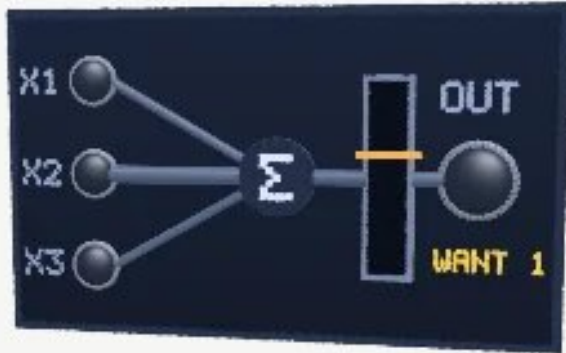
SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/pythagoras-proof/

CREDITS

- Built from a recipe in the browser; no outside files.



Perceptron

Tap it to watch it try an example, get it wrong, and learn.



Scan to play with it.

A perceptron is the simplest artificial neuron, described by Frank Rosenblatt in 1958. It takes a few inputs, multiplies each by a weight (how much that input counts), and adds them up. If the sum passes a threshold, it fires: its answer is 1, otherwise 0. It learns by nudging its weights each time it gets an answer wrong.

Tap it: inputs X1 and X3 light up and send pulses along wires as thick as their weights. The sum fills the gauge but stays under the threshold line, so the lamp flashes red. The two live wires thicken, the pulses go again, the gauge passes the line and the lamp lights gold. Pick the flat poster or a 3D model in the Toy tab. This repeats a scripted learning example; it does not keep training on your taps.

HOW TO PLAY

- **Tap:** Three input lights send pulses along wires whose thickness shows their weights into one node; the sum fills a gauge, and if it passes the threshold the output lamp snaps on.
- **Sound:** A blip for each input, a click as the lamp decides, a rising tone when it learns.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

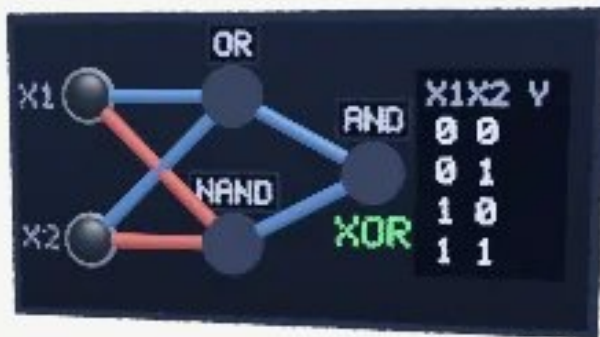
VIEWS

- **View:** Poster (2D), 3D model

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/perceptron/

CREDITS

- Built from a recipe in the browser; no outside files.



Multilayer perceptron

Tap it to try all four inputs and fill in the XOR truth table.



Scan to play with it.

One perceptron can only split its inputs with a single straight line, so it cannot learn XOR, “exclusive or”, which is 1 when exactly one of two inputs is 1. Put neurons in layers and it can. Here an OR neuron and a NAND (“not both”) neuron feed an AND neuron, and together they give XOR. Blue wires add to a neuron’s sum and red wires subtract.

Tap it and it tries the inputs 00, 01, 10 and 11 in turn, lighting each neuron that fires, and fills in the truth table: 0, 1, 1, 0. Pick the poster or a 3D model in the Toy tab.

HOW TO PLAY

- **Tap:** A two-layer perceptron solving XOR, which a single perceptron cannot: each input pair lights in turn, pulses run through an OR and a NAND neuron to an AND neuron, and the truth table fills in.
- **Sound:** A blip for each input, a ding when the answer is 1, a knock when it is 0.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

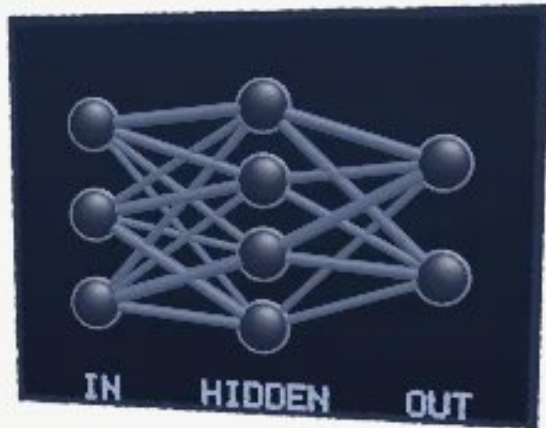
VIEWS

- **View:** Poster (2D), 3D model

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/multilayer-perceptron/

CREDITS

- Built from a recipe in the browser; no outside files.



Neural network

Tap it to send signals forward, then learn backward. Set its size in the Toy tab.



Scan to play with it.

A neural network is made of simple artificial neurons in layers. Each neuron adds up the signals coming in, each multiplied by a weight, and passes on a signal of its own; the weights are what the network learns. In training, a forward pass makes a guess, and backpropagation sends the error back through the network, finding how each weight affects the error; an optimizer then adjusts the weights.

Tap it: pulses run from the inputs through the hidden layer to the outputs, each neuron glowing as strongly as it fires, and one output wins. Then red pulses run back and the wires thicken or thin as the weights change. In the Toy tab, set the number of inputs, hidden layers, neurons and outputs, and pick the poster or a 3D model. The pulses and changing wires illustrate training; this toy repeats a prepared example.

HOW TO PLAY

- **Tap:** A forward pass: a wave of light runs left to right through three layers, each neuron glowing as bright as its activation; then a red backprop pulse runs right to left and the weights shift a little (4.5 s).
- **Sound:** More intense, an electric surge as each layer lights over a dark chord, a deep hit at the answer and crackling current as the pulses run back.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

VIEWS

- **View:** Poster (2D), 3D model

SETTINGS

- **Inputs:** slider
- **Hidden layers:** slider
- **Neurons per hidden layer:** slider
- **Outputs:** slider

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/neural-network/

CREDITS

- Built from a recipe in the browser; no outside files.



Convolutional network

Tap it to read the digit. Draw your own digit in the Toy tab.



Scan to play with it.

A convolutional network, or CNN, is a neural network built for pictures. It slides small filters across the picture: each filter is a little grid of weights that lights up where it finds its pattern, such as a slanted stroke, and makes a feature map. Pooling shrinks the map, keeping the strongest signal in each patch, and later layers combine the features to recognize the whole shape.

Tap it: a glowing 3-by-3 filter slides over a handwritten 7, stamping a feature map tile by tile; the tiles pool into a smaller map, and the scores for the digits 0 to 9 rise with 7 on top. In the Toy tab, pick a 3D view, or draw a digit and a small trained network reads it. Each square on the drawing pad is one pixel of the small 8-by-8 picture the network reads: the lighter its gray, the more ink it holds, and the pad starts with the sample 7.

HOW TO PLAY

- **Tap:** A glowing 3×3 filter slides across a pixel picture of a handwritten 7, stamping a feature map tile by tile; pooling shrinks the map, and a bar chart of the digits 0 to 9 rises with 7 on top (5 s).
- **Sound:** A tick at each step of the filter, a chime at the answer.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

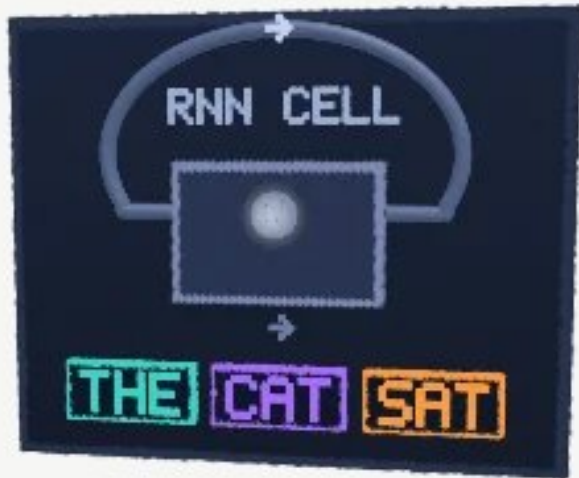
VIEWS

- **View:** Poster (2D), 3D model, 3D, draw a digit

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/cnn/

CREDITS

- **Handwritten digits:** [Optical Recognition of Handwritten Digits \(UCI Machine Learning Repository\)](#) by E. Alpaydin and C. Kaynak (1998), CC BY 4.0.



Recurrent network

Tap it to read a sentence one word at a time. Try the LSTM style in the Toy tab.



Scan to play with it.

A recurrent network, or RNN, reads a sequence, like the words of a sentence, one piece at a time. It keeps a memory called the hidden state: after each word, the state loops back in with the next word, so what it knows builds up as it reads. An LSTM (long short-term memory) adds gates that decide what to forget, what to take in and what to pass on.

Tap it: THE, CAT and SAT rise into the cell one at a time. The cell flashes, and the glowing orb, the hidden state, takes on the word's color mixed with what it carried and runs around the loop. In the Toy tab, pick the LSTM style to see its three gates open and shut like shutters, or a 3D model. This is a scripted sequence, not a trained sentence reader.

HOW TO PLAY

- **Tap:** Word blocks feed in one at a time; the hidden state, a glowing orb, loops back through the cell after each one and changes color as it carries what came before.
- **Sound:** A soft pulse of current as each word goes in and the loop's chord growing a note each time (no rising hum).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

VIEWS

- **Style:** Simple, LSTM
- **View:** Poster (2D), 3D model

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/rnn/

CREDITS

- Built from a recipe in the browser; no outside files.



Transformer

Tap it to watch a prepared next-word example. Try the Encoder–decoder diagram in the Toy tab.



Scan to play with it.

A transformer is a neural network first described in 2017, for translating text. It splits text into tokens, and in each layer every token uses attention: it looks at all the others and weighs how much each one matters to it. Language models built this way predict the next token.

Tap it: arcs jump between THE CAT SAT ON, thicker where attention is stronger, one color for each of two heads; the tiles rise through the layers, and MAT drops in. The Encoder–decoder diagram in the Toy tab is the classic design. Its key: an eye is attention; a lidded eye, masked attention (it only looks back); +, add and norm; », feed forward; a vector, embedding; a wave, positional encoding; a slash, linear; bars, softmax; N×, repeated N times. HELLO WORLD goes in, and given START HOLA, the prepared example ends with MUNDO. The words and attention strengths are prepared; this view does not calculate attention or predict text.

HOW TO PLAY

- **Tap:** A row of token tiles; arcs of light jump between them, thicker where attention is stronger and a different color for each head; the tiles rise through a feed-forward block and up the stack of layers, and a new token tile appears at the end of the row (5 s).
- **Sound:** The first two sounds are warm swells of current instead of electronic shimmers; the last two are unchanged.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

VIEWS

- **View:** Poster (2D), 3D model

SETTINGS

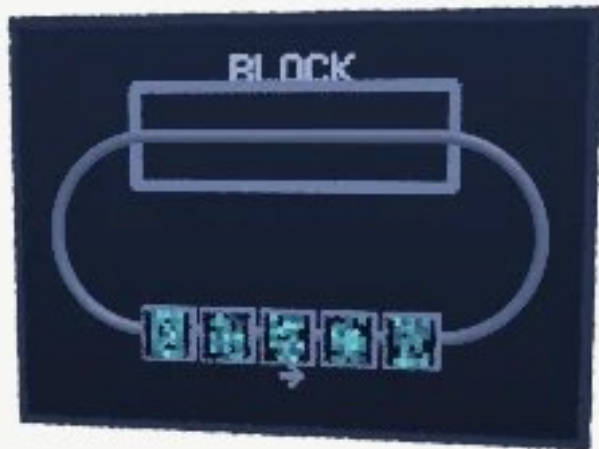
- **Diagram:** Token flow, Encoder–decoder

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/transformer/

CREDITS

- Built from a recipe in the browser; no outside files.



Looped transformer

Tap it: the tiles loop through one block, sharper each lap, until $3 + 4 = 7$.



Scan to play with it.

A looped transformer runs the same transformer block again and again, feeding its output back in as its next input, instead of stacking many different layers. Each pass through the loop can refine the answer a little more, a bit like checking your work, so a small model can spend longer thinking about a harder problem.

Tap it: five tiles ride three laps around a track through one block. Each pass sharpens every tile one step, from noise to a blocky mosaic, to nearly right, to exact, until " $3 + 4 = 7$ " settles with the 7 in gold. Pick the poster or a 3D model in the Toy tab.

HOW TO PLAY

- **Tap:** One transformer block with a track looping through it: the row of tokens rides round the loop several times, going from blurry to sharp on each pass, until the answer settles (5 s).
- **Sound:** A tone that climbs a step on each loop, then resolves.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

VIEWS

- **View:** Poster (2D), 3D model

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/looped-transformer/

CREDITS

- Built from a recipe in the browser; no outside files.

Diffusion model

Tap it: the noise clears, step by step, into a rubber duck.



Scan to play with it.

A diffusion model makes pictures out of noise. It is trained by taking real pictures, adding random noise to them a little at a time until only specks are left, and learning to undo each step. To make a new picture, it starts from pure noise and takes a little away at each of many steps, until a clear picture appears.

Tap it: a cloud of random specks clears in ten steps into a rubber duck while the STEP counter runs down from 50 to 0, then the noise washes back over it. Pick the poster or a 3D model in the Toy tab.

HOW TO PLAY

- **Tap:** A cloud of random specks clears step by step into a crisp small toy (a rubber duck) while a step counter runs down from 50 to 0; then the noise washes back over it (5 s).
- **Sound:** A soft dark swell as the noise clears (no sandy hiss) and a chord rolled on the piano as the duck appears.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

VIEWS

- **View:** Poster (2D), 3D model

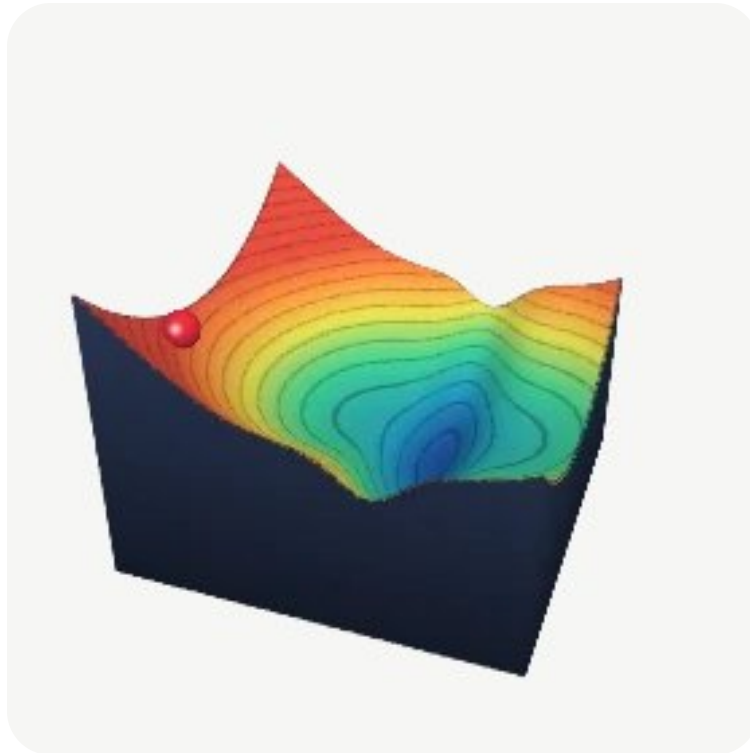
Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/diffusion-model/

CREDITS

- Built from a recipe in the browser; no outside files.

Gradient descent

Tap it to roll the ball downhill.
Try the Learning rate in the Toy tab.



Scan to play with it.

Gradient descent is how most neural networks learn. Picture the network's error as a hilly landscape: the lower the ground, the better its answers. At each step, it works out the gradient, which points toward the steepest increase, and steps in the opposite direction. The learning rate sets how big each step is, and momentum lets it keep some speed from step to step.

Tap it: the ball takes 21 hopping steps downhill, leaving a trail of dots, overshoots the valley and settles in it. In the Toy tab, set the Learning rate: too low, and it creeps down the slope; too high, and it bounces from wall to wall; just right, and it settles. This prepared path illustrates the three settings; descent is not guaranteed to find the lowest valley of every landscape.

HOW TO PLAY

- **Tap:** A ball rolls down a hilly loss landscape, overshoots and settles into a valley.
- **Sound:** A soft tap as the ball lands each of its 21 hops down, in sync, and one as it lands home (the falling tone and whoosh are gone).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Learning rate:** Too low, Just right, Too high

In the Toy tab of the app.

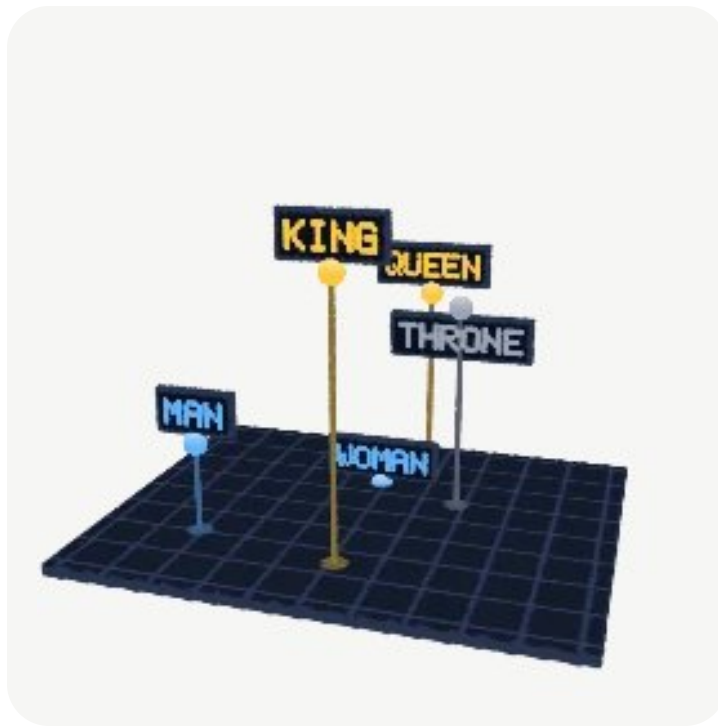
Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/gradient-descent/

CREDITS

- Built from a recipe in the browser; no outside files.

Word vectors

Tap it to work out king – man + woman. Type your own words in the Toy tab.



Scan to play with it.

Word vectors turn each word into a list of numbers, a point in a space with many directions, so that words used in similar ways land near each other. These have 50 numbers per word, learned from a huge amount of text. Directions in the space can carry meaning: the step from man to woman is much like the step from king to queen.

Tap it: an arrow runs out to KING, the step from MAN to WOMAN is added on from there, and it lands right next to QUEEN, which lights up. Type your own $A - B + C$ in the Toy tab, with any of 24,000 common words, and it finds the nearest of the 10,000 most common words, with the runner-up in gray. A text file containing $A - B + C$ works too. These relationships are approximate and can reflect biases in the training text.

HOW TO PLAY

- **Tap:** Word points float in 3D, each with a small sign; arrows add up, king minus man plus woman, and land right next to queen, which lights up (4 s).
- **Sound:** Three chimes, then a bright ding at the answer.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/word-vectors/

CREDITS

- **Word vectors:** GloVe: Global Vectors for Word Representation (Wikipedia 2014 + Gigaword 5, 50 dimensions) by Jeffrey Pennington, Richard Socher and Christopher D. Manning, ODC Public Domain Dedication and License 1.0.



Sorting machine

Tap it to sort the bars. Pick one of eight ways to sort, and how to watch it, in the Toy tab.



Scan to play with it.

A sorting algorithm is a step-by-step recipe a computer follows to put things in order. Bubble sort keeps swapping neighbors that are the wrong way around; quicksort picks one item and splits the rest into smaller and bigger piles; merge sort sorts small groups, then merges them. On a long list, some need far fewer steps than others.

Tap it: eight bars sort themselves, each gliding to its new place, while a counter counts the swaps, or moves for merge sort; then they shuffle back. Pick one of eight algorithms in the Toy tab. Listen too: each comparison plays the two bars' notes softly, each swap the moving bar's note, every algorithm on its own instrument, so you hear how it works. The View option shows the same steps as crates of different sizes, a ring of colored pucks that sorts into a color wheel, or the classic dots. The steps come from each algorithm's real code, checked step by step against separate reference code.

HOW TO PLAY

- **Tap:** A row of colored bars of different heights sorts itself, the bars swapping as solid pieces, by bubble sort, quicksort or merge sort (an option), with a swap counter; then it shuffles again (5 s).
- **Sound:** Each algorithm has its own voice: every comparison plays the two compared bars' notes softly, every swap the moving bar's note, each timed to its step; then the sorted bars play their scale.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

VIEWS

- **View:** Bars, Crates by size, Ring of colors (3D), Dots

SETTINGS

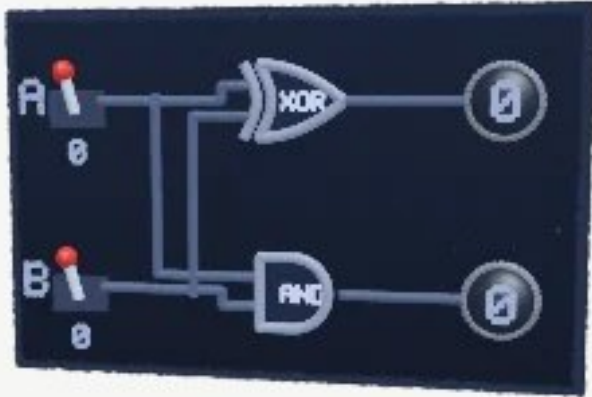
- **Algorithm:** Bubble sort, Quicksort, Merge sort, Insertion sort, Selection sort, Cocktail shaker sort, Shell sort, Heap sort

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/sorting-machine/

CREDITS

- Built from a recipe in the browser; no outside files.



Half adder

Tap it to add $1 + 1$ in binary: the answer is 10.



Scan to play with it.

A half adder adds two bits, each 0 or 1. Its XOR gate gives the sum bit: 1 when exactly one input is 1. Its AND gate gives the carry: 1 when both inputs are 1. With carry written first, $0 + 0$ gives 00, $0 + 1$ and $1 + 0$ give 01, and $1 + 1$ gives 10.

Tap to watch the fixed $1 + 1$ example. Both switches flip to 1, light travels through the gates, and only the carry lamp lights: binary 10 means two. Then the switches reset. There is no incoming carry on a half adder. Two half adders plus an OR gate can make a full adder, which also accepts that carry; chaining full adders adds longer binary numbers.

HOW TO PLAY

- **Tap:** Two input switches flip; light flows along the wires through an XOR gate and an AND gate, whose shapes glow as they fire, and the sum and carry lamps show $1 + 1 = 10$ in binary (3.5 s).
- **Sound:** The buzz near the end is gone; the rest is unchanged.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/half-adder/

CREDITS

- Built from a recipe in the browser; no outside files.

Turing machine

Tap it to run the program on the tape. Type your own number in the Toy tab.



Scan to play with it.

In 1936 the mathematician Alan Turing imagined the simplest possible computer: a long tape of squares, a head that reads and writes one square at a time, and a short table of rules. Each rule says: in this state, reading this symbol, write this, move left or right, and switch to that state. Anything a modern computer can work out, a machine like this can too, given enough tape and time.

Tap it: the head reads a tile, flips it to write, and the tape slides, following the lit row of the rule card, until the machine halts with a bell. Add one turns 1011 into 1100 as the carry ripples left; tap again to count on. The busy beavers write as many 1s as a machine of their size can and still halt.

HOW TO PLAY

- **Tap:** A tape of 0 and 1 tiles runs under a read-write head with a state lamp: the head reads, flips tiles to write and the tape slides, following the rule card, until it halts with a bell.
- **Sound:** A relay click as the head reads, a wooden clack as each tile flips, a short whir as the tape slides, and a bell at the halt.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Program:** Add one, Busy beaver, 2 states, Busy beaver, 3 states

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/turing-machine/

CREDITS

- Built from a recipe in the browser; no outside files.

Difference engine

Tap to turn the crank once; tap again to keep cranking. Type your own polynomial in the Toy tab.



Scan to play with it.

Charles Babbage designed Difference Engine No. 2 in the 1840s to make mathematical tables. Its calculating section was built from his drawings in London in 1991. It uses finite differences: for squares 1, 4, 9, 16, the gaps are 3, 5, 7, and their gaps are always 2. Repeated addition can therefore find the next square without multiplying.

Tap to turn the crank once; tap again for another value. Wheels turn and carries ripple between digits while each difference adds into its neighbor. The starting example is x^2 . In the Toy tab, choose the starting x and type a polynomial in x or n , up to degree three, that gives whole numbers, such as x^3 or $n(n + 1)/2$. The toy has five-digit columns, so values wrap at 100,000; it is a small working illustration, not the full historical engine.

HOW TO PLAY

- **Tap:** Babbage's engine: columns of numbered wheels and a crank.
- **Sound:** The ratchet of the crank, a brass click for each wheel step, a sharper snap for each carry, and a small bell when a new result is ready.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Start at $x =$:** slider

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/difference-engine/

CREDITS

- Built from a recipe in the browser; no outside files.



Enigma machine

Tap the keys or type to code letters; tap the machine to decode. Set the rotors in the Toy tab.



Scan to play with it.

The Enigma was a cipher machine German forces used to keep radio messages secret in World War II. Each key press steps the rotors, including an occasional double step, and a current runs through the plugboard, three wired rotors and a reflector, lighting a different letter. Because the rotors keep turning, the same letter comes out differently each time. Polish mathematicians first broke it in 1932.

Tap a key, or type: the rotors step (with the middle rotor's real double step) and the coded letter lights and goes on the pad. Tap the machine off the keys to decode, or to type the stored message. Tap the pad for a clean sheet; tap the rotors, or press Backspace, to take a letter back. On a keyboard, hold Shift for P and R. Set it in the Toy tab as from a key sheet: three of five rotors, reflector B or C, rings, start letters and up to ten plugboard pairs, shown on the pad. Or pick Barbarossa, 1941, and decode a real German Army message with its published key.

HOW TO PLAY

- **Tap:** An Enigma cipher machine in its wooden box: tap and it types your message, each key going down, the rotors stepping like an odometer and the coded letter lighting on the lampboard; a second tap types the coded text back and gets the message again (about 4 s).
- **Sound:** Heavy key clacks, the ratchet of the rotors stepping, and a faint click as each lamp lights.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Setting:** Set it yourself, Barbarossa, 1941, part 1, Barbarossa, 1941, part 2
- **Left rotor:** Rotor I, Rotor II, Rotor III, Rotor IV, Rotor V
- **Middle rotor:** Rotor I, Rotor II, Rotor III, Rotor IV, Rotor V
- **Right rotor:** Rotor I, Rotor II, Rotor III, Rotor IV, Rotor V
- **Reflector:** B, C
- **Left ring:** A (01), B (02), C (03), D (04), E (05), F (06), G (07), H (08) and 18 more
- **Middle ring:** A (01), B (02), C (03), D (04), E (05), F (06), G (07), H (08) and 18 more
- **Right ring:** A (01), B (02), C (03), D (04), E (05), F (06), G (07), H (08) and 18 more
- **Left start:** A, B, C, D, E, F, G, H and 18 more
- **Middle start:** A, B, C, D, E, F, G, H and 18 more
- **Right start:** A, B, C, D, E, F, G, H and 18 more
- **Plugboard:** type your own
- **Key:** button
- **Clean sheet:** button
- **Step back:** button

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/enigma-machine/

CREDITS

- Built from a recipe in the browser; no outside files.



Bombe

Tap it to search for the Enigma setting. Type a message to break in the Toy tab.



Scan to play with it.

The Bombe was the codebreaking machine of Bletchley Park in England. Alan Turing designed it in 1939, building on a Polish machine, and Gordon Welchman added the diagonal board that made it far faster. Its rows of drums copy Enigma rotors. From a crib, a few words the codebreakers guessed were in a message, it tried rotor settings at great speed and stopped when the logic held. About 200 were built, run day and night mostly by members of the Women's Royal Naval Service, the Wrens.

Tap it: the drums spin through the settings, stop, and the lamp lights; then the message reads out. Simplified here: the plugboard is taken as known, and the drums turn far slower than the real ones.

HOW TO PLAY

- **Tap:** The Bletchley Park codebreaking machine: rows of colored drums spin in their sets through rotor settings, stop together on a possible setting and a lamp lights; then the coded message reads out in plain text (about 5 s).
- **Sound:** The drums' clatter without the low motor buzz; the switch, the stop and the bell are unchanged.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/bombe/

CREDITS

- Built from a recipe in the browser; no outside files.



Jack-o'-lantern

Tap to lift the lid; tap again to put it back.



Scan to play with it.

Learn more: [Jack-o'-lantern on Wikipedia](#)

A jack-o'-lantern is a pumpkin carved with a face and lit from inside by a candle, a symbol of Halloween. The custom comes from Ireland and Britain, where people once carved faces into turnips and potatoes; in North America, the big, soft pumpkin turned out to be much easier to carve.

This one glows through its carved eyes and grin. Tap to lift the lid off the top, and tap again to put it back. The name comes from an old Irish tale of a man called Stingy Jack, who wandered the night carrying a lantern.

HOW TO PLAY

- **Tap:** Lift the lid
- **Sound:** Spooky flicker whoosh.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/jack-o-lantern/

CREDITS

- Built from a recipe in the browser; no outside files.



Snowman

Tap to melt it and build it again. Try the Warmth slider in the Toy tab.



Scan to play with it.

Learn more:
[Snowman on Wikipedia](#)

A snowman is built from big balls of snow rolled across the ground, stacked up and given a face of coal, a carrot nose, stick arms, a scarf and a hat. Snow packs best when it is just at the melting point and a little wet, so the flakes stick together. Fresh snow is mostly air, which is why it is so light and fluffy.

Tap it and it melts: the snowballs shrink, the head first, drips fall, and the hat, nose, coals, arms and scarf drop off one by one, and the meltwater spreads into a puddle around it. Then it builds itself again from the bottom up. The Warmth slider in the Toy tab melts it the same way.

HOW TO PLAY

- **Tap:** Melts into a puddle, then rebuilds itself.
- **Sound:** Drips, then a crunch of snow and a twinkle as it rebuilds.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

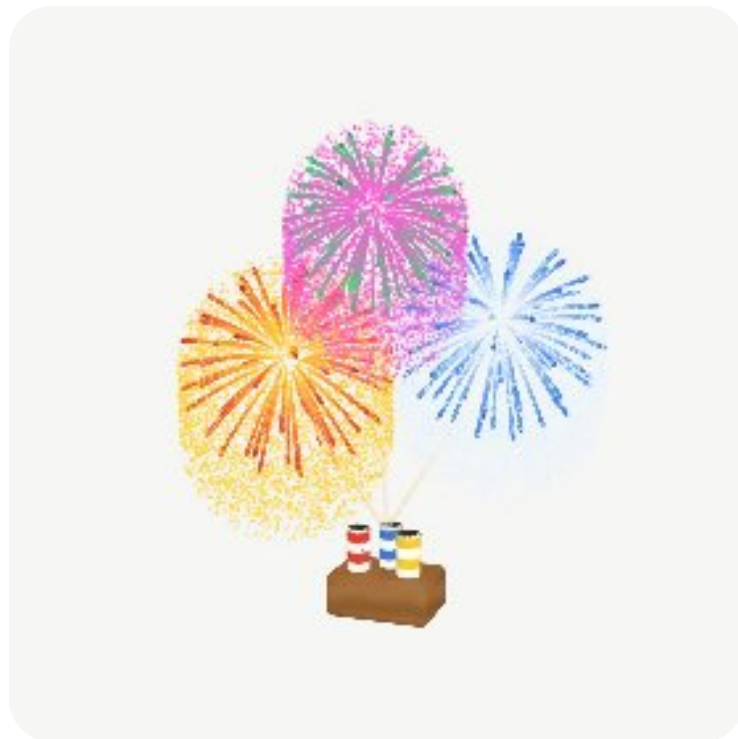
- **Warmth:** slider

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/snowman/

CREDITS

- Built from a recipe in the browser; no outside files.



Fireworks

Tap it to launch a firework; tap quickly for a few at once.



Scan to play with it.

Learn more: [Fireworks on Wikipedia](#)

Fireworks were invented in China more than a thousand years ago. A firework shell is shot high into the air, where it bursts and throws out little pellets called stars, which burn in bright colors. The colors come from metals mixed in: strontium burns red, barium green, copper blue and sodium yellow.

Tap it to launch: a rocket rises from one of the tubes in the crate and bursts in that tube's color. Each tap fires a different tube and a different shell: a round peony, a ring, a drooping willow or a star, so no two in a row look alike. Tap quickly and up to three fly at once, one from each tube.

HOW TO PLAY

- **Tap:** Random shell types each tap; burst colour matches its tube.
- **Sound:** A real firework, the launch's thump and rush (no whistle), a deep boom and the stars' soft crackle; real CC0 recordings now (fireworks-burst.mp3), with the synthesized sound as a fallback.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/fireworks/

CREDITS

- **Sound:** [Firework Explosion 3](#) by a deleted Freesound account (deleted_user_3544904), CC0 1.0.



Decorated tree

Tap to switch the lights on or off.



Scan to play with it.

Learn more: [Christmas tree on Wikipedia](#)

A decorated tree is an evergreen tree, such as a fir or a pine, brought indoors and hung with lights, ornaments and a star on top for the Christmas season. The custom began in Germany about 500 years ago and spread around the world. Evergreens stay green all winter, so they became a sign of life in the darkest time of the year.

The tree starts with its lights off. Tap to switch them on: they sweep up the tree, stay on and cycle through chasing, rippling and steady patterns, while the star glows. Tap again to switch them off. The first electric tree lights were made in 1882.

HOW TO PLAY

- **Tap:** Lights come on and chase in patterns and stay on; the star glows.
- **Sound:** Jingle Bells on a real glockenspiel with a shake of real sleigh bells, at the same tempo.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/decorated-tree/

CREDITS

- **Sound:** [Glockenspiel - C5 \(glock_medium_C5.wav\)](#) by sgossner, CC0 1.0.
- **Sound:** [Sleigh bells hit](#) by Selector, CC0 1.0.



Patterned egg

Tap it to spin it. Pick a pattern and its colors in the Toy tab.



Scan to play with it.

Learn more: [Easter egg on Wikipedia](#)

Decorating eggs with bright patterns is a spring tradition in many countries, especially at Easter. In Ukraine and nearby lands, painted eggs called pysanky are made by drawing lines in melted wax, dipping the egg in dye, and repeating with darker colors; the wax keeps each color where it was drawn.

Tap it to spin it. Pick a folk, striped, dotted, zigzag, flower or star pattern and its two colors in the Toy tab. Try spinning a real hard-boiled egg: it spins smoothly, while a raw one wobbles and stops, because the runny inside sloshes.

HOW TO PLAY

- **Tap:** Spin
- **Sound:** Painted-shell tap.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Pattern:** Folk, Stripes, Dots, Zigzag, Flowers, Stars
- **Colour 1:** pick a color
- **Colour 2:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/patterned-egg/

CREDITS

- Built from a recipe in the browser; no outside files.



Paper lantern

Tap it to set it swinging on its string; tap again for another push.



Scan to play with it.

Learn more: [Paper lantern on Wikipedia](#)

A paper lantern is a light made of thin paper stretched over a frame of ribs, glowing from a light inside. In China and many other places, red lanterns stand for luck and happiness, and thousands are hung up for the Lantern Festival, which ends the Lunar New Year celebrations on the first full moon of the year.

Tap it and it swings on its string like a pendulum, back and forth, slowing a little on each swing until it hangs still again. Tap while it swings to give it another push. Pick a red, gold, teal or purple lantern in the Toy tab.

HOW TO PLAY

- **Tap:** Swing
- **Sound:** Paper rustle.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colour:** Red, Gold, Teal, Purple

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/paper-lantern/

CREDITS

- Built from a recipe in the browser; no outside files.



Diya

Tap to light the ring of diyas; tap again to put them out.



Scan to play with it.

Learn more: [Diya \(lamp\) on Wikipedia](#)

A diya is a small clay lamp that holds oil or butter and a cotton wick. Rows of diyas are lit for Diwali, the festival of lights celebrated by Hindus, Sikhs and Jains, to welcome light and good fortune. Homes are also decorated with rangoli, bright patterns of colored powder or flowers on the floor.

This diya sits on a rangoli. Tap it and its flame flares and grows, and eight small diyas around the pattern light one after another and stay lit. Tap again to put them out. In the Toy tab, set the flame's size or use Blow to make it dip; this is a button, not microphone input.

HOW TO PLAY

- **Tap:** The flame grows and a ring of small diyas lights around it.
- **Sound:** Soft flame whoosh.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Flame:** slider
- **Blow:** button

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/diya/

CREDITS

- Built from a recipe in the browser; no outside files.



Menorah

Tap to put out the candles; tap again to light them one by one.



Scan to play with it.

Learn more:

[Hanukkah menorah](#)
on [Wikipedia](#)

A Hanukkah menorah is a candle holder with nine branches, lit during Hanukkah, the Jewish festival of lights. Hanukkah lasts eight nights, and one more candle is lit each night. The ninth candle, the helper called the shamash, stands apart and is used to light the others. The festival remembers the rededication of the Temple in Jerusalem, when, the story says, one day's oil burned for eight days.

The menorah starts with every candle lit. Tap to put them out, and tap again to light them: the helper candle first, then the others one by one.

HOW TO PLAY

- **Tap:** Light the candles
- **Sound:** A match struck, then each wick catching with a soft flame as it lights, one by one; going out, a small puff as each flame goes out in turn (no big gust).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/menorah/

CREDITS

- Built from a recipe in the browser; no outside files.

Acoustic guitar

Tap it to strum a few chords.



Scan to play with it.

Learn more: [Acoustic guitar on Wikipedia](#)

An acoustic guitar has six strings stretched over a hollow wooden body. Plucking a string makes it vibrate, and the body and the round soundhole make the sound bigger and warmer. Pressing a string down against the metal frets on the neck makes it shorter, and a shorter string plays a higher note.

Tap it to strum: the guitar rocks with the stroke, the strings bend and shake one after another, and rings of light pulse out of the soundhole as it plays four chords. Pick a sunburst, natural or cherry finish in the Toy tab. The six strings are usually tuned, from lowest to highest, to E, A, D, G, B and E.

HOW TO PLAY

- **Tap:** Keep the strum; strings vibrate visibly.
- **Sound:** A real strummed chord progression (synth plucks).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Finish:** Sunburst, Natural, Cherry

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/guitar/

CREDITS

- Built from a recipe in the browser; no outside files.



Snare drum

Tap it for a drum roll; tap again quickly for a faster, longer roll.



Scan to play with it.

Learn more: [Snare drum on Wikipedia](#)

A snare drum is a shallow drum with a skin, called a head, stretched across each side. Under the bottom head runs a set of thin metal wires, the snares, which rattle against it every time the top is hit, giving the drum its crisp, buzzing crack. A drum roll is many fast strokes, played one stick after the other so quickly that they blur into one sound.

Tap it and the sticks play a roll. Tap again quickly, and each tap steps the roll up to a faster speed and makes it last longer, through four speeds in all. Pick the color of the shell in the Toy tab.

HOW TO PLAY

- **Tap:** Sticks play a roll; fast repeated taps make faster, longer rolls.
- **Sound:** Snare hits with rattle.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

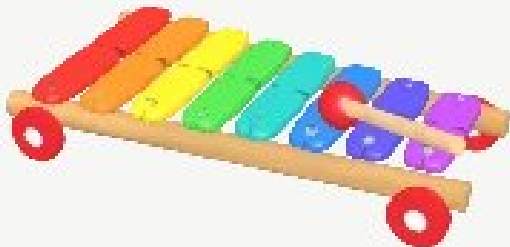
- **Shell:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/drum/

CREDITS

- Built from a recipe in the browser; no outside files.



Xylophone

Tap a bar, or drag across the bars for a glissando. Tap the mallet or frame to play a scale.



Scan to play with it.

Learn more:
[Xylophone on Wikipedia](#)

A xylophone is a row of tuned wooden bars that you strike with a mallet. Each bar's length sets its note: the shorter the bar, the higher it sounds. The name comes from the Greek for "wood" and "sound".

This toy one has eight colored bars, a scale from C up to the next C. Tap a bar and the mallet swings over and strikes it, and the bar jumps and plays its note, so you can play a tune. Tap the mallet, a wheel or the ends of the frame instead, and it plays the whole scale. Drag along the bars for a run of notes, called a glissando.

HOW TO PLAY

- **Tap:** Play a scale
- **Sound:** The scale it already plays, as mallet tones.
- **Drag:** plays with the toy itself; a drag beside it turns the view.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Strike:** button

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/xylophone/

CREDITS

- Built from a recipe in the browser; no outside files.



Toy piano

Tap or drag along the keys to play them, or tap the case to play Twinkle, Twinkle.



Scan to play with it.

Learn more: [Toy piano on Wikipedia](#)

A toy piano is a small piano with no strings. Each key works a tiny hammer, and the hammer strikes a metal rod held at one end, which rings like a little bell. Shorter rods sound higher, so the rods get shorter from left to right. Toy pianos were first made in the 1800s, and the composer John Cage wrote a whole suite for one in 1948.

This one has 18 keys, from C up to the F an octave and a half higher, and its back and lid are open so you can see inside. Tap a key and its hammer swings up and strikes its rod, which shivers as the note rings. Tap anywhere else to hear “Twinkle, Twinkle, Little Star.” Pick its color in the Toy tab.

HOW TO PLAY

- **Tap:** Tap a key: it dips, its hammer swings up and strikes its rod, and the rod shivers as the note rings.
- **Sound:** The plinky, bell-like tone of struck metal rods, made in the page.
- **Drag:** plays with the toy itself; a drag beside it turns the view.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Color:** pick a color
- **Strike:** button

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/toy-piano/

CREDITS

- Built from a recipe in the browser; no outside files.



Grand piano

Tap or drag along the keys to play them. Tap elsewhere for a song; the song bar plays it all.



Scan to play with it.

Learn more: [Piano on Wikipedia](#)

A grand piano is a keyboard instrument whose strings lie flat under a lid shaped like a wing. Each of its 88 keys throws a felt hammer up against its strings, and most notes have dampers that lift to let them ring until the key comes up. The right pedal lifts every damper at once. Bartolomeo Cristofori built the first pianos in Italy around 1700.

Tap a key and it dips, its hammer strikes and its damper lifts. Tap anywhere else for the opening of Für Elise, or pick Clair de lune, Gymnopédie No. 1 or Ode to Joy in the Toy tab. The song bar plays, pauses, loops and slows a song, and you can open a MIDI file of your own or paste a tune in ABC notation.

HOW TO PLAY

- **Tap:** Tap a key: it dips, its hammer flies up under the open lid and strikes the strings, and the damper lifts while the note rings.
- **Sound:** A real acoustic piano: recorded CC0 notes (one every four semitones, each played at its key's pitch), with the damper falling when a key comes up.
- **Drag:** plays with the toy itself; a drag beside it turns the view.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Play a key:** button

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/grand-piano/

CREDITS

- **Sound:** [A0.ogg \(88 piano keys, long reverb\)](#) by TEDAgame, CC0 1.0.
- **Sound:** [C#1.ogg \(88 piano keys, long reverb\)](#) by TEDAgame, CC0 1.0.
- **Sound:** [F1.ogg \(88 piano keys, long reverb\)](#) by TEDAgame, CC0 1.0.
- **Sound:** [A1.ogg \(88 piano keys, long reverb\)](#) by TEDAgame, CC0 1.0.
- **Sound:** [C#2.ogg \(88 piano keys, long reverb\)](#) by TEDAgame, CC0 1.0.
- **Sound:** [F2.ogg \(88 piano keys, long reverb\)](#) by TEDAgame, CC0 1.0.
- **Sound:** [A2.ogg \(88 piano keys, long reverb\)](#) by TEDAgame, CC0 1.0.
- **Sound:** [C#3.ogg \(88 piano keys, long reverb\)](#) by TEDAgame, CC0 1.0.
- **Sound:** [F3.ogg \(88 piano keys, long reverb\)](#) by TEDAgame, CC0 1.0.
- **Sound:** [A3.ogg \(88 piano keys, long reverb\)](#) by TEDAgame, CC0 1.0.
- **Sound:** [C#4.ogg \(88 piano keys, long reverb\)](#) by TEDAgame, CC0 1.0.
- **Sound:** [F4.ogg \(88 piano keys, long reverb\)](#) by TEDAgame, CC0 1.0.
- **Sound:** [A4.ogg \(88 piano keys, long reverb\)](#) by TEDAgame, CC0 1.0.
- **Sound:** [C#5.ogg \(88 piano keys, long reverb\)](#) by TEDAgame, CC0 1.0.
- **Sound:** [F5.ogg \(88 piano keys, long reverb\)](#) by TEDAgame, CC0 1.0.
- **Sound:** [A5.ogg \(88 piano keys, long reverb\)](#) by TEDAgame, CC0 1.0.
- **Sound:** [C#6.ogg \(88 piano keys, long reverb\)](#) by TEDAgame, CC0 1.0.
- **Sound:** [F6.ogg \(88 piano keys, long reverb\)](#) by TEDAgame, CC0 1.0.
- **Sound:** [A6.ogg \(88 piano keys, long reverb\)](#) by TEDAgame, CC0 1.0.
- **Sound:** [C#7.ogg \(88 piano keys, long reverb\)](#) by TEDAgame, CC0 1.0.
- **Sound:** [F7.ogg \(88 piano keys, long reverb\)](#) by TEDAgame, CC0 1.0.
- **Sound:** [A7.ogg \(88 piano keys, long reverb\)](#) by TEDAgame, CC0 1.0.
- **Sound:** [C8.ogg \(88 piano keys, long reverb\)](#) by TEDAgame, CC0 1.0.



Upright piano

Tap or drag along the keys to play them. Tap elsewhere for a ragtime; the song bar plays it all.



Scan to play with it.

Learn more: [Piano on Wikipedia](#)

An upright piano stands its strings on end, so a full piano fits against a wall. Its hammers swing forward onto the strings instead of flying up, and a spring brings them back. Uprights filled homes, schools and dance halls, and a slightly out-of-tune one gives the bright, jangly honky-tonk sound of ragtime.

This one has its upper front panel off, so you see the row of hammers, the dampers above them and the strings behind. Tap a key and its hammer strikes. Tap anywhere else for the opening of The Entertainer, a rag Scott Joplin published in 1902. Open a MIDI file of your own, or paste a tune in ABC notation, in the Toy tab.

HOW TO PLAY

- **Tap:** Tap a key to play it: the front panel is off, so you see the row of hammers swing forward onto the upright strings.
- **Sound:** A brighter, slightly honky-tonk upright voice, so it sounds different from the grand.
- **Drag:** plays with the toy itself; a drag beside it turns the view.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Play a key:** button

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/upright-piano/

CREDITS

- Built from a recipe in the browser; no outside files.



Harpsichord

Tap a key to pluck its string, or drag along the keys. Tap anywhere else for a minuet.



Scan to play with it.

Learn more:
[Harpsichord on Wikipedia](#)

A harpsichord plucks its strings instead of striking them. Each key lifts a thin wooden jack, and a small quill on the jack catches the string on the way up. Because a pluck sounds the same however hard you press, players shape their music by timing. Harpsichords were the main keyboard of the 1600s and 1700s.

This one is painted in the French style, with black naturals and pale sharps. Tap a key and its jack rises, plucks, and the string quivers until the jack's felt stops it. Tap anywhere else for the Minuet in G from the notebook Bach kept for his wife, Anna Magdalena. Open a MIDI file of your own in the Toy tab.

HOW TO PLAY

- **Tap:** Tap a key: its jack rises and the quill plucks the string, which you see quiver.
- **Sound:** A bright, plucked harpsichord voice, made in the page.
- **Drag:** plays with the toy itself; a drag beside it turns the view.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Play a key:** button

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/harpsichord/

CREDITS

- Built from a recipe in the browser; no outside files.

Electronic keyboard

Tap or drag along the keys; a colored button changes the voice. Tap elsewhere for a song.



Scan to play with it.

Learn more:

[Electronic keyboard on Wikipedia](#)

An electronic keyboard makes its sounds with circuits instead of strings. It can sound like a piano, an organ, a synthesizer or vibes, and many can play a drum beat along with you. Learning keyboards light up the key to play next.

This one has 61 keys. The four colored buttons pick its voice: piano, organ, synth and vibes. Tap anywhere else and a song starts: each key lights up just before its note, the little screen scrolls the title and the drum pads flash with the beat. Pick Ode to Joy, Twinkle, Twinkle, Little Star or Frère Jacques in the Toy tab, or open a MIDI file of your own.

HOW TO PLAY

- **Tap:** Tap a key to play it.
- **Sound:** Four synth voices and a simple drum beat, all made in the page.
- **Drag:** plays with the toy itself; a drag beside it turns the view.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Play a key:** button
- **Voice:** button

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/electronic-keyboard/

CREDITS

- Built from a recipe in the browser; no outside files.



Rocket

Tap it to launch the rocket.



Scan to play with it.

Learn more: [Rocket on Wikipedia](#)

A rocket flies by pushing hot gas out of its engine very fast. As the gas rushes down, it pushes the rocket up, just as a balloon zooms off when you let its air out. A rocket carries its own oxygen to burn its fuel, so unlike a plane it can fly where there is no air at all, out in space.

Tap it to launch: the arm of the tower swings away, the engine lights, clouds of smoke billow across the pad, and the rocket shakes, lifts off and climbs faster and faster until it is gone. Then a new one stands ready on the pad. Pick its color in the Toy tab.

HOW TO PLAY

- **Tap:** Launch
- **Sound:** The flickering roar of the exhaust over a deep rumble.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colour:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/rocket/

CREDITS

- Built from a recipe in the browser; no outside files.



Helicopter

Tap to take off; tap again to land.



Scan to play with it.

Learn more:
[Helicopter on Wikipedia](#)

A helicopter lifts itself with a big rotor on top: its long blades are thin, spinning wings. By tilting the blades, the pilot can fly up, down, forward, backward or sideways, or hover in one place. The small rotor on the tail stops the body from spinning around the other way.

Its rotors are always turning. Tap to take off: it lifts into the air, tips its nose down a little and hovers, bobbing gently. Tap again to land. Pick its color in the Toy tab. Helicopters can land in places planes cannot, so they are used for rescues at sea and in the mountains.

HOW TO PLAY

- **Tap:** Take off or land
- **Sound:** Real rotor blades chopping; landing, the chop slowing as it fades.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colour:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/helicopter/

CREDITS

- **Sound:** [Helicopter Flyby / Pass](#) by mil0001, CC0 1.0.



Hot-air balloon

Tap it to fire the burner: the balloon swells and climbs.



Scan to play with it.

Learn more: [Hot air balloon on Wikipedia](#)

A hot-air balloon flies because hot air is lighter than the cool air around it. A burner under the big fabric envelope heats the air inside, and the balloon floats up; as the air cools, it sinks again. Pilots can only go up and down, so they steer by finding winds at different heights that blow the way they want to go. The first people to fly in a hot-air balloon took off in Paris in 1783.

Tap it to fire the burner: a big flame roars, the envelope swells and glows warm, and the balloon climbs well up, then drifts back down. Pick stripes or a rainbow, and two colors, in the Toy tab.

HOW TO PLAY

- **Tap:** Burner roars, the envelope swells and it rises higher.
- **Sound:** Burner roar.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

VIEWS

- **Style:** Stripes, Rainbow

SETTINGS

- **Colour 1:** pick a color
- **Colour 2:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/hot-air-balloon/

CREDITS

- Built from a recipe in the browser; no outside files.



Steam train

Tap it to blow the whistle.



Scan to play with it.

Learn more: [Steam locomotive on Wikipedia](#)

A steam locomotive burns coal or wood to boil water in a long boiler. The steam pushes pistons back and forth, and rods turn that push into the turning of the big driving wheels. The first steam locomotive to pull a train on rails ran in Wales in 1804, and for more than a hundred years steam trains carried people and goods around the world.

The engine puffs smoke from its chimney. Tap it to blow the whistle: a jet of white steam shoots up from the whistle and a big cloud billows from the chimney. Pick the color of the engine in the Toy tab.

HOW TO PLAY

- **Tap:** Blow the whistle
- **Sound:** A real steam locomotive chuffing, fading with its steam; no bell, horn or whistle.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colour:** pick a color

In the Toy tab of the app.

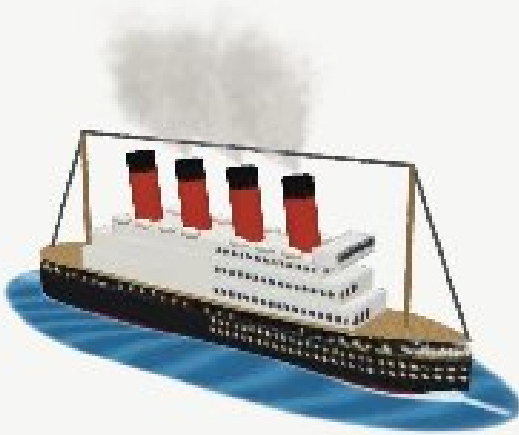
Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/steam-train/

CREDITS

- **Sound:** [d_s478_underbridge.wav](#) by relwin, CC0 1.0.

Ocean liner

Tap it to sound the horn.



Scan to play with it.

Learn more: [Ocean liner on Wikipedia](#)

An ocean liner is a big passenger ship that sailed on a regular route, or line, across an ocean. Before airliners, liners were the way to cross the Atlantic, and the fastest took about five days. The largest carried thousands of people, with dining rooms, lounges and decks for walking.

The liner rocks gently on the waves, smoke drifting from its funnels. Tap it to sound the horn: steam blasts from the horn and the funnels smoke harder. Pick the color of the funnels in the Toy tab. A ship's horn is deep and loud so that other ships can hear it from miles away, even in fog.

HOW TO PLAY

- **Tap:** Sound the horn
- **Sound:** Deep ship horn.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Funnels:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/ocean-liner/

CREDITS

- Built from a recipe in the browser; no outside files.



Sports car

Tap it to rev the engine.
Hands-on: push it and it rolls on its wheels.



Scan to play with it.

Learn more: [Sports car on Wikipedia](#)

A sports car is a small, light car with a powerful engine, built for speed and for taking corners well. It sits low to the ground, which helps it stay steady, and the wing on its back works like an upside-down airplane wing: at speed, the air pushes the car down onto the road, so its tires grip better.

Tap it to rev the engine: the car rocks on its springs, the wheels spin and a puff of exhaust shoots out of the back. In the Toy tab, pick its color and turn its racing stripes on or off.

HOW TO PLAY

- **Tap:** Rev the engine
- **Sound:** The same real rev, now fading out with the exhaust (gone at 1.8 s).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colour:** pick a color
- **Stripes:** switch

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/sports-car/

CREDITS

- **Sound:** [Car Engine Revving.WAV](#) by DigPro120, CC0 1.0.



Bus

Tap it to stop for passengers: the lights flash and the doors open.



Scan to play with it.

Learn more: [Bus on Wikipedia](#)

A bus carries many people along a set route, stopping to let them on and off. In the United States and Canada, school buses are painted a bright yellow that is easy to see, and when they stop, red lights flash and a stop sign swings out so that traffic waits while children cross. Double-decker buses have two floors, and they are a famous sight in London.

Tap it: the school bus stops, its warning lights flash, the stop sign swings out and the doors open onto the lit doorway, then everything folds away again. Pick the school bus or a double-decker, which flashes its lights and opens its middle door, in the Toy tab.

HOW TO PLAY

- **Tap:** Stop sign swings out, lights flash, doors open.
- **Sound:** Horn beep and door hiss.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

VIEWS

- **Style:** School bus, Double-decker

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/bus/

CREDITS

- Built from a recipe in the browser; no outside files.



Propeller plane

Tap it to loop the loop.



Scan to play with it.

Learn more:
[Airplane on Wikipedia](#)

This propeller plane is a biplane: it has two sets of wings, one above the other, held together by struts and wires. Two wings give a lot of lift for a light, slow plane. The spinning propeller at the front pulls it through the air. The Wright brothers' Flyer made a sustained, controlled powered flight in 1903; it was a biplane too.

Tap it to loop the loop: it climbs, flies up and over on its back in a big circle, and comes out level again. Pick the colors of the body and the wings in the Toy tab.

HOW TO PLAY

- **Tap:** Loop the loop
- **Sound:** A real piston engine with the propeller's beat; real CC0 recordings now (propeller-plane-engine.mp3), with the synthesized sound as a fallback.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Body:** pick a color
- **Wings:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/propeller-plane/

CREDITS

- **Sound:** [Aeroplane Passing Close.wav](#) by paulprit, CC0 1.0.



Jet airliner

Tap it to climb and bank left, then right.



Scan to play with it.

Learn more: [Jet airliner on Wikipedia](#)

A jet airliner is a big passenger plane driven by jet engines. Each engine sucks in air, squeezes it, burns fuel in it and blasts it out of the back, pushing the plane forward. Airliners cruise about 10 to 12 kilometers (6 to 7 miles) high, where the thin air lets them fly fast while burning less fuel.

Tap it and it climbs nose up while banking hard to the left, then hard to the right. Banking means tilting the wings: a plane turns by leaning into the turn, just as a bicycle does. Pick the color of the tail in the Toy tab.

HOW TO PLAY

- **Tap:** Climbs and banks hard left then right.
- **Sound:** No whistle; a softer, natural rush of the engines.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Tail:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/jet/

CREDITS

- Built from a recipe in the browser; no outside files.



Sailboat

Tap it for a gust of wind: it heels over, then rocks back upright.



Scan to play with it.

Learn more:
[Sailboat on Wikipedia](#)

A sailboat is pushed along by the wind in its sails. A sail works like a wing, so a sailboat can even sail at an angle toward the wind. Under the water, a keel resists sideways motion; its weight helps a ballasted boat stay upright.

Tap it for a big gust: the boat heels over, leaning far to one side, with spray flying at the bow and the stern, then it rocks back upright. Sailors lean out over the high side to help hold the boat up. Pick the colors of the sails and the stripe in the Toy tab.

HOW TO PLAY

- **Tap:** A big gust heels it over with spray.
- **Sound:** Wind and water spray.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

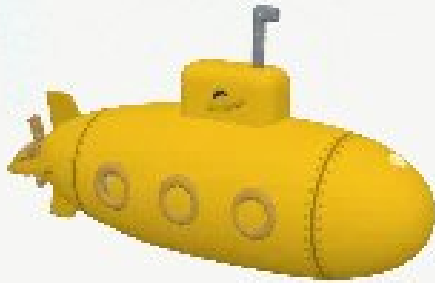
- **Sails:** pick a color
- **Stripe:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/sailboat/

CREDITS

- Built from a recipe in the browser; no outside files.



Submarine

Tap it to dive; then it surfaces and raises its periscope.



Scan to play with it.

Learn more:
[Submarine on Wikipedia](#)

A submarine is a ship that can travel underwater. To dive, it lets seawater into big tanks, making it heavy enough to sink; to come back up, it blows the water out with compressed air. A periscope, a long tube with mirrors inside, lets the crew see above the surface while the submarine stays hidden below.

Tap it to dive: the periscope drops, bubbles rush from the hull and it sinks under the water. Then it rises back to the surface and raises its periscope again. Pick its color in the Toy tab.

HOW TO PLAY

- **Tap:** Dives with bubbles, then surfaces and raises the periscope.
- **Sound:** Sonar ping.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colour:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/submarine/

CREDITS

- Built from a recipe in the browser; no outside files.



Bicycle

Tap it to ring the bell and spin the wheels.



Scan to play with it.

Learn more: [Bicycle on Wikipedia](#)

A bicycle has two wheels, one behind the other, and pedals that turn a chain, which turns the back wheel. It stays up while it moves, because the rider keeps steering the front wheel a little under the bike, balancing it without thinking. The first bicycles with pedals were built in France in the 1860s.

Tap it: the bell shakes and rings, the wheels spin through five fast turns, and the pedals go around twice. Bicycles are among the most efficient ways to travel ever invented: a rider uses less energy to go a mile than a walker does. Pick the color of the frame in the Toy tab.

HOW TO PLAY

- **Tap:** Pedals spin, the wheels turn and the bell rings clearly.
- **Sound:** A real bicycle bell's ding-ding.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Frame:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/bicycle/

CREDITS

- **Sound:** [Bicycle Bell.wav](#) by PanosA, CC0 1.0.



Tractor

Tap it to make it chug: it shakes and puffs smoke.



Scan to play with it.

Learn more: [Tractor on Wikipedia](#)

A tractor is a strong farm machine for pulling heavy things: plows, trailers and all kinds of tools. Its big back wheels have deep treads to grip soft, muddy ground, and its small front wheels steer. Tractors are built for pulling power, not speed.

Its wheels turn slowly and smoke rises from its exhaust pipe. Tap it to make it chug: the engine shakes the tractor, the wheels turn faster and thick puffs of smoke rise from the pipe. Pick its color in the Toy tab. Before tractors, farmers used horses and oxen to pull their plows.

HOW TO PLAY

- **Tap:** Chug chug
- **Sound:** A slow diesel's putt-putt with its clatter; real CC0 recordings now (tractor-engine.mp3), with the synthesized sound as a fallback.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Colour:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/tractor/

CREDITS

- **Sound:** [Tractor whirring](#) by vonis22, CC0 1.0.

Flying saucer

Tap to switch the beam off or on.



Scan to play with it.

Learn more: [Flying saucer on Wikipedia](#)

A flying saucer is an imaginary spaceship from science fiction, shaped like a round, flat saucer. The name was first used in 1947, after a pilot said he had seen shiny objects skimming across the sky. Anything seen in the sky that no one can identify is called a UFO, an unidentified flying object; most turn out to be planes, balloons, birds or bright planets.

This saucer hovers over a field with its rim lights turning, and its glowing beam lifts a cow into the air. Tap to switch the beam off, and the cow settles back on the grass; tap again to switch it on.

HOW TO PLAY

- **Tap:** Beam on or off
- **Sound:** A deep pulsing hum and a heavy rush as it lifts (no whistling theremin).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/ufo/

CREDITS

- Built from a recipe in the browser; no outside files.



Eiffel Tower

Tap it for its night show of lights and fireworks.



Scan to play with it.

Learn more: [Eiffel Tower on Wikipedia](#)

The Eiffel Tower is an iron lattice tower in Paris, built by the engineer Gustave Eiffel's company for the World's Fair of 1889. It is about 330 meters (1,083 feet) tall, and it was the tallest structure in the world until 1930.

Tap it for a night show: the ironwork lights up gold, sparkling white lights climb the tower, and four fireworks burst around it. The real tower sparkles like this for five minutes every hour after dark.

HOW TO PLAY

- **Tap:** Sparkling lights run up the tower and fireworks burst around it.
- **Sound:** Real fireworks, four bursts with booms and soft crackle (no twinkling); real CC0 recordings now (fireworks-burst.mp3), with the synthesized sound as a fallback.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/eiffel-tower/

CREDITS

- **Sound:** [Firework Explosion 3](#) by a deleted Freesound account (deleted_user_3544904), CC0 1.0.



Washington Monument

Tap it for a day in a few seconds: the sun crosses and the shadow swings around.



Scan to play with it.

Learn more:
[Washington Monument on Wikipedia](#)

The Washington Monument is a tall stone obelisk in Washington, D.C., built to honor George Washington, the first president of the United States. It is about 169 meters (555 feet) tall. Building began in 1848, stopped for more than 20 years, and was finished in 1884; the stone above the point where work paused is a slightly different color.

Tap it for a whole day in a few seconds, like a giant sundial: the sun rises behind the monument, arcs over and sets, the obelisk's shadow swings around the lawn in front, and the flags ripple in the breeze.

HOW TO PLAY

- **Tap:** The reflecting pool shimmers and the flags wave.
- **Sound:** Only the soft chime at noon (no wind).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

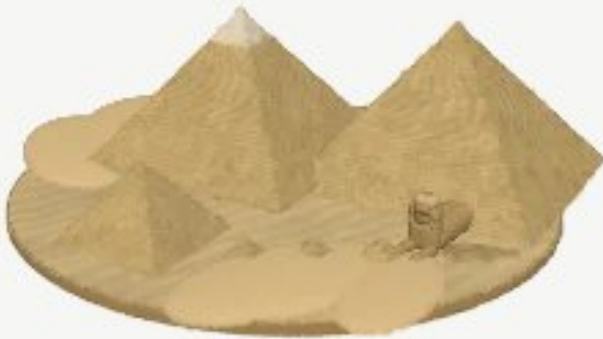
SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/washington-monument/

CREDITS

- Built from a recipe in the browser; no outside files.



Pyramids of Giza

Tap it: a tiny flying saucer beams up a camel made of sand.



Scan to play with it.

Learn more: [Giza pyramid complex on Wikipedia](#)

The Pyramids of Giza stand at the edge of the desert near Cairo, in Egypt. They were built about 4,500 years ago as tombs for three kings, or pharaohs: Khufu, Khafre and Menkaure. The Great Pyramid of Khufu was first about 146 meters (481 feet) tall, and for almost 4,000 years it was the tallest building in the world. It is the only one of the Seven Wonders of the Ancient World still standing.

Tap it for a playful visitor: a tiny flying saucer glides in and switches on its beam, sand streams up into the shape of a camel, which floats up into the saucer, and the saucer wobbles happily and zips away.

HOW TO PLAY

- **Tap:** A tiny UFO (the Flying saucer) hovers over the Great Pyramid and beams up a camel-shaped cloud of sand.
- **Sound:** Theremin warble, sand hiss and a zip away.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/pyramids/

CREDITS

- Built from a recipe in the browser; no outside files.



Twisting supertall

Tap it to twist the tower further and send light up its glass.



Scan to play with it.

A supertall is a skyscraper more than 300 meters (about 1,000 feet) tall. Some are built to twist as they rise: each floor sits turned a little from the one below. The twist is not just for looks: wind flowing past a tall building makes it sway, and a twisted shape breaks the wind up, so the tower sways less.

Tap it and the floors wring around further, in twelve solid bands with the top turning most, while a ring of light runs up the glass; then they unwind with a little sway as a second ring runs up. Pick the color of the glass in the Toy tab.

HOW TO PLAY

- **Tap:** The tower twists further and its lights run upward.
- **Sound:** The lift's low whirl as the floors wring round and a soft low bell each time the light reaches the top (no rising tone or bright dings).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Glass:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/supertall/

CREDITS

- Built from a recipe in the browser; no outside files.



Lighthouse

Tap to switch the light off or on.



Scan to play with it.

Learn more:
[Lighthouse on Wikipedia](#)

A lighthouse is a tower with a bright light at the top that guides ships at night and warns them away from rocks and dangerous coasts. Each lighthouse flashes in its own pattern and is painted in its own stripes or colors, so sailors can tell which one they are looking at, by night or by day. Many use a special lens, invented by Augustin Fresnel in the 1820s, that gathers the light into a strong beam.

This lighthouse starts with its beam sweeping around. Tap to switch the light off, and tap again to switch it on. Pick the color of its stripes in the Toy tab.

HOW TO PLAY

- **Tap:** Light on or off
- **Sound:** Foghorn.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Stripes:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/lighthouse/

CREDITS

- Built from a recipe in the browser; no outside files.



Statue of Liberty

Tap it to make the torch flare and send up golden sparks.



Scan to play with it.

Learn more: [Statue of Liberty on Wikipedia](#)

The Statue of Liberty stands on Liberty Island in New York Harbor. It was a gift from the people of France to the United States and was dedicated in 1886. The sculptor Frédéric Auguste Bartholdi designed it, and Gustave Eiffel, the engineer of the Eiffel Tower, designed its iron frame. Its skin is thin copper, which slowly turned green in the weather.

Tap it and the torch flares: the flame leaps to twice its size in a halo of light, a warm glow spreads down the statue, and golden sparks drift up and away on the breeze and burn out.

HOW TO PLAY

- **Tap:** The torch flame flares bright and sparkles drift from it.
- **Sound:** The whoosh and the bell, no crackling clicks.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/statue-of-liberty/

CREDITS

- Built from a recipe in the browser; no outside files.

White House

Tap it: the fountain shoots up and the windows light up one by one.



Scan to play with it.

Learn more: [White House on Wikipedia](#)

The White House, in Washington, D.C., is the home and office of the president of the United States. It was designed by the architect James Hoban and built from 1792 to 1800, and John Adams was the first president to live there. It has 132 rooms, and its walls are painted white.

Tap it for an evening scene: the fountain on the south lawn shoots up a tall jet that falls back as spray, the lights come on in the windows one by one and the flag ripples at the top of its pole; then the jet sinks and the lights go out again.

HOW TO PLAY

- **Tap:** The fountain on the lawn sprays higher and the flag raises.
- **Sound:** The spray's splash and a soft, steady rush of water falling back (no patter of clicks).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/white-house/

CREDITS

- Built from a recipe in the browser; no outside files.



Leaning Tower of Pisa

Tap to drop two balls from the top. Set the lean in the Toy tab.



Scan to play with it.

Learn more: [Leaning Tower of Pisa on Wikipedia](#)

The Leaning Tower of Pisa, in Italy, is the bell tower of the city's cathedral. Building began in 1173, and the tower started to lean while it was still being built, because the ground under it is soft; it took about 200 years to finish. Engineers straightened it a little from 1990 to 2001, and it now leans about 4 degrees.

The story goes that Galileo dropped two balls of different weights from the top to show that they fall together. Tap it and try it: a big iron ball and a small bronze one roll off the top and land at the same moment. Set the lean with the Lean slider in the Toy tab.

HOW TO PLAY

- **Tap:** Two balls of different sizes drop from the top and land together.
- **Sound:** The tower grinds, then two thuds at the same moment.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Lean:** slider

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/leaning-tower/

CREDITS

- Built from a recipe in the browser; no outside files.

Colosseum

Tap it for a chariot race around the arena.



Scan to play with it.

Learn more: [Colosseum on Wikipedia](#)

The Colosseum is a huge oval arena in Rome, Italy, built of stone and concrete about 2,000 years ago, and finished in the year 80. It could seat about 50,000 people, who came to watch shows and games. Much of it still stands, and it is one of the most visited places in the world.

Tap it for a chariot race: a crowd fills the seats, and four chariots in the old racing teams' colors, red, white, blue and green, race around the arena, swapping the lead, while the crowd jumps and cheers. In ancient Rome, the big chariot races were really held at the Circus Maximus, a long racetrack nearby.

HOW TO PLAY

- **Tap:** Non-violent: a tiny chariot race loops the arena while the crowd cheers (no fighting, per the no-gore rule).
- **Sound:** A real crowd in the stands and horses' hooves trotting; real CC0 recordings now (colosseum-crowd.mp3, colosseum-hooves.mp3), with the synthesized sound as a fallback.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/colosseum/

CREDITS

- **Sound:** [cheering and clapping crowd 1](#) by AlaskaRobotics, CC0 1.0.
- **Sound:** [Horse and Carriage.mp3](#) by maadmacs, CC0 1.0.



Parthenon

Tap it for a procession. Pick the temple today or as it was in the Toy tab.



Scan to play with it.

Learn more:
[Parthenon on Wikipedia](#)

The Parthenon is a marble temple on the Acropolis, the rocky hill above Athens, in Greece. It was built from 447 to 432 BC for the goddess Athena, the city's protector. Its rows of columns look perfectly straight, but they bulge and lean very slightly, so that from a distance they look straighter.

Tap it for a procession like the great festival the people of Athens held for Athena: eight robed figures, some carrying baskets and jars on their heads, walk in single file along the temple and across its steps. In the Toy tab, see the temple as it stands today or as it looked when it was new.

HOW TO PLAY

- **Tap:** Tiny robed figures walk between the columns in procession.
- **Sound:** A lyre plays a walking tune.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

VIEWS

- **Style:** Today, Ancient

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/parthenon/

CREDITS

- Built from a recipe in the browser; no outside files.



Stonehenge

Tap it for the midsummer sunrise shining through the stones.



Scan to play with it.

Learn more:
[Stonehenge on Wikipedia](#)

Stonehenge is a ring of great standing stones on Salisbury Plain, in England. It was built in stages from about 5,000 years ago; some of its smaller stones, the bluestones, were brought from Wales, more than 200 kilometers (140 miles) away. Its biggest stones stand in pairs with a stone laid across the top, called trilithons.

Stonehenge is lined up with the sun: on the summer solstice, the longest day of the year, the sun rises in line with its entrance. Tap it: the sun comes up over the far bank, framed by the great trilithon, a golden beam shines through the stones, and they glow gold.

HOW TO PLAY

- **Tap:** Solstice sunrise: the sun rises through the stones and lights them gold.
- **Sound:** A low drone and a bell as the stones light.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/stonehenge/

CREDITS

- Built from a recipe in the browser; no outside files.



Big Ben

Tap it: the hands spin around, the dials glow and the bell swings.



Scan to play with it.

Learn more: [Big Ben on Wikipedia](#)

Big Ben is the nickname of the great bell in the clock tower of the Houses of Parliament in London, England, and people often use it for the whole tower. The tower was finished in 1859 and, since 2012, has been called the Elizabeth Tower. The bell weighs about 13.7 tonnes, and the clock has four huge dials, one on each side.

The hands show the real time where you are. Tap it: the hands spin all the way around and land back on the right time, all four dials glow, and the bell swings in the open belfry.

HOW TO PLAY

- **Tap:** Hands spin to the hour, the clock face glows and the bell swings.
- **Sound:** Big Ben chimes (Westminster quarters, simplified).
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/big-ben/

CREDITS

- Built from a recipe in the browser; no outside files.



Taj Mahal

Tap it for moonlight: night falls and the white dome glows.



Scan to play with it.

Learn more: [Taj Mahal on Wikipedia](#)

The Taj Mahal is a white marble tomb in Agra, in India. The emperor Shah Jahan had it built in memory of his wife, Mumtaz Mahal; the work took from about 1632 to 1653, and about 20,000 workers built it. Its white marble is inlaid with patterns of colored stone, and it stands in a garden with a long reflecting pool.

Tap it for moonlight: night falls over the garden and the buildings, the moon rises behind the Taj, the great dome glows white against the night sky, and moonlight shimmers on the rippling pool; then the day comes back.

HOW TO PLAY

- **Tap:** Moonlight: the dome glows and the reflecting pool ripples.
- **Sound:** A slow sitar phrase.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/taj-mahal/

CREDITS

- Built from a recipe in the browser; no outside files.

Castle

Tap to lower the drawbridge; tap again to raise it.



Scan to play with it.

Learn more: [Castle on Wikipedia](#)

A castle is a fortified residence. Many medieval castles had stone walls and towers; some had moats, while earlier ones were built partly or entirely of wood. The drawbridge across the moat could be pulled up on chains, so the gate was closed off. Castles were small towns inside: kitchens, stables, a hall for feasts and a well for water.

The drawbridge starts up. Tap to lower it, and five small knights march out, the leader carrying a banner; tap again and they march back in and the drawbridge rises. Pick the color of the roofs in the Toy tab.

HOW TO PLAY

- **Tap:** The drawbridge drops and tiny knights march out with a banner.
- **Sound:** Chain rattle and a trumpet.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Roofs:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/castle/

CREDITS

- Built from a recipe in the browser; no outside files.



Pagoda

Tap it to ring the wind chimes; lanterns and doors light up.



Scan to play with it.

Learn more: [Pagoda on Wikipedia](#)

A pagoda is a tower of many roofs, one above the other, built at Buddhist temples across China, Korea, Japan and much of Asia. The idea came from the stupa, a mound that held holy relics in ancient India. Tall wooden pagodas in Japan have stood through many earthquakes for centuries, partly thanks to a great central pillar that helps them sway without falling.

Tap it: wind chimes at the corners of every roof swing, one tier after another, and two stone lanterns by the path light up while the doors glow. Pick the color of the timber in the Toy tab.

HOW TO PLAY

- **Tap:** Wind chimes swing on every roof tier and lanterns light up.
- **Sound:** Wind chimes.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

- **Timber:** pick a color

In the Toy tab of the app.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/pagoda/

CREDITS

- Built from a recipe in the browser; no outside files.



Windmill

Tap it for a gust of wind that spins the sails hard.



Scan to play with it.

Learn more:
[Windmill on Wikipedia](#)

A windmill uses the wind to do work. The wind turns its big sails, and gears inside turn that into the turning of heavy millstones, which grind grain into flour; some windmills pump water instead. In the Netherlands, windmills pumped water out of low land for hundreds of years, so that people could live and farm there.

Its sails always turn in the breeze. Tap it for a gust of wind that spins the sails up hard for three extra turns, then they slow back to their steady pace.

HOW TO PLAY

- **Tap:** Faster sails; the gust spins them up hard.
- **Sound:** Creaking sails and wind.
- **Drag:** turns it.
- **Pinch or scroll** to zoom; **double-tap** to reset the view.

SETTINGS

None: tap it, turn it, share it.

Play with it on its page: ryanjosephkamp.github.io/splashery/site/toys/windmill/

CREDITS

- Built from a recipe in the browser; no outside files.